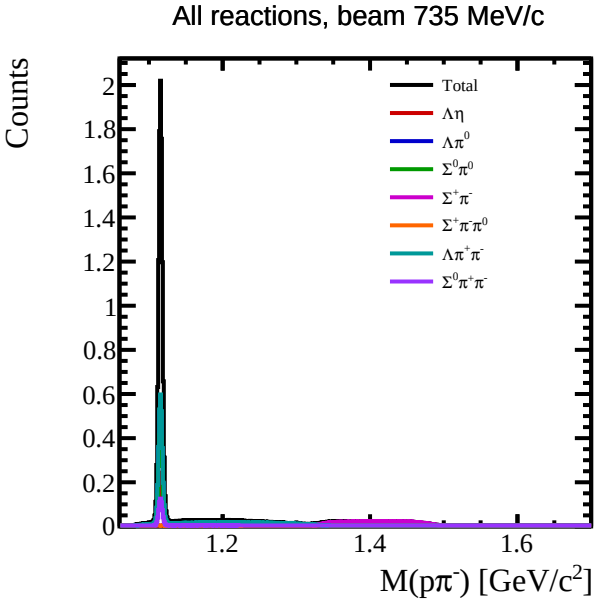
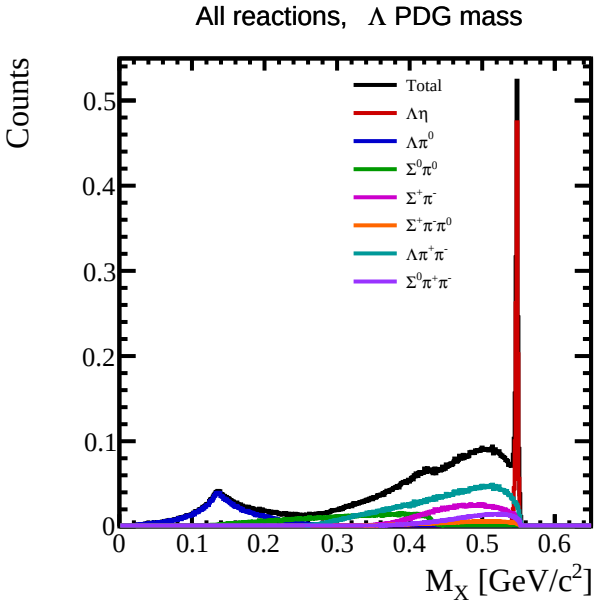
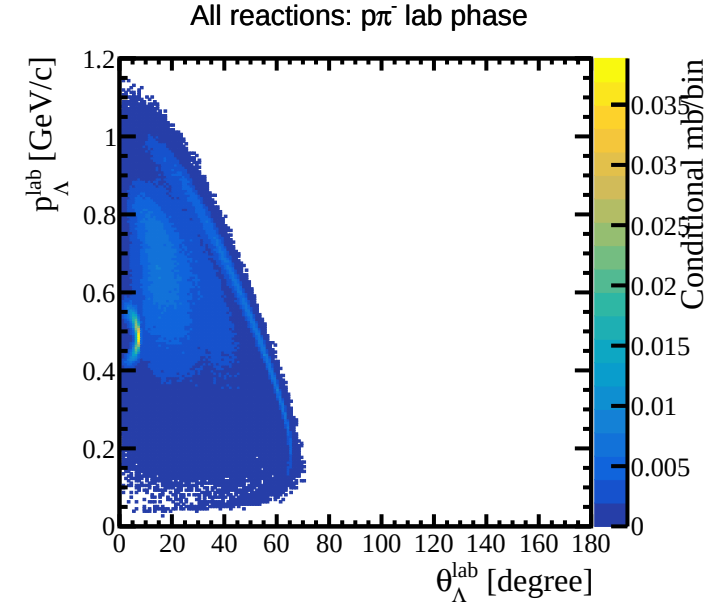
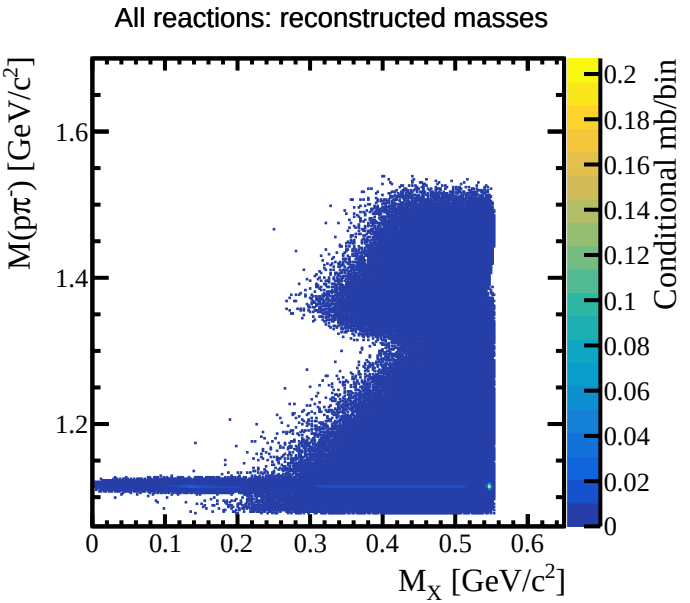
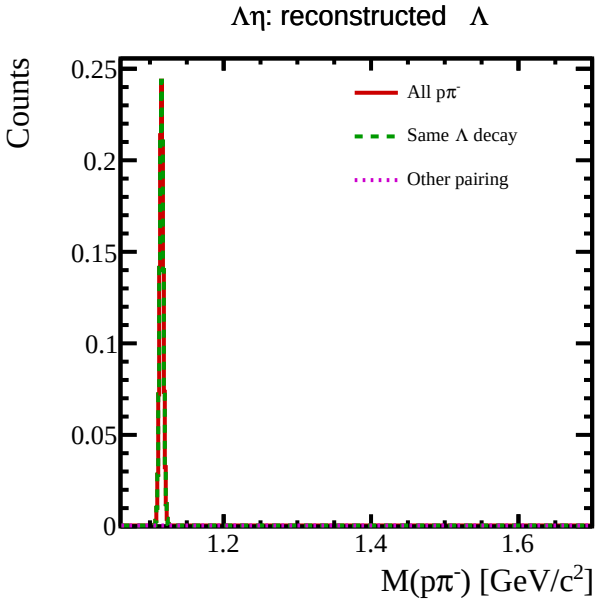
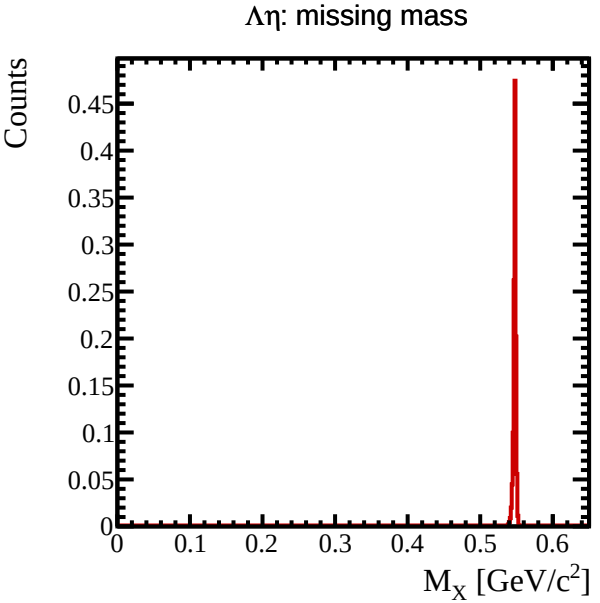

All reactions, beam 735 MeV/c
Counts
M(pπ⁻) [GeV/c²]
Total
Λη
Λπ⁰
Σ⁰π⁰
Σ⁺π⁻
Σ⁺π⁻π⁰
Λπ⁺π⁻
Σ⁰π⁺π⁻

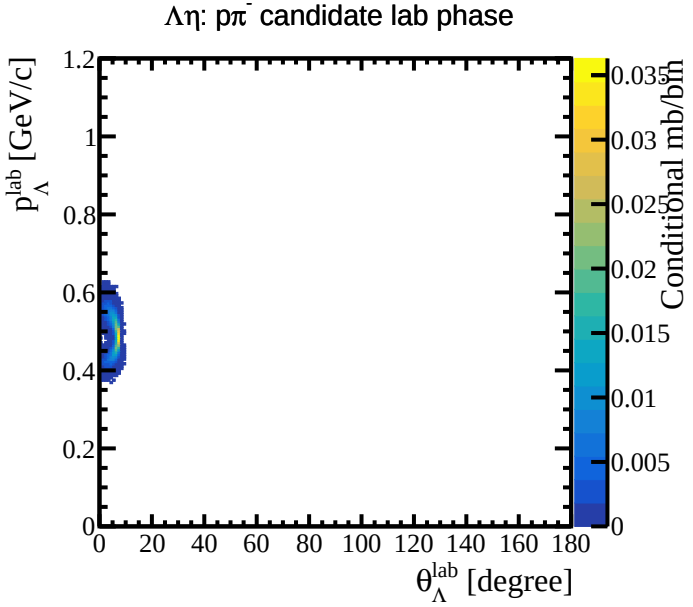

All reactions, Λ PDG mass
Counts
M_X [GeV/c²]
Total
Λη
Λπ⁰
Σ⁰π⁰
Σ⁺π⁻
Σ⁺π⁻π⁰
Λπ⁺π⁻
Σ⁰π⁺π⁻

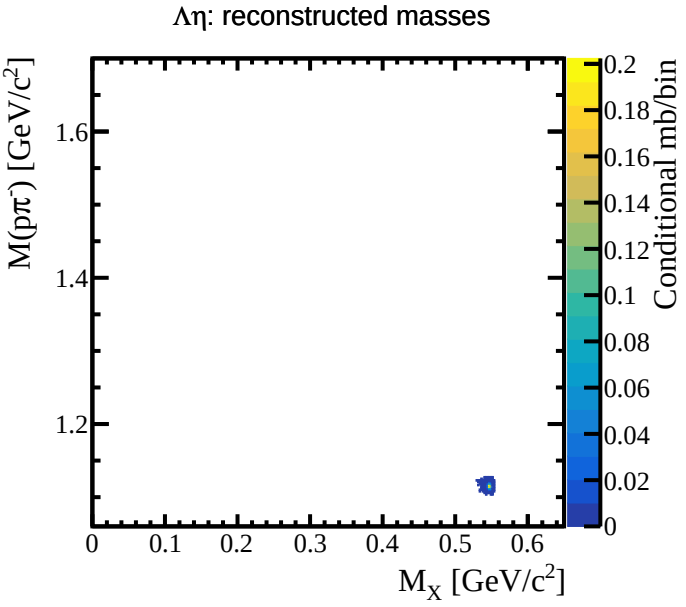

All reactions: pπ⁻ lab phase
p_Λ^lab [GeV/c]
θ_Λ^lab [degree]
Conditional mb/bin

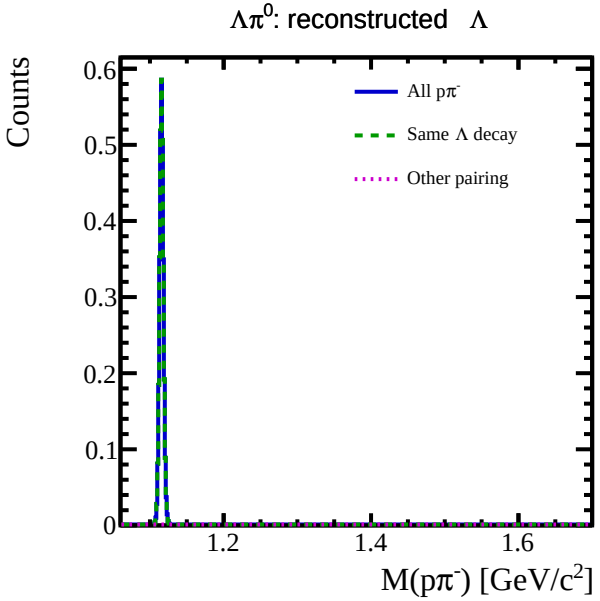

All reactions: reconstructed masses
M(pπ⁻) [GeV/c²]
M_X [GeV/c²]
Conditional mb/bin

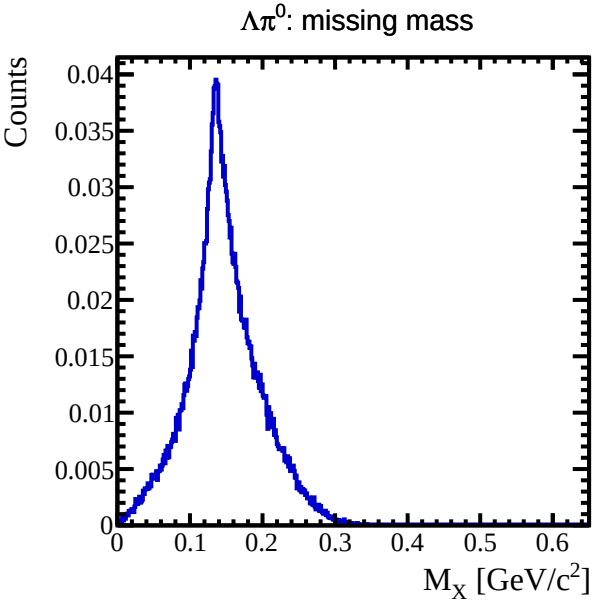

Λη: reconstructed Λ
Counts
All pπ-
Same Λ decay
Other pairing
M(pπ-) [GeV/c2]

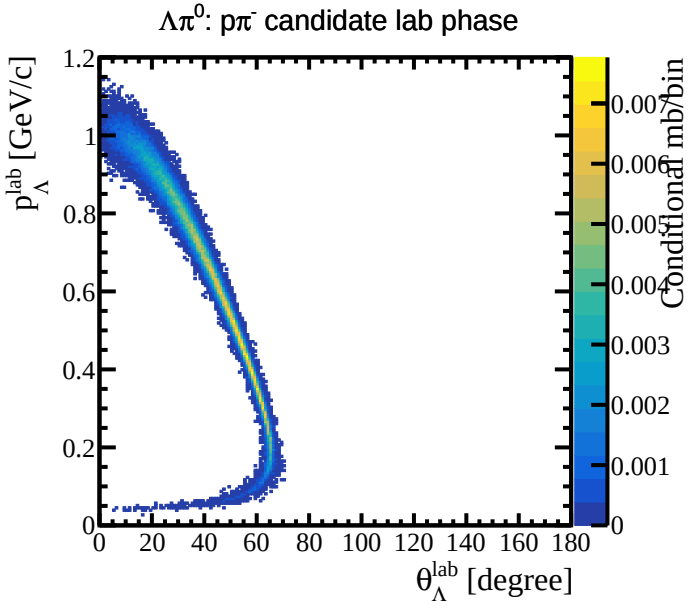

Λη: missing mass
Counts
MX [GeV/c2]

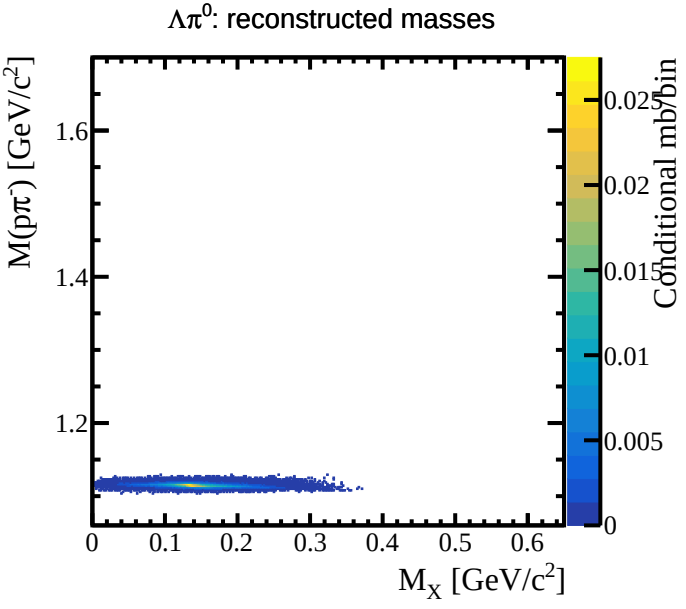

Λη: pπ- candidate lab phase
pΛlab [GeV/c]
θΛlab [degree]
Conditional mb/bin

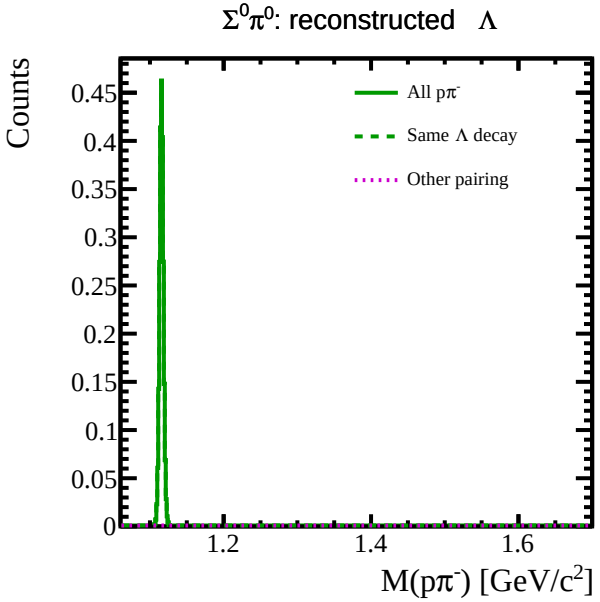

Λη: reconstructed masses
M(pπ-) [GeV/c2]
MX [GeV/c2]
Conditional mb/bin

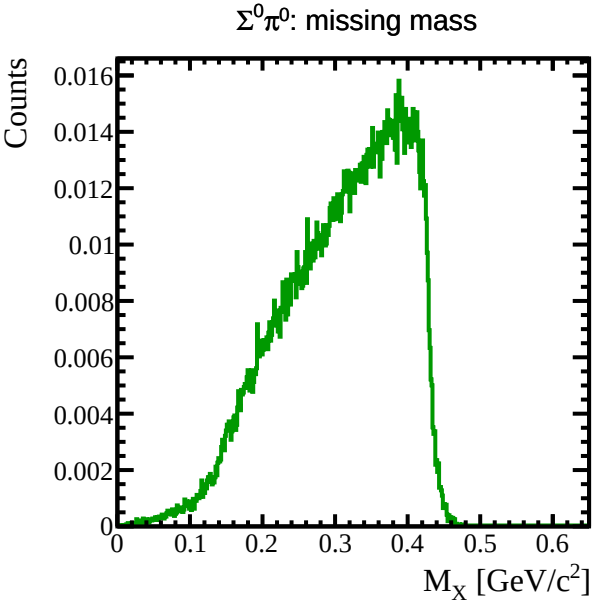

Λπ0: reconstructed Λ
Counts
All pπ-
Same Λ decay
Other pairing
M(pπ-) [GeV/c2]

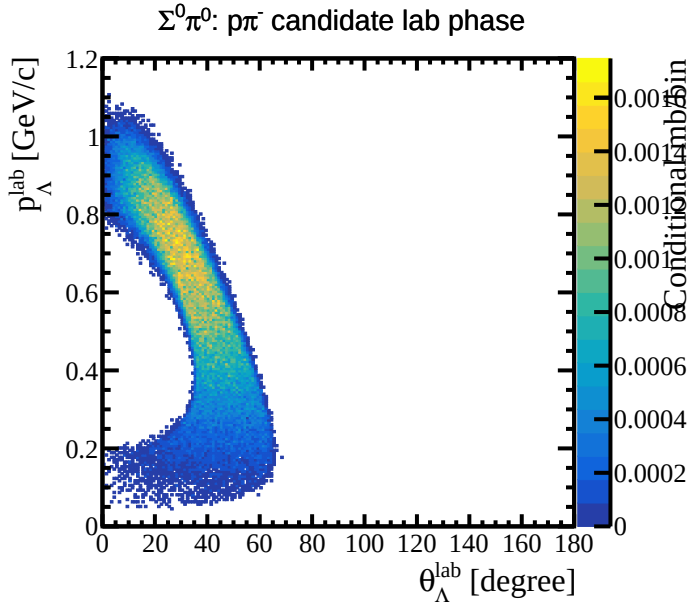

Λπ0: missing mass
Counts
MX [GeV/c2]

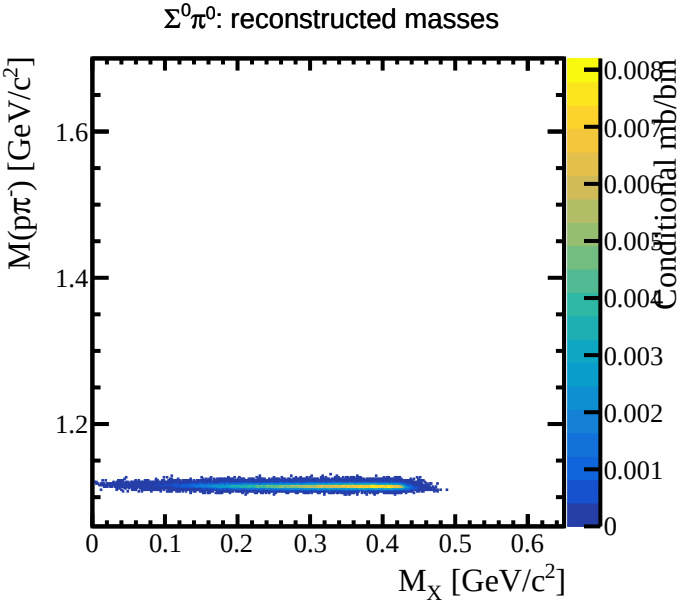

Λπ0: pπ- candidate lab phase
pΛlab [GeV/c]
θΛlab [degree]
Conditional mb/bin

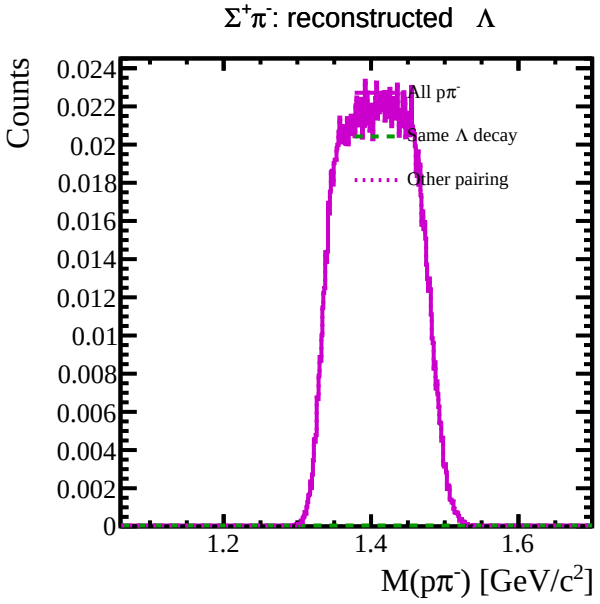

Λπ0: reconstructed masses
M(pπ-) [GeV/c2]
MX [GeV/c2]
Conditional mb/bin

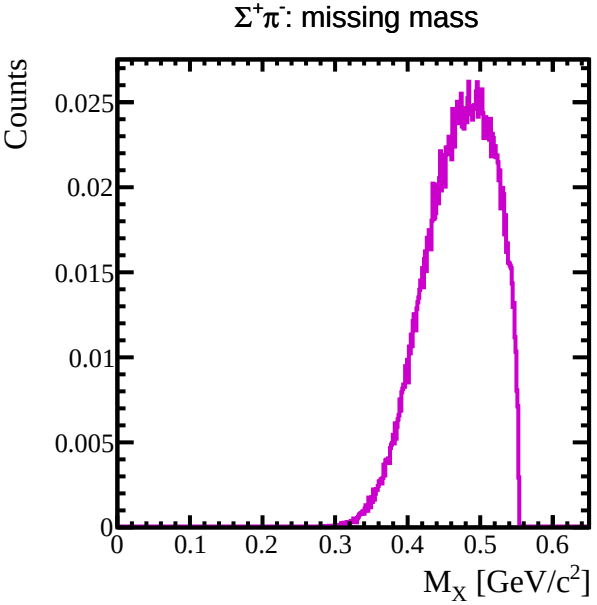

$\Sigma^0\pi^0$: reconstructed Λ
Counts
All $p\pi^-$
Same Λ decay
Other pairing
$M(p\pi^-)$ [GeV/c^2]

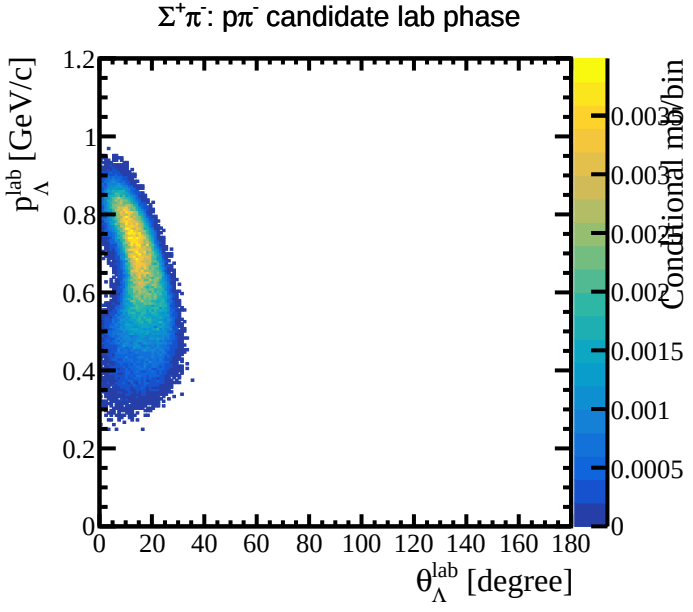

$\Sigma^0\pi^0$: missing mass
Counts
M_X [GeV/c^2]

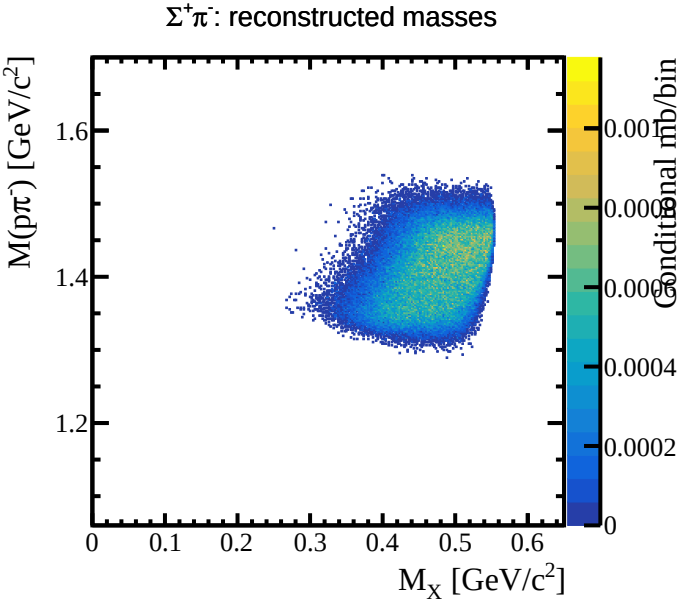

$\Sigma^0\pi^0$: $p\pi^-$ candidate lab phase
p_Λ^{lab} [GeV/c]
θ_Λ^{lab} [degree]
Conditional mb/bin

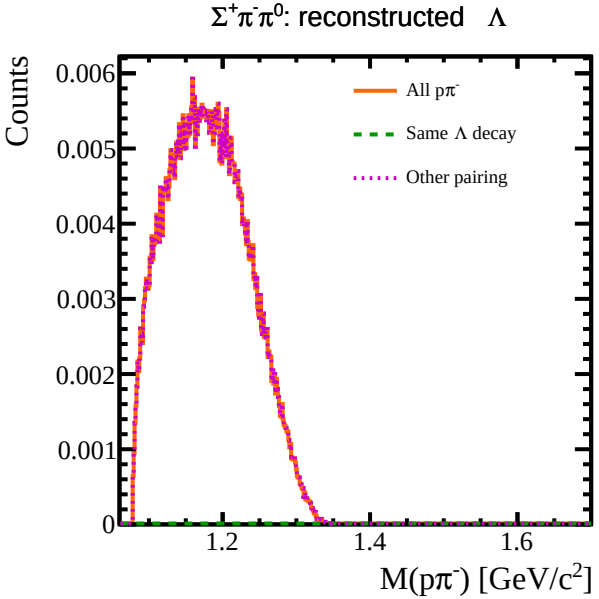

$\Sigma^0\pi^0$: reconstructed masses
$M(p\pi^-)$ [GeV/c^2]
M_X [GeV/c^2]
Conditional mb/bin

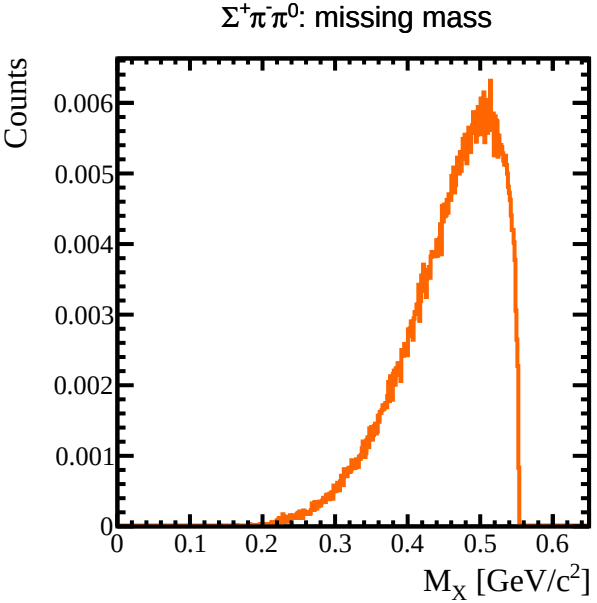

Σ+π-: reconstructed Λ
Counts
All pπ-
Same Λ decay
Other pairing
M(pπ-) [GeV/c2]

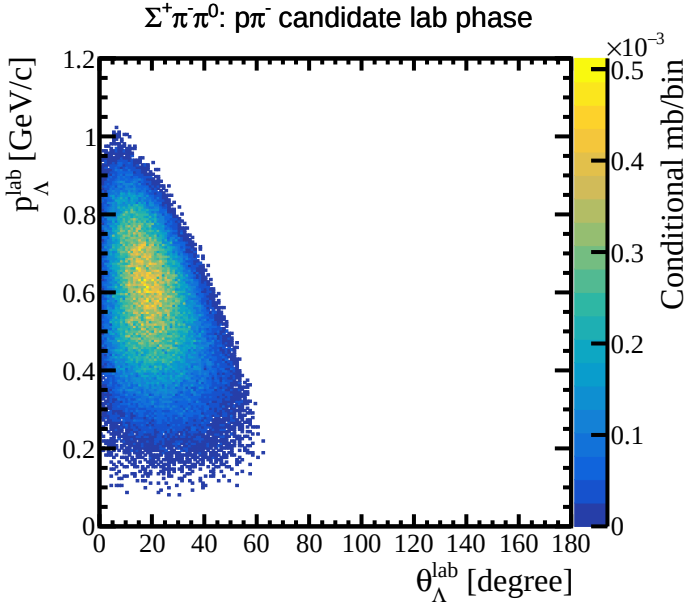

Σ+π-: missing mass
Counts
MX [GeV/c2]

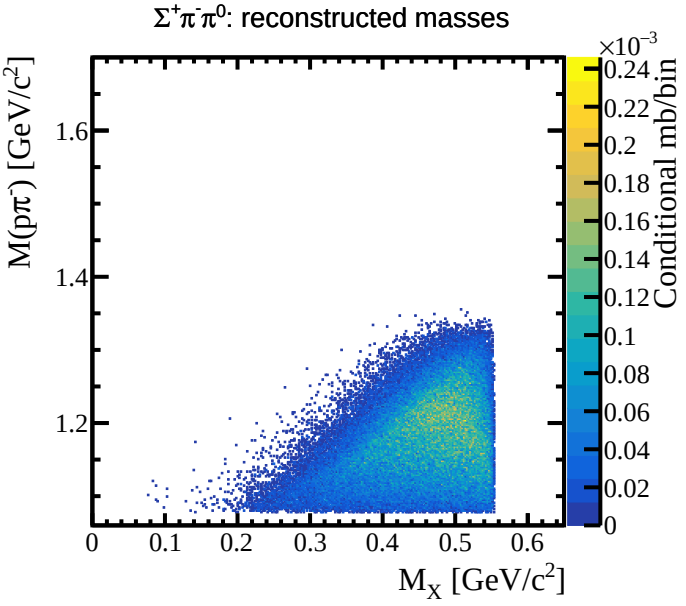

Σ+π-: pπ- candidate lab phase
pΛlab [GeV/c]
θΛlab [degree]
Conditional mb/bin

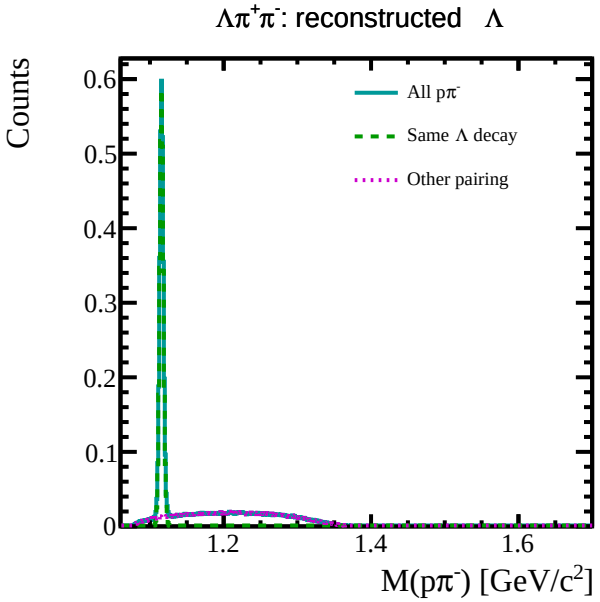

Σ+π-: reconstructed masses
M(pπ-) [GeV/c2]
MX [GeV/c2]
Conditional mb/bin

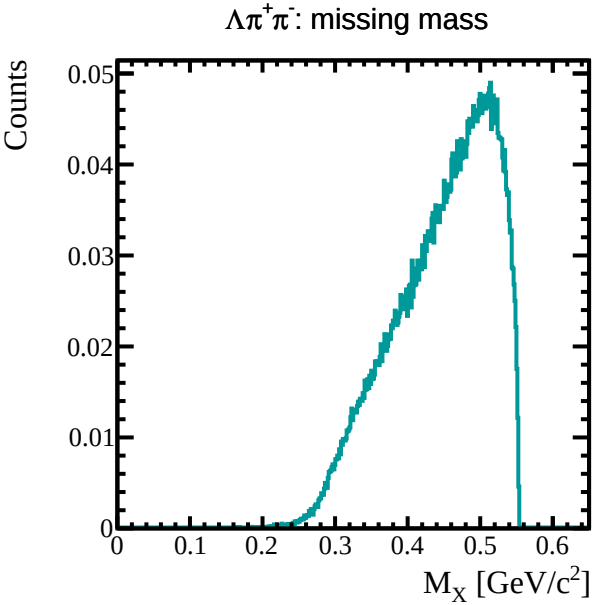

Σ+π-π0: reconstructed Λ
Counts
All pπ-
Same Λ decay
Other pairing
M(pπ-) [GeV/c2]

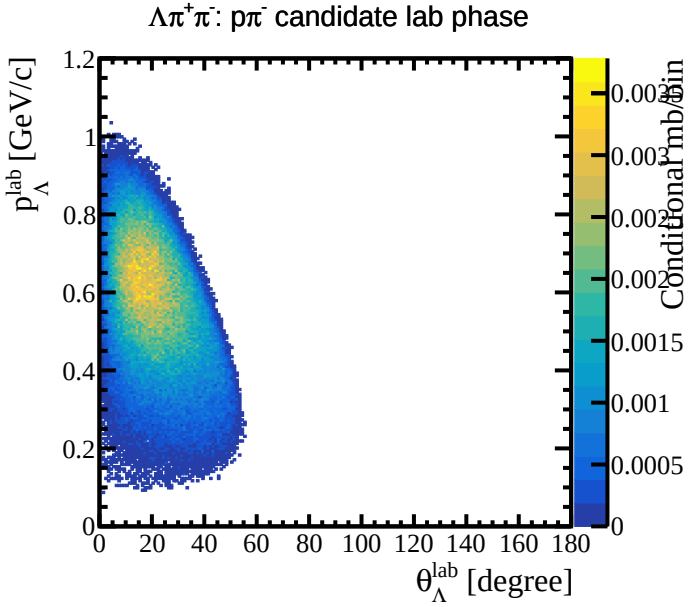

Σ+π-π0: missing mass
Counts
MX [GeV/c2]

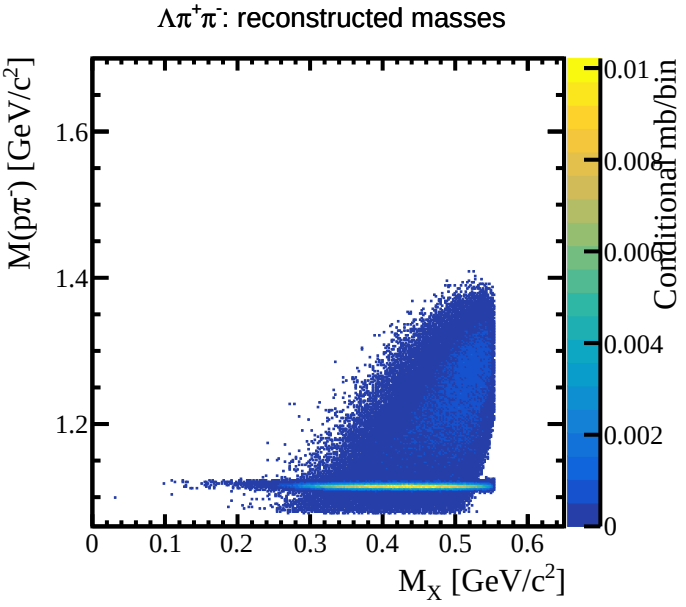

Σ+π-π0: pπ- candidate lab phase
pΛlab [GeV/c]
θΛlab [degree]
Conditional mb/bin

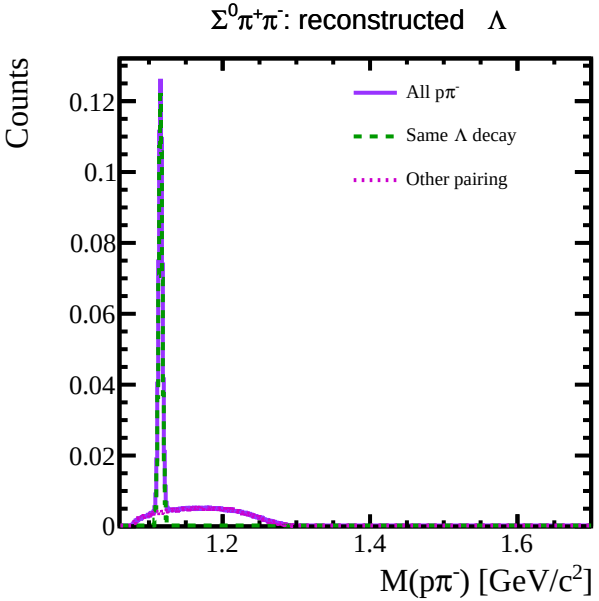

Σ+π-π0: reconstructed masses
M(pπ-) [GeV/c2]
MX [GeV/c2]
Conditional mb/bin

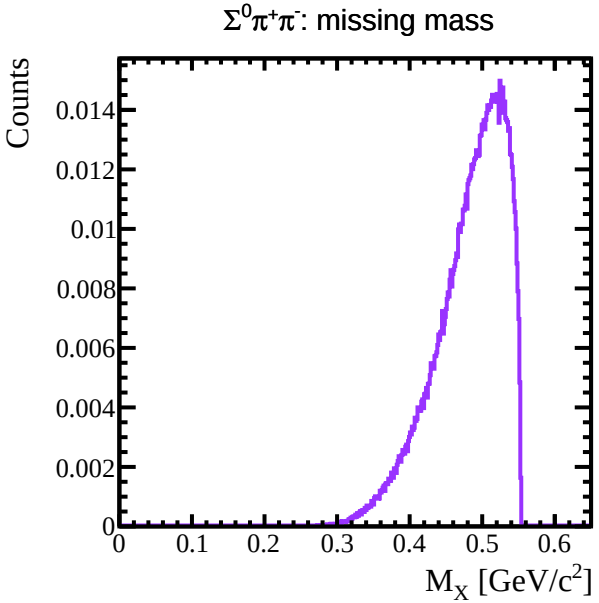

Λπ+π-: reconstructed Λ
Counts
All pπ-
Same Λ decay
Other pairing
M(pπ-) [GeV/c2]

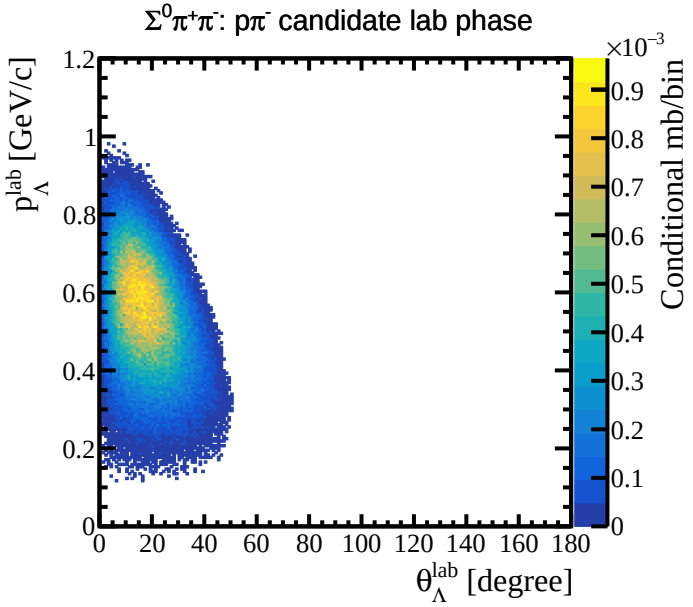

Λπ+π-: missing mass
Counts
MX [GeV/c2]

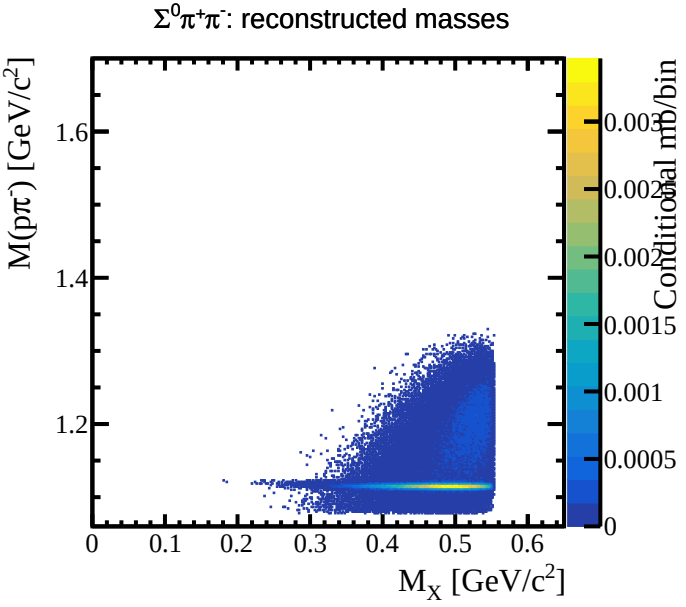

Λπ+π-: pπ- candidate lab phase
pΛlab [GeV/c]
θΛlab [degree]
Conditional mb/bin

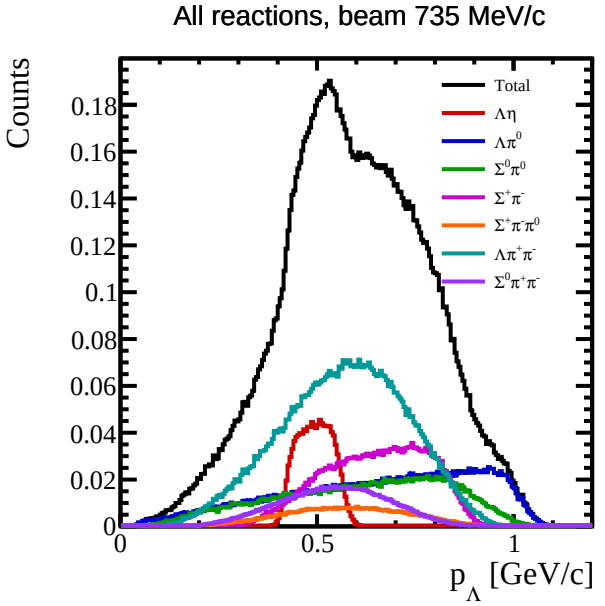

Λπ+π-: reconstructed masses
M(pπ-) [GeV/c2]
MX [GeV/c2]
Conditional mb/bin

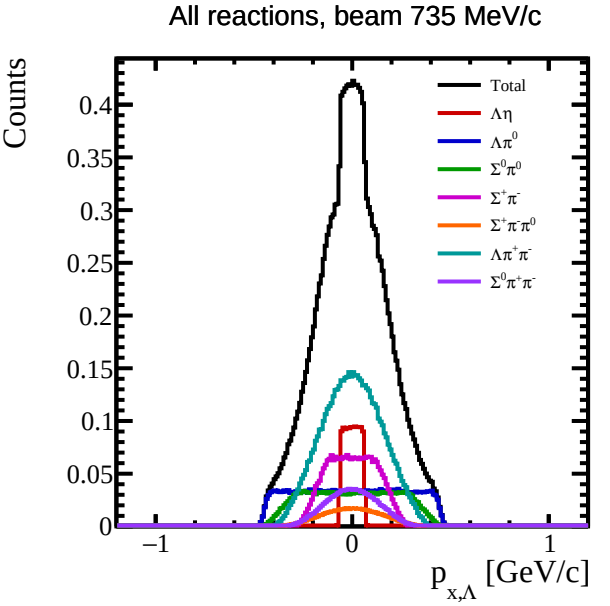

$\Sigma^0\pi^+\pi^-$: reconstructed Λ
Counts
All $p\pi^-$
Same Λ decay
Other pairing
$M(p\pi^-)$ [GeV/c^2]

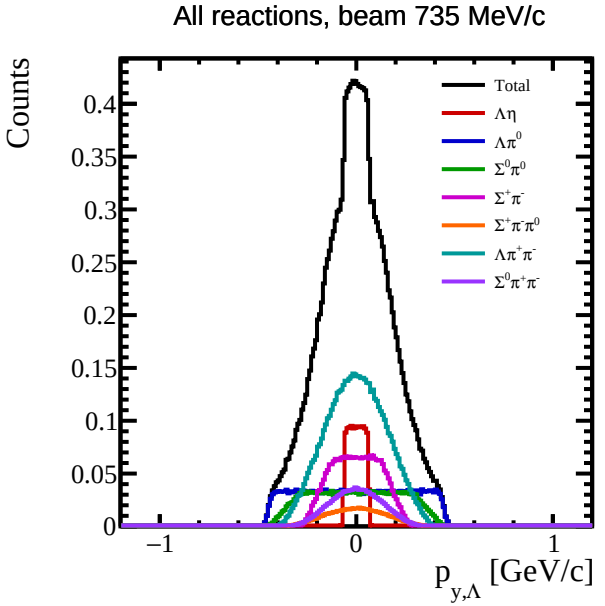

$\Sigma^0\pi^+\pi^-$: missing mass
Counts
M_X [GeV/c^2]

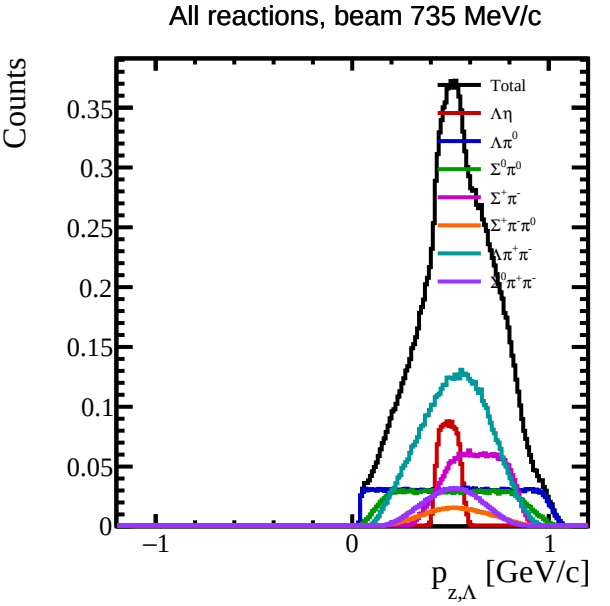

$\Sigma^0\pi^+\pi^-$: $p\pi^-$ candidate lab phase
p_Λ^{lab} [GeV/c]
θ_Λ^{lab} [degree]
Conditional mb/bin

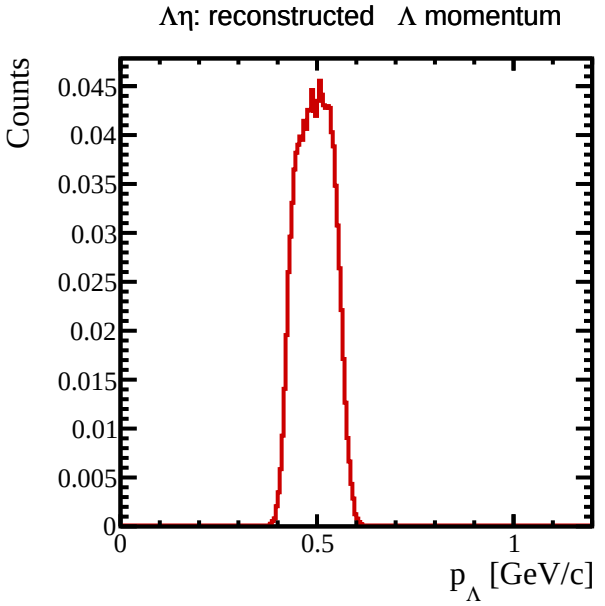

$\Sigma^0\pi^+\pi^-$: reconstructed masses
$M(p\pi^-)$ [GeV/c^2]
M_X [GeV/c^2]
Conditional mb/bin

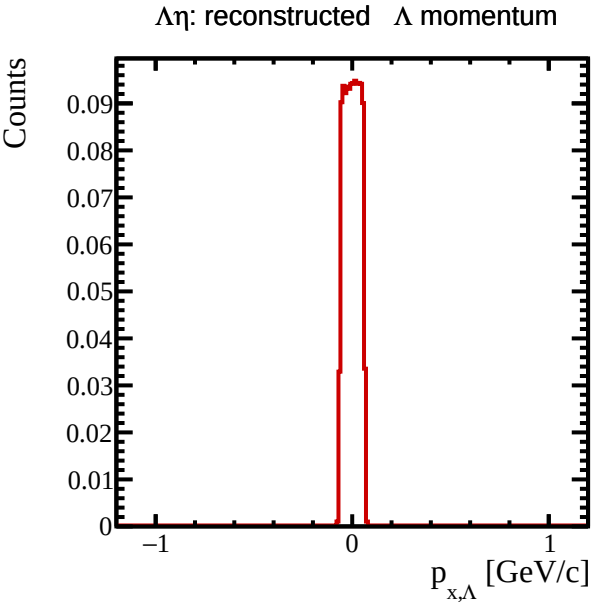

All reactions, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
p Λ [GeV/c]

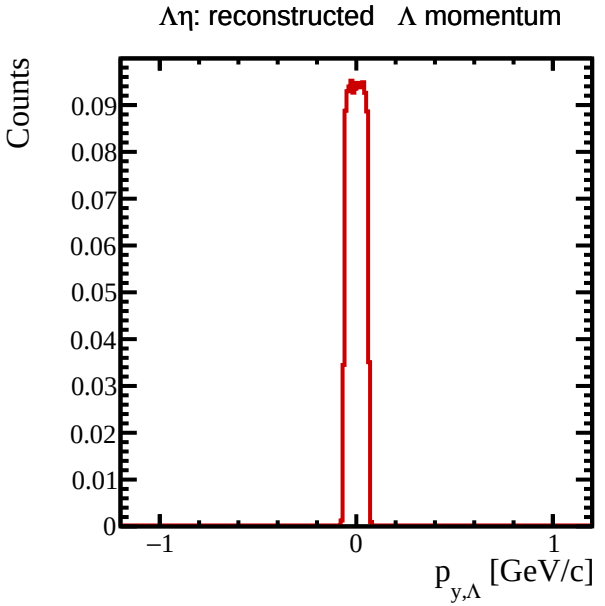

All reactions, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
p x,Λ [GeV/c]

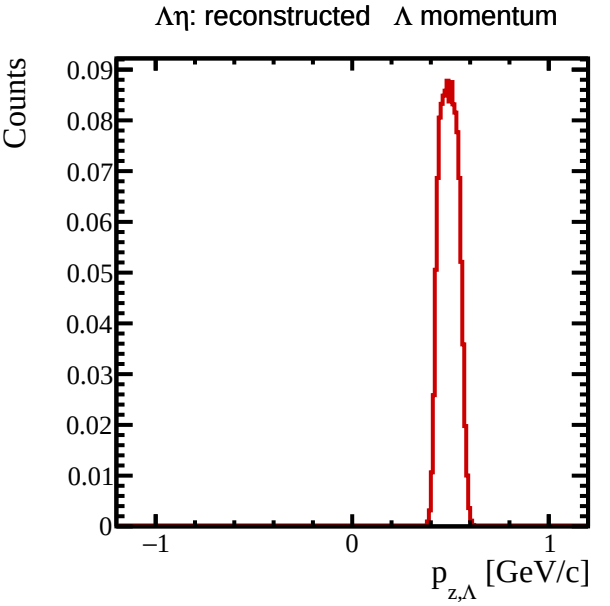

All reactions, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
p y,Λ [GeV/c]

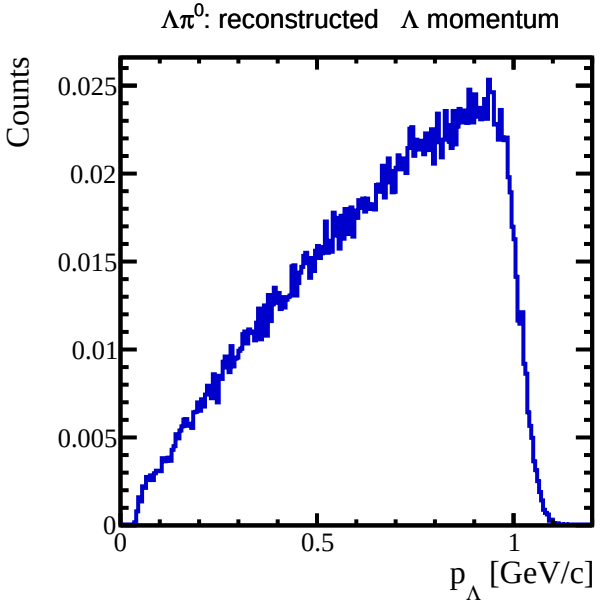

All reactions, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
p z,Λ [GeV/c]

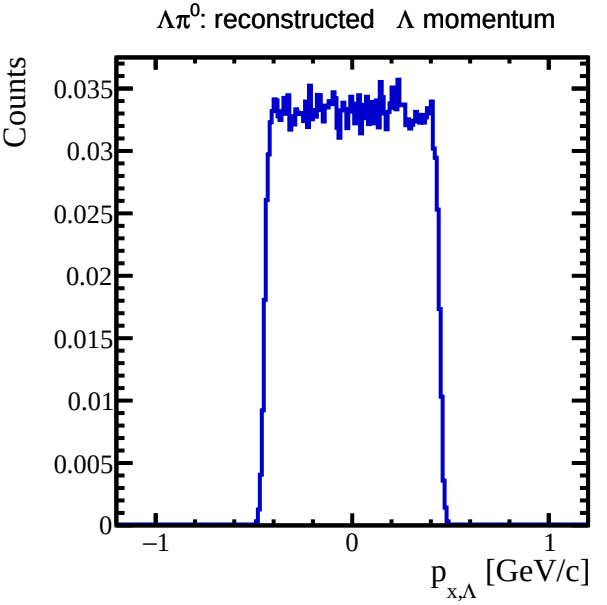

Λη: reconstructed Λ momentum
Counts
0.045
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
0
0.5
1
p_Λ [GeV/c]

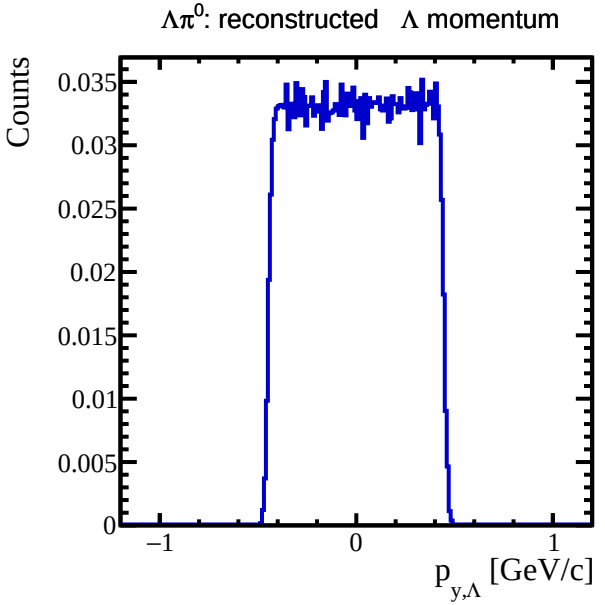

Λη: reconstructed Λ momentum
Counts
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_x,Λ [GeV/c]

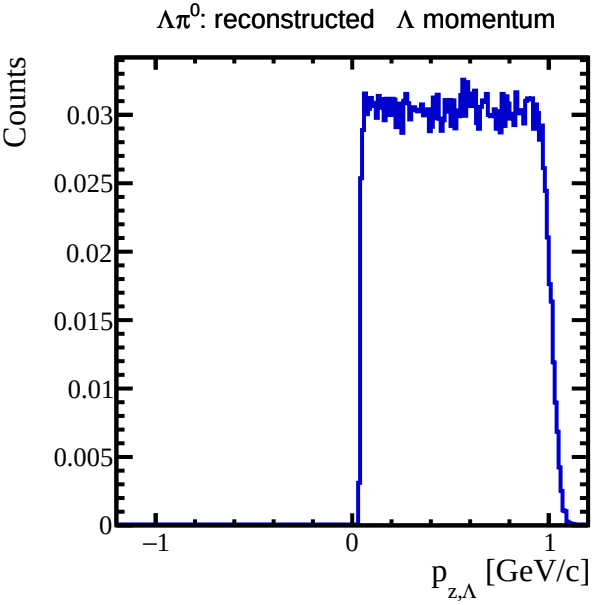

Λη: reconstructed Λ momentum
Counts
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_y,Λ [GeV/c]

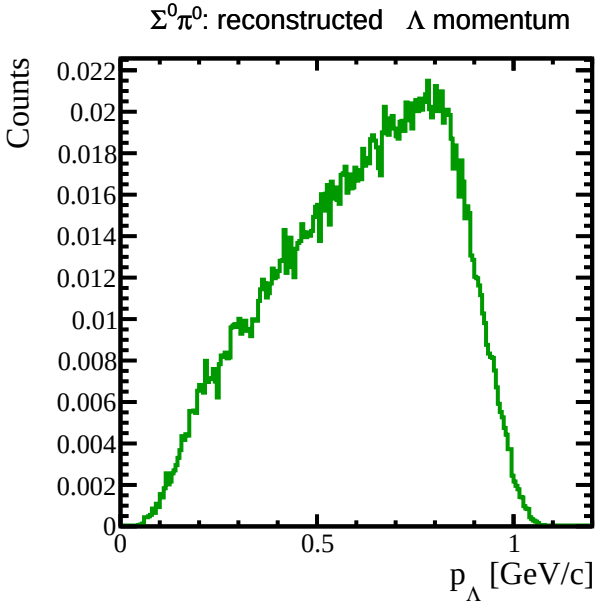

Λη: reconstructed Λ momentum
Counts
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_z,Λ [GeV/c]

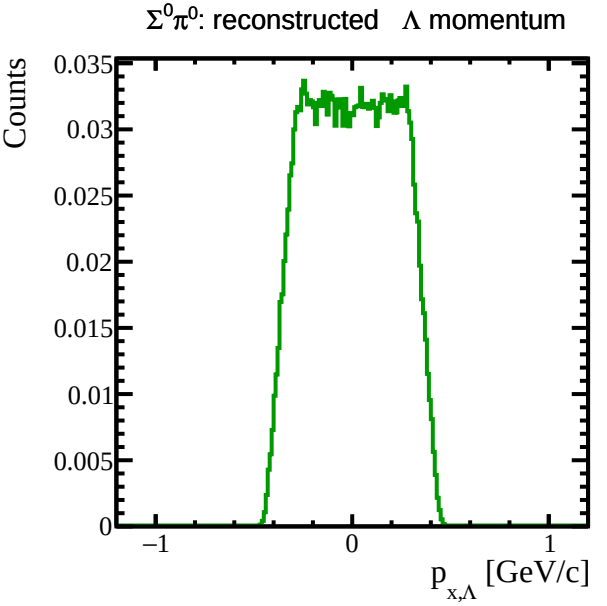

$\Lambda\pi^0$: reconstructed Λ momentum
Counts
0.025
0.02
0.015
0.01
0.005
0
0
0.5
1
p_Λ [GeV/c]

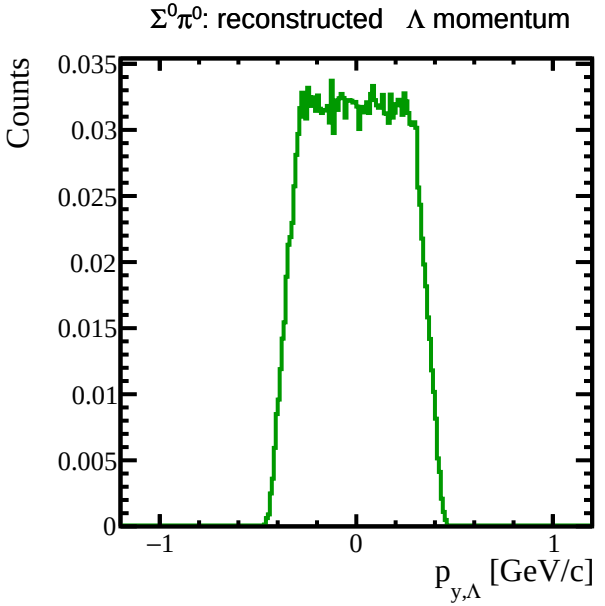

$\Lambda\pi^0$: reconstructed Λ momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
$p_{x,\Lambda}$ [GeV/c]

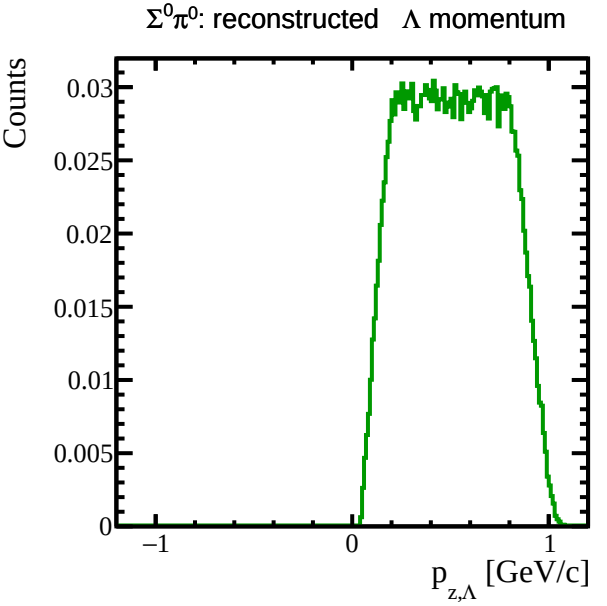

$\Lambda\pi^0$: reconstructed Λ momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
$p_{y,\Lambda}$ [GeV/c]

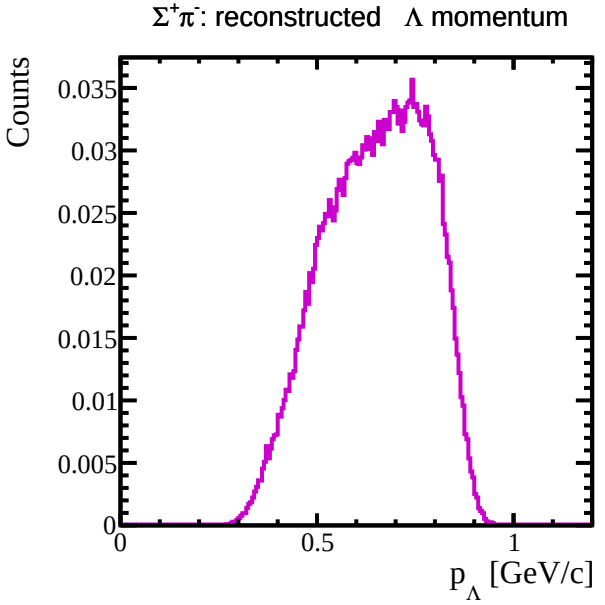

$\Lambda\pi^0$: reconstructed Λ momentum
Counts
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
$p_{z,\Lambda}$ [GeV/c]

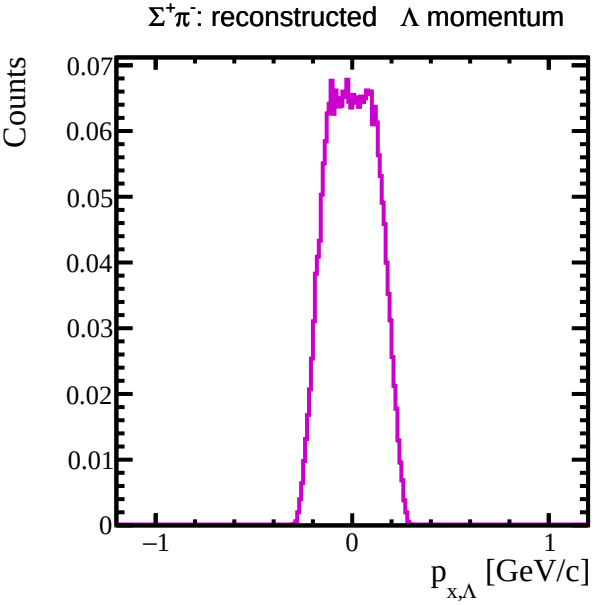

Σ⁰π⁰: reconstructed Λ momentum
Counts
0.022
0.02
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
0
0.5
1
p_Λ [GeV/c]

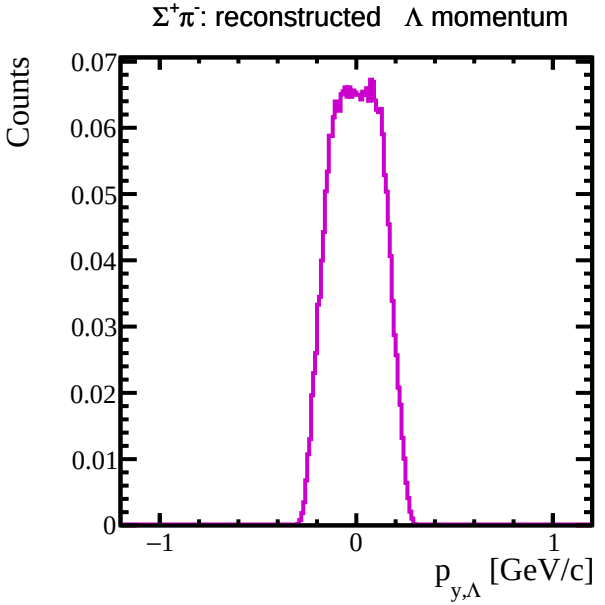

Σ⁰π⁰: reconstructed Λ momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_x,Λ [GeV/c]

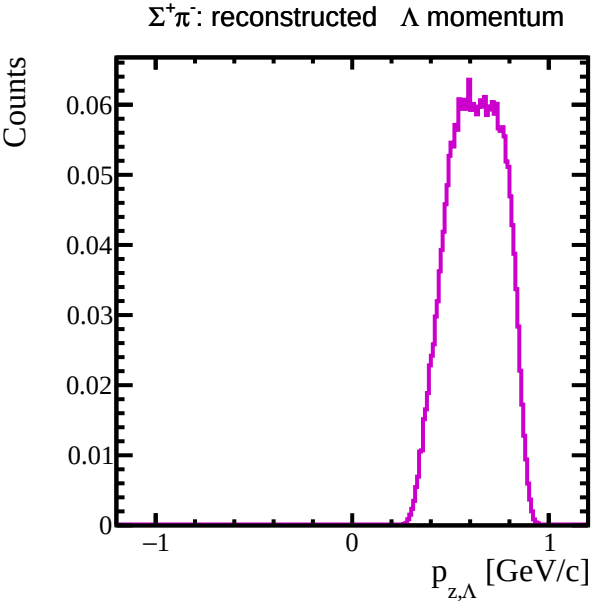

Σ⁰π⁰: reconstructed Λ momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_y,Λ [GeV/c]

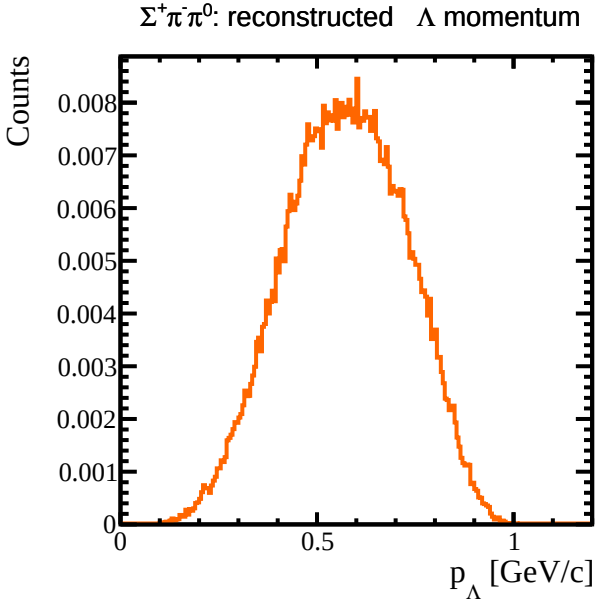

Σ⁰π⁰: reconstructed Λ momentum
Counts
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_z,Λ [GeV/c]

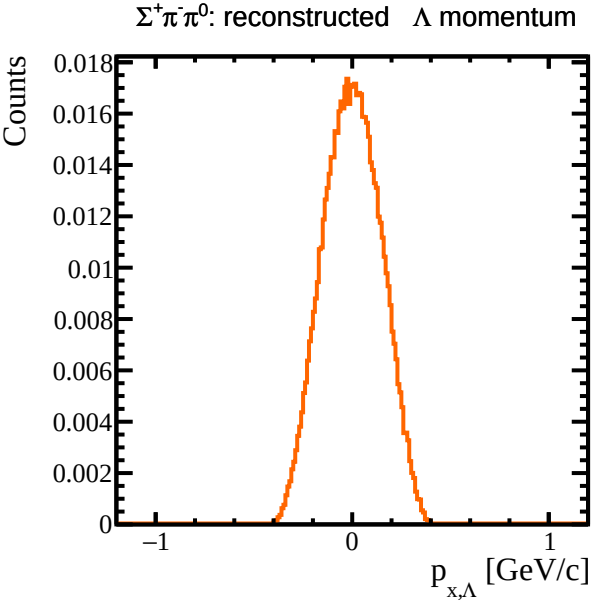

Σ+π-: reconstructed Λ momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
0
0.5
1
p_Λ [GeV/c]

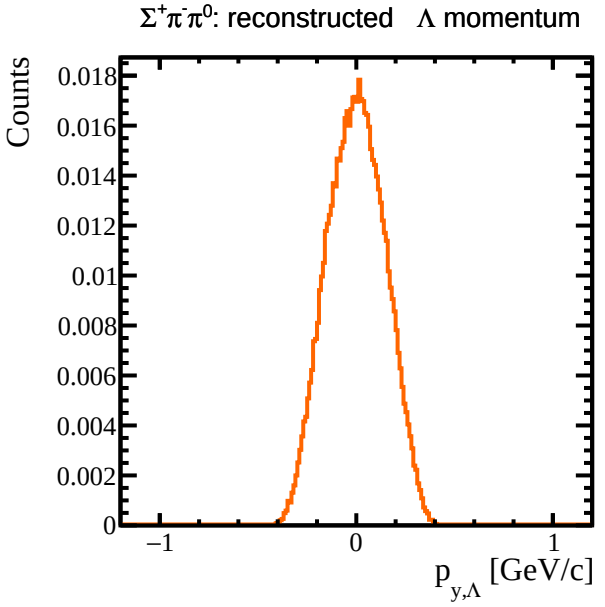

Σ+π-: reconstructed Λ momentum
Counts
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_x,Λ [GeV/c]

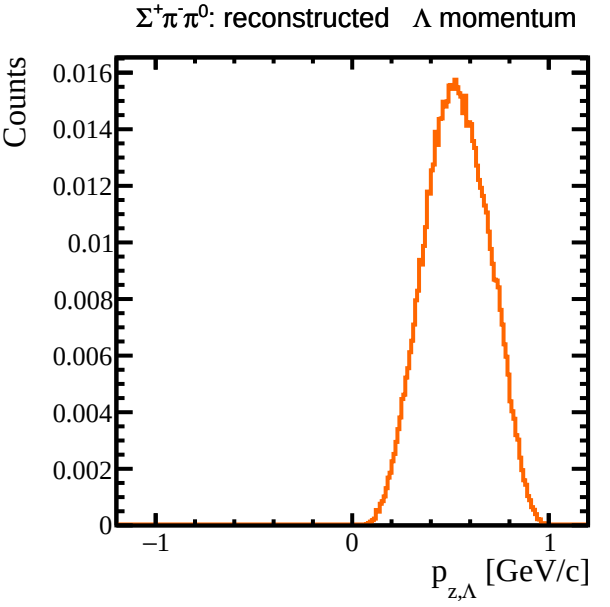

Σ+π-: reconstructed Λ momentum
Counts
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_y,Λ [GeV/c]

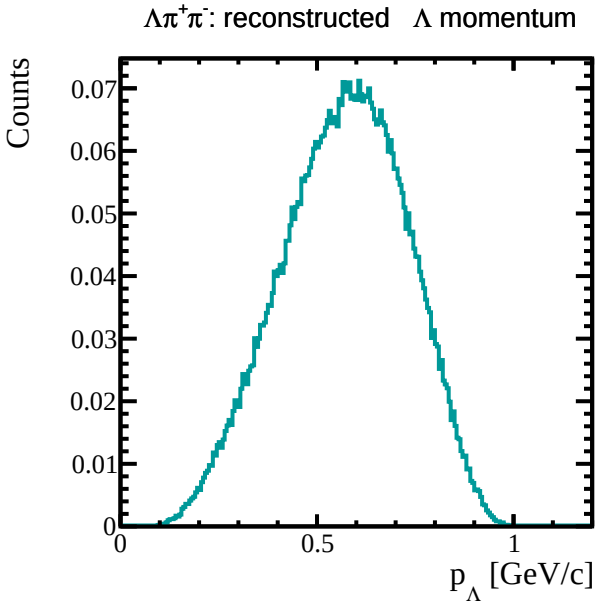

Σ+π-: reconstructed Λ momentum
Counts
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_z,Λ [GeV/c]

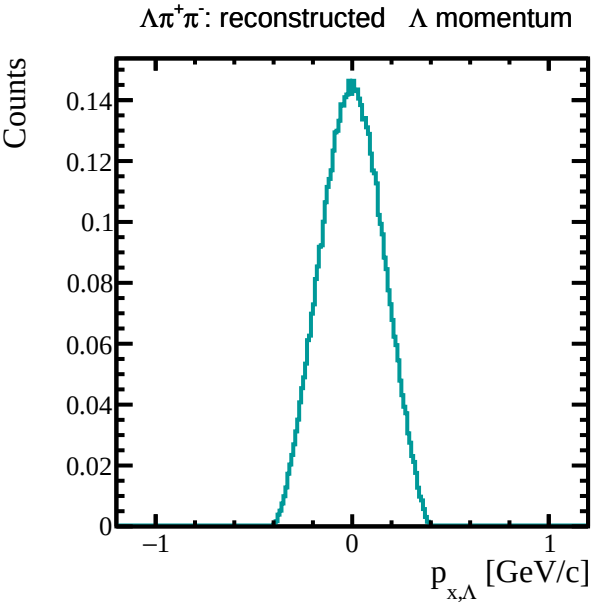

$\Sigma^+\pi^-\pi^0$: reconstructed Λ momentum
Counts
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0
0
0.5
1
p_Λ [GeV/c]

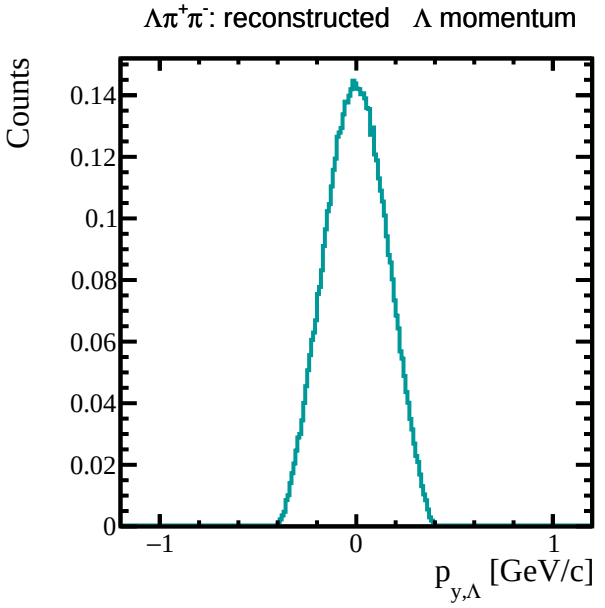

$\Sigma^+\pi^-\pi^0$: reconstructed Λ momentum
Counts
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
$p_{x,\Lambda}$ [GeV/c]

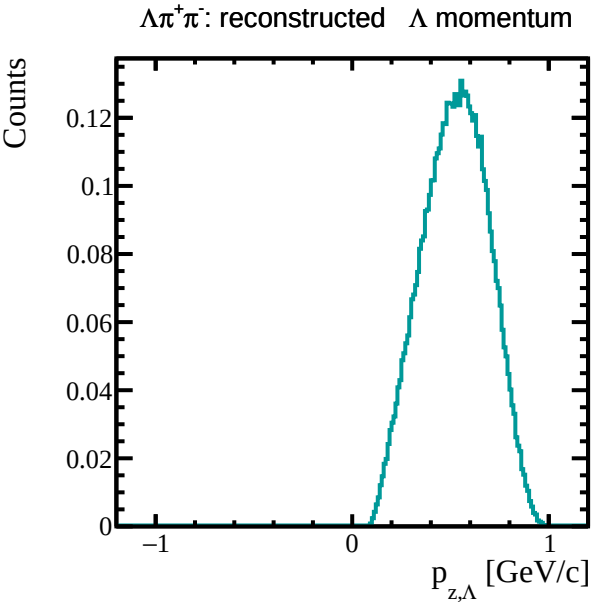

$\Sigma^+\pi^-\pi^0$: reconstructed Λ momentum
Counts
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
$p_{y,\Lambda}$ [GeV/c]

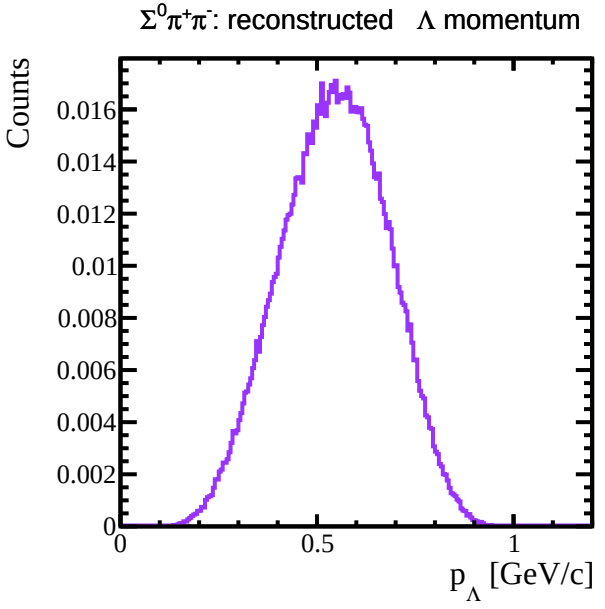

$\Sigma^+\pi^-\pi^0$: reconstructed Λ momentum
Counts
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
$p_{z,\Lambda}$ [GeV/c]

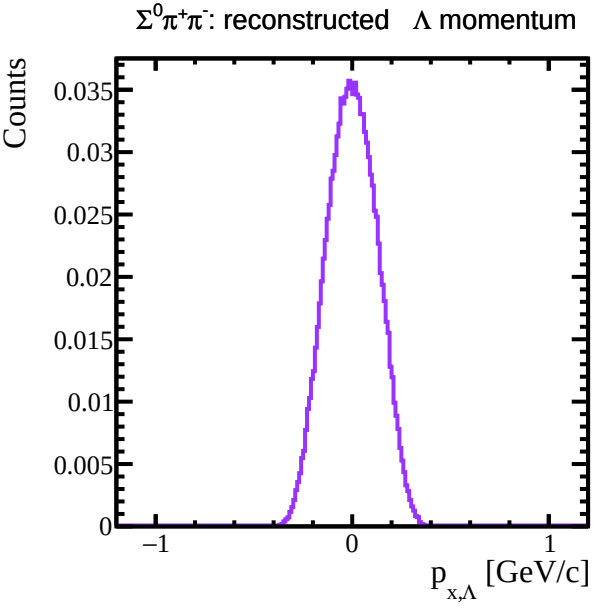

Λπ+π-: reconstructed Λ momentum
Counts
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
0
0.5
1
pΛ [GeV/c]

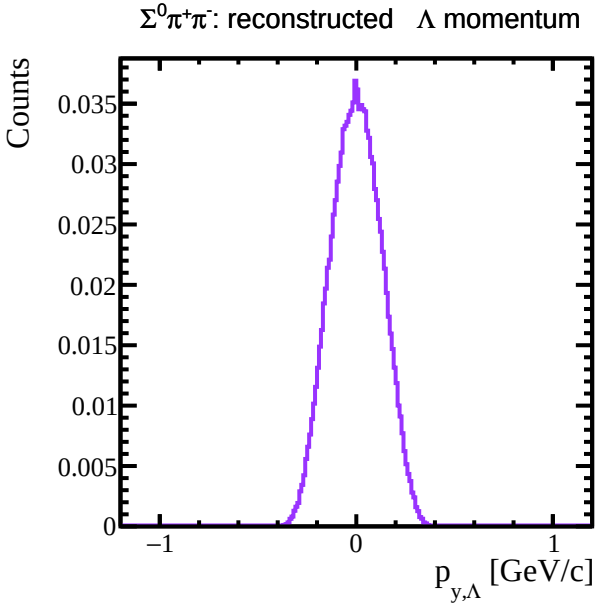

Λπ+π-: reconstructed Λ momentum
Counts
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
px,Λ [GeV/c]

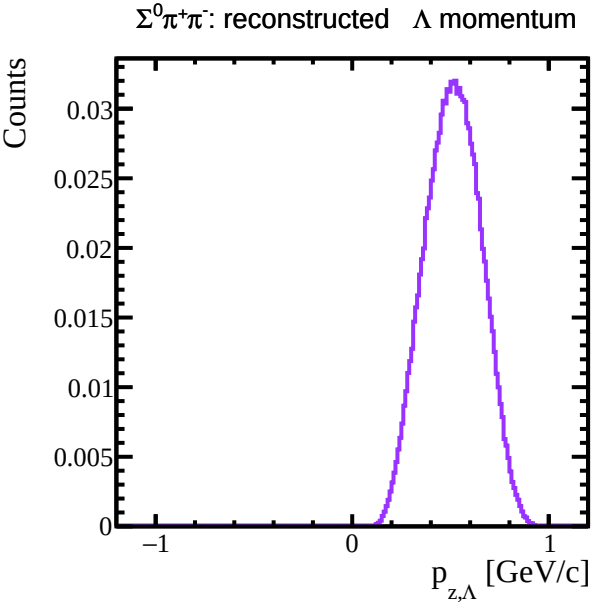

Λπ+π-: reconstructed Λ momentum
Counts
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
py,Λ [GeV/c]

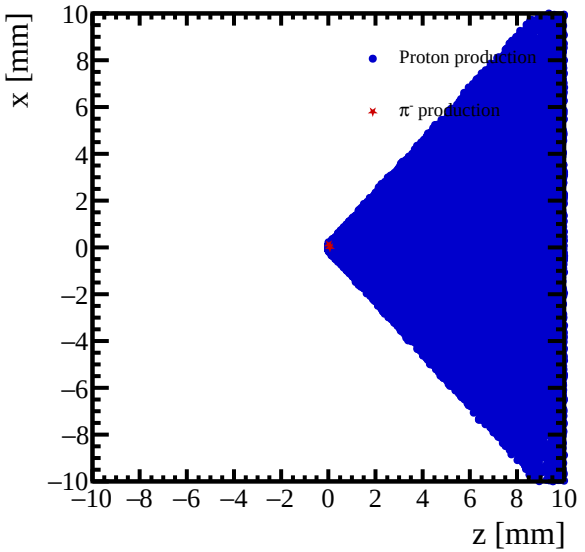

Λπ+π-: reconstructed Λ momentum
Counts
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
pz,Λ [GeV/c]

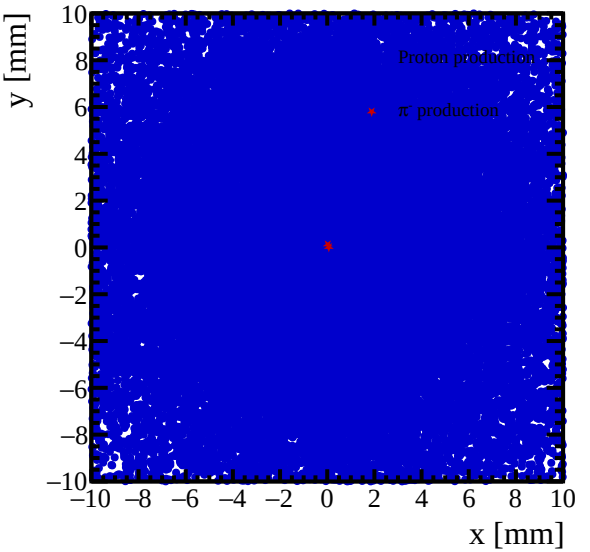

$\Sigma^0\pi^+\pi^-$: reconstructed Λ momentum
Counts
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
0
0.5
1
p_Λ [GeV/c]

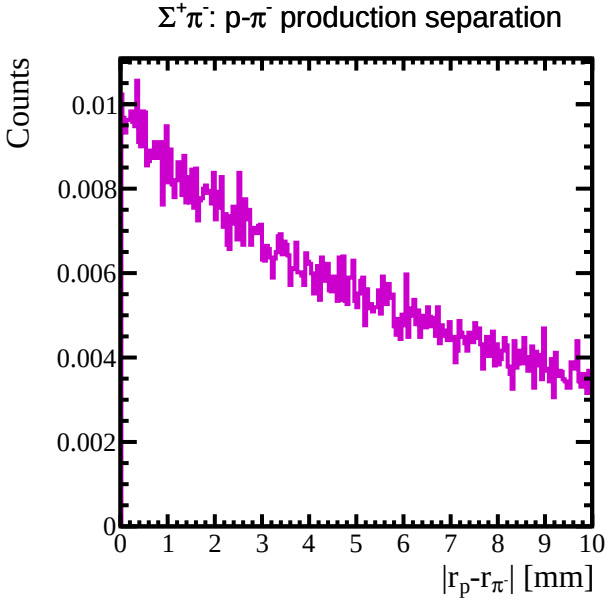

$\Sigma^0\pi^+\pi^-$: reconstructed Λ momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
$p_{x,\Lambda}$ [GeV/c]

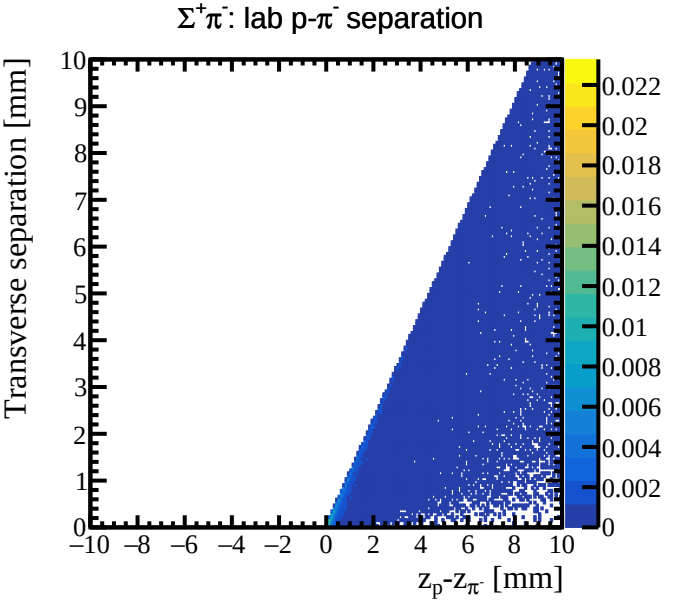

$\Sigma^0\pi^+\pi^-$: reconstructed Λ momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
$p_{y,\Lambda}$ [GeV/c]

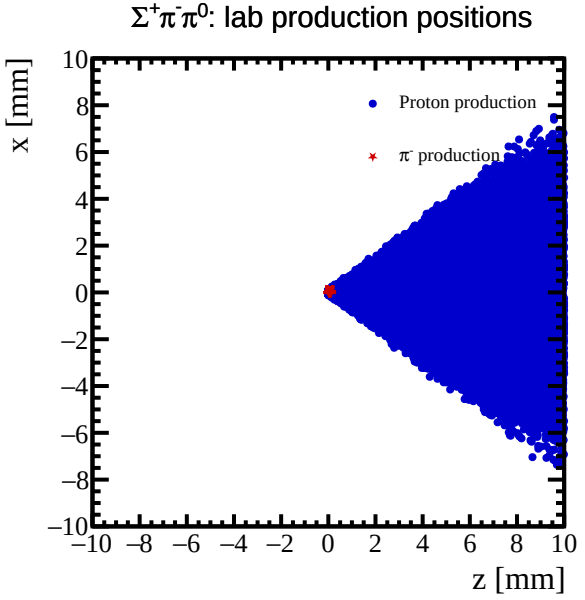

$\Sigma^0\pi^+\pi^-$: reconstructed Λ momentum
Counts
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
$p_{z,\Lambda}$ [GeV/c]

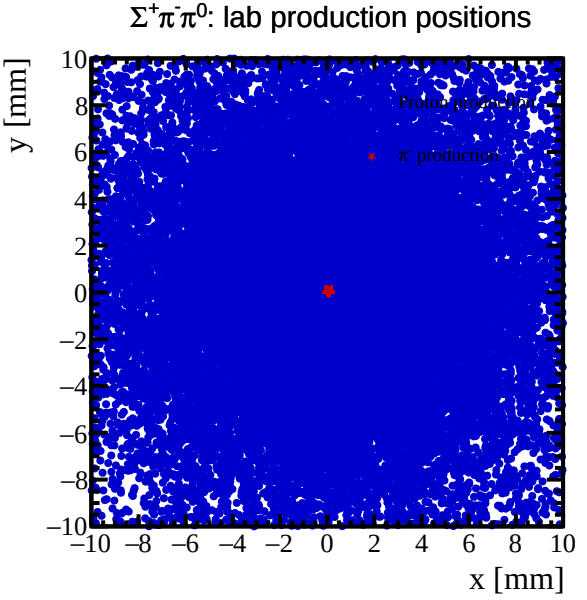

Σ+π-: lab production positions
Proton production
π- production
x [mm]
z [mm]

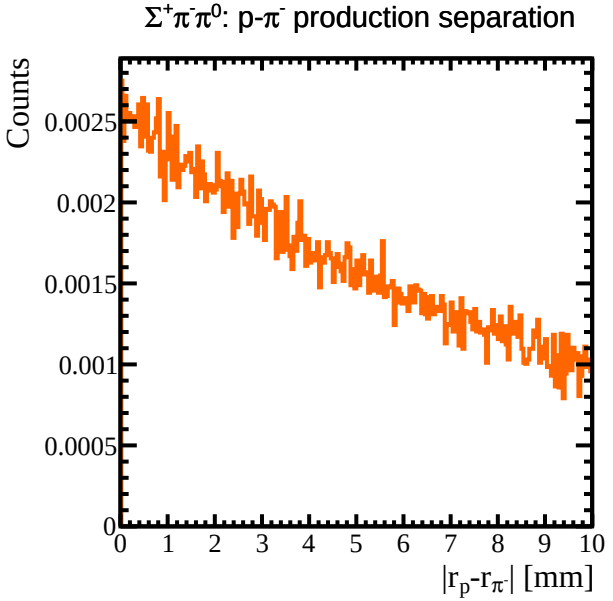

Σ+π-: lab production positions
Proton production
π- production
y [mm]
x [mm]

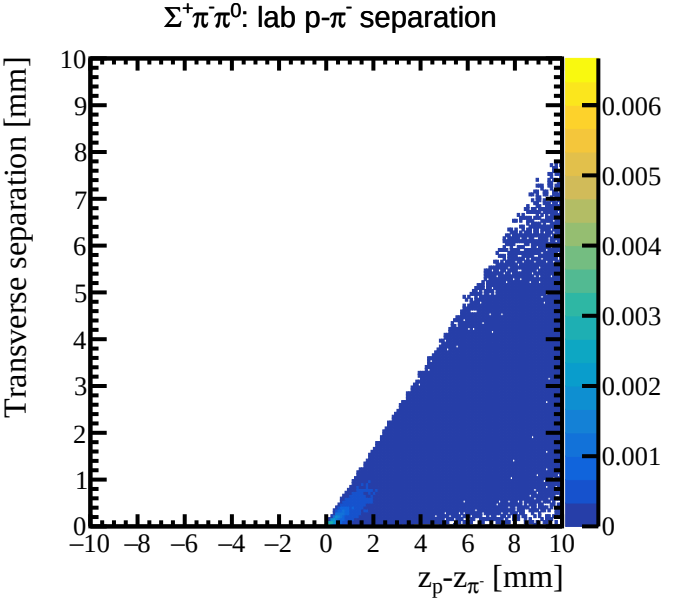

Σ+π-: p-π- production separation
Counts
|rp-rπ-| [mm]

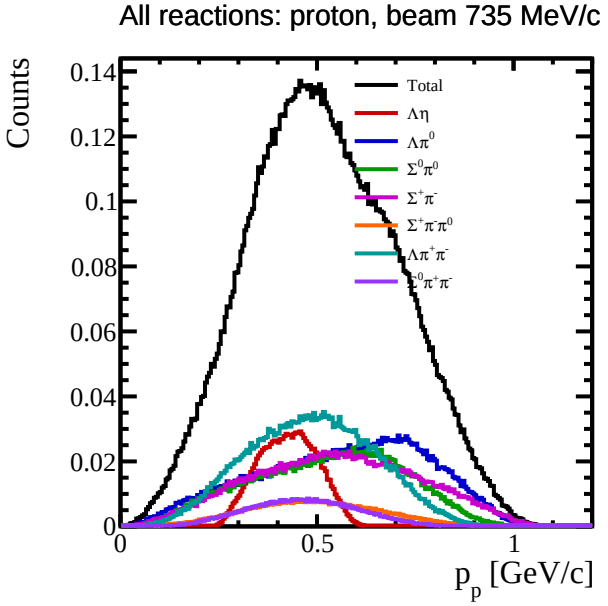

Σ+π-: lab p-π- separation
Transverse separation [mm]
zp-zπ- [mm]

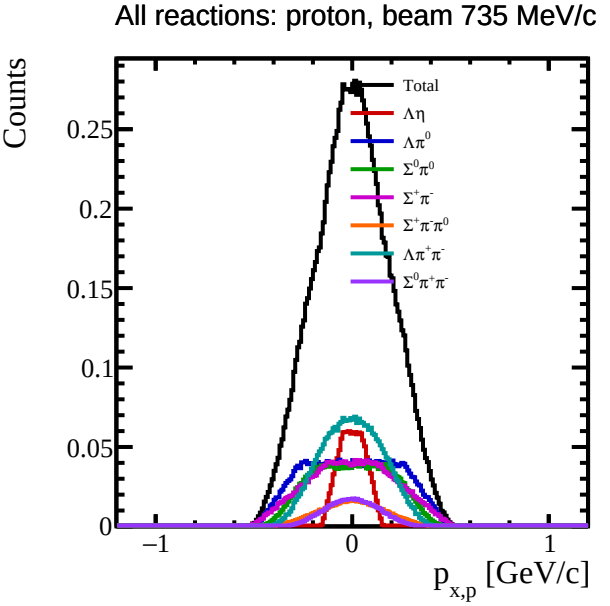

Σ+π-π0: lab production positions
Proton production
π- production
x [mm]
z [mm]

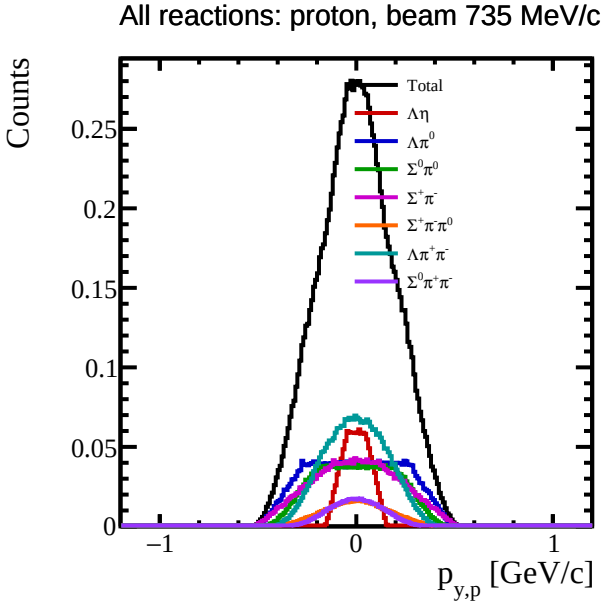

Σ+π-π0: lab production positions
Proton production
π- production
y [mm]
x [mm]

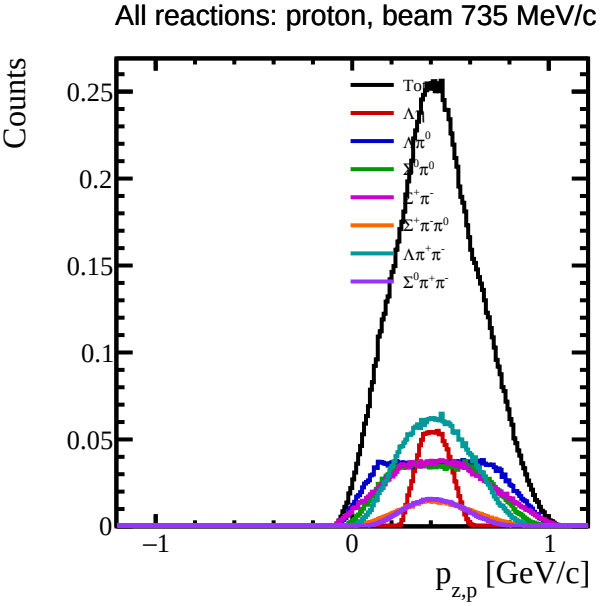

Σ+π-π0: p-π- production separation
Counts
|rp-rπ-| [mm]

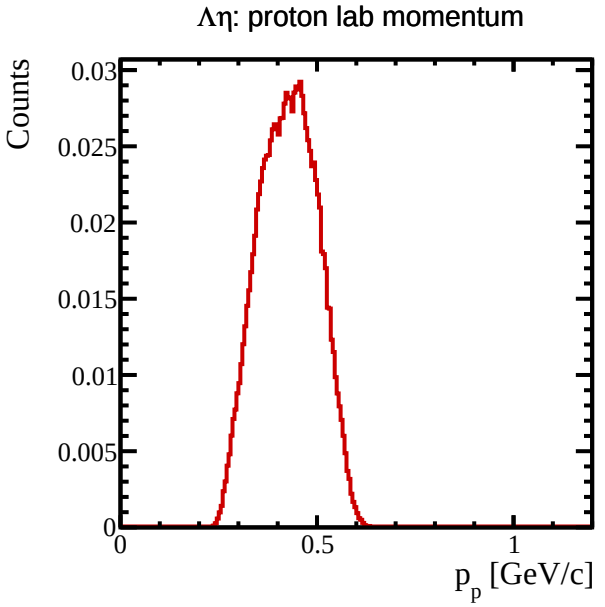

Σ+π-π0: lab p-π- separation
Transverse separation [mm]
zp-zπ- [mm]

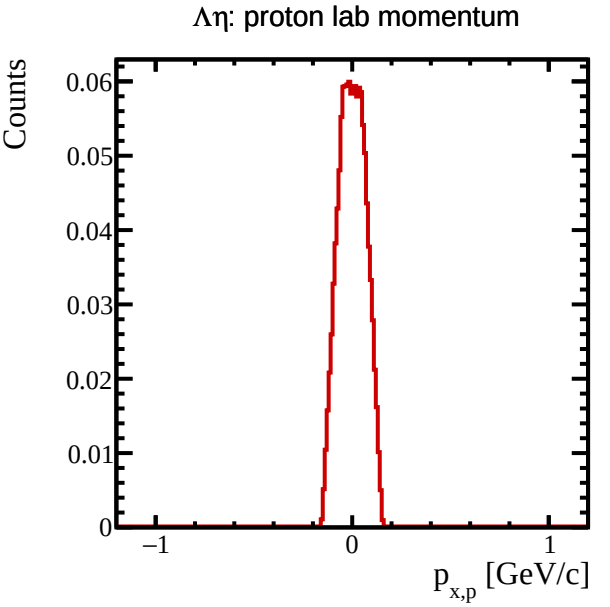

All reactions: proton, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0
0.5
1
p_p [GeV/c]

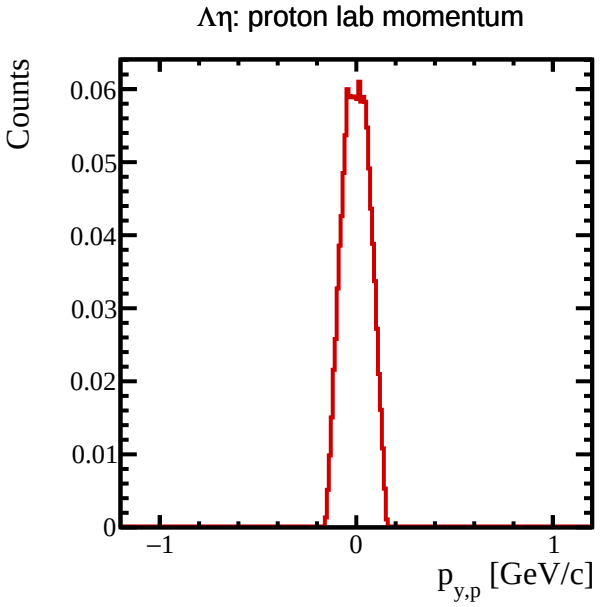

All reactions: proton, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
0.25
0.2
0.15
0.1
0.05
0
−1
0
1
p_x,p [GeV/c]

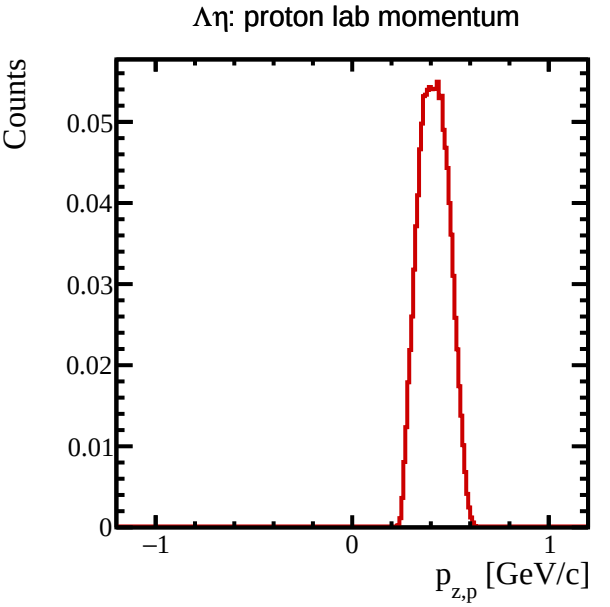

All reactions: proton, beam 735 MeV/c
Counts
Total
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
0.25
0.2
0.15
0.1
0.05
0
−1
0
1
p_y,p [GeV/c]

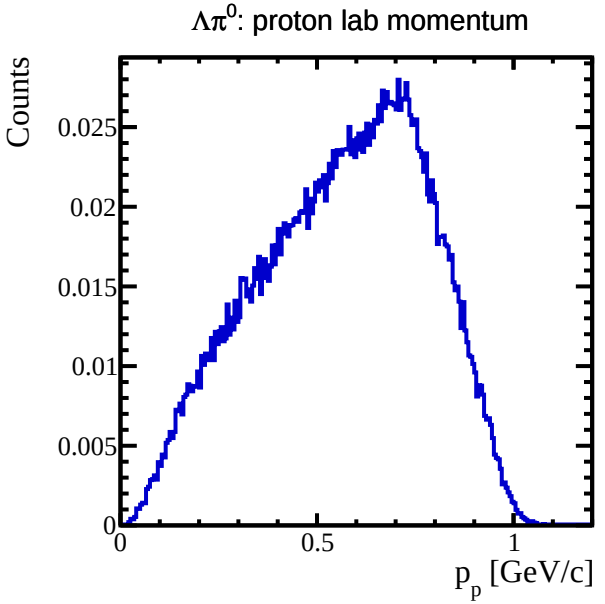

All reactions: proton, beam 735 MeV/c
Counts
Λη
Λπ0
Σ0π0
Σ+π-
Σ+π-π0
Λπ+π-
Σ0π+π-
0.25
0.2
0.15
0.1
0.05
0
−1
0
1
p_z,p [GeV/c]

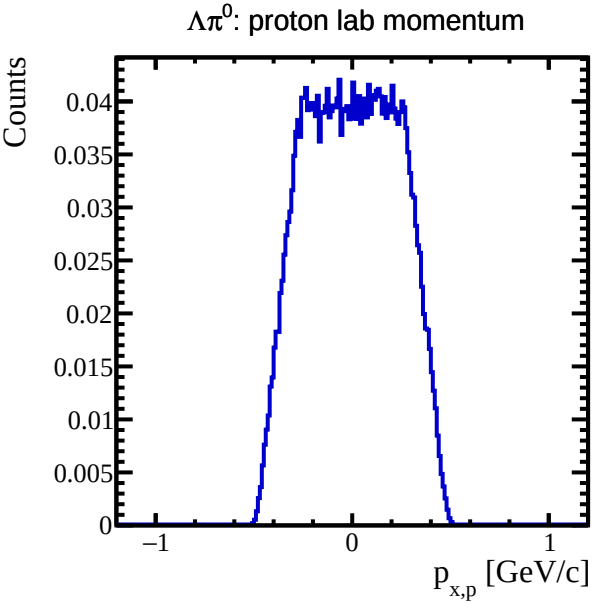

Λη: proton lab momentum
Counts
0.03
0.025
0.02
0.015
0.01
0.005
0
0
0.5
1
p_p [GeV/c]

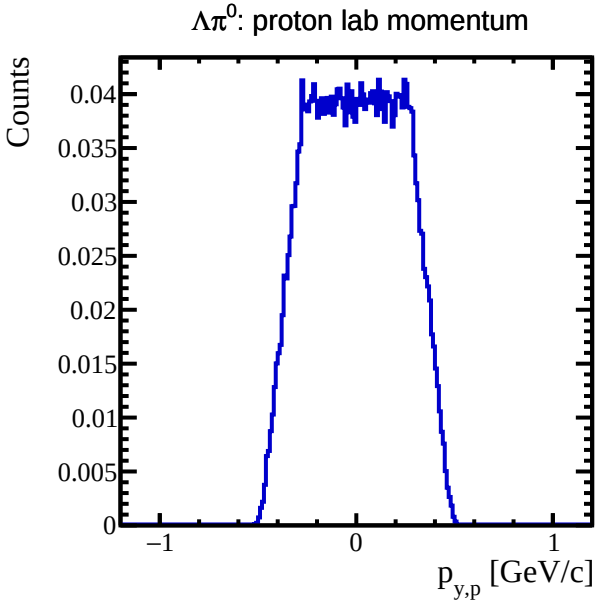

Λη: proton lab momentum
Counts
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_x,p [GeV/c]

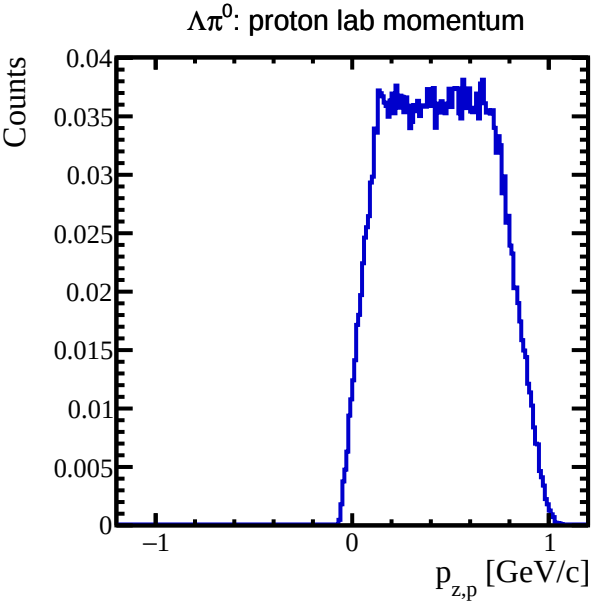

Λη: proton lab momentum
Counts
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_y,p [GeV/c]

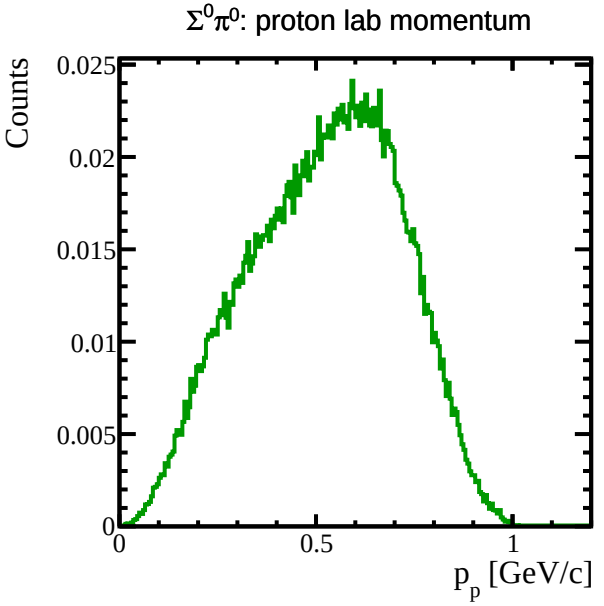

Λη: proton lab momentum
Counts
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_z,p [GeV/c]

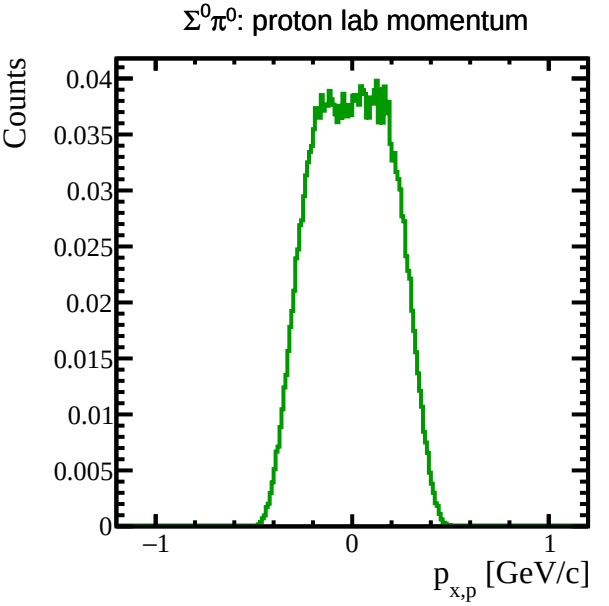

Λπ0: proton lab momentum
Counts
0.025
0.02
0.015
0.01
0.005
0
0
0.5
1
p_p [GeV/c]

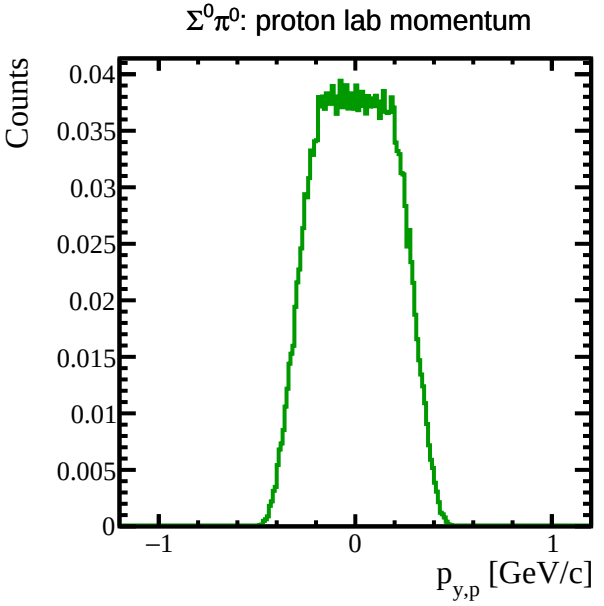

Λπ0: proton lab momentum
Counts
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_x,p [GeV/c]

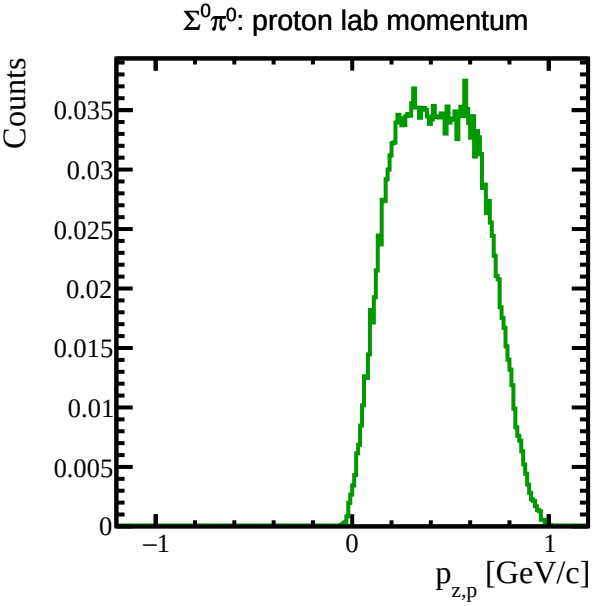

Λπ0: proton lab momentum
Counts
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_y,p [GeV/c]

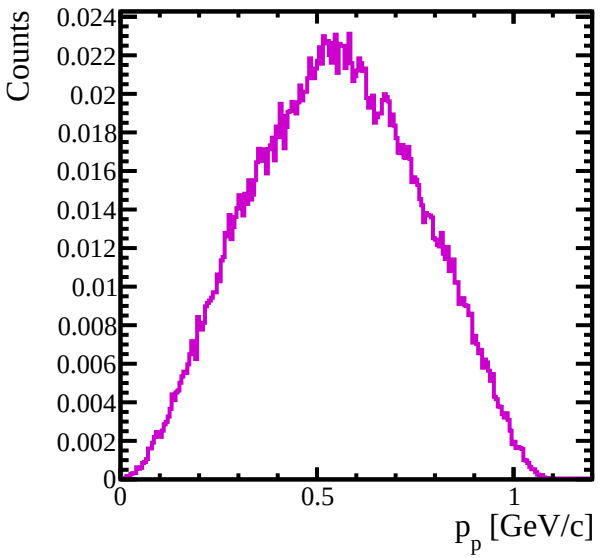

Λπ0: proton lab momentum
Counts
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_z,p [GeV/c]

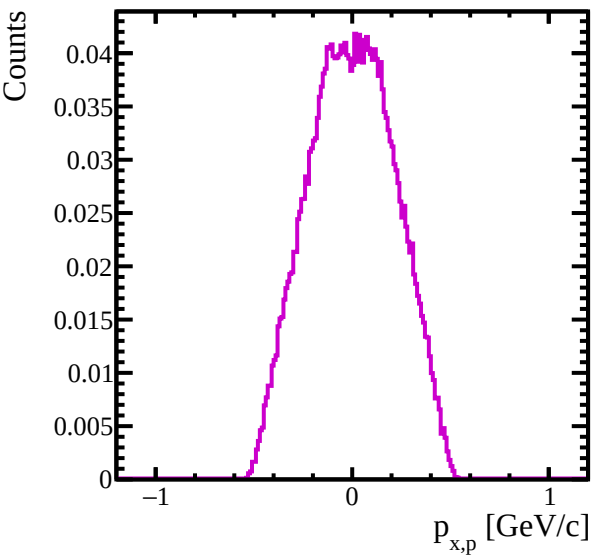

Σ⁰π⁰: proton lab momentum
Counts
0.025
0.02
0.015
0.01
0.005
0
0
0.5
1
p_p [GeV/c]

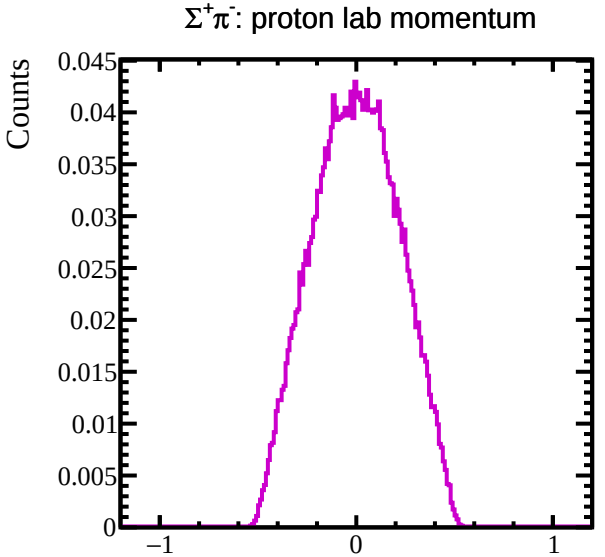

Σ⁰π⁰: proton lab momentum
Counts
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_x,p [GeV/c]

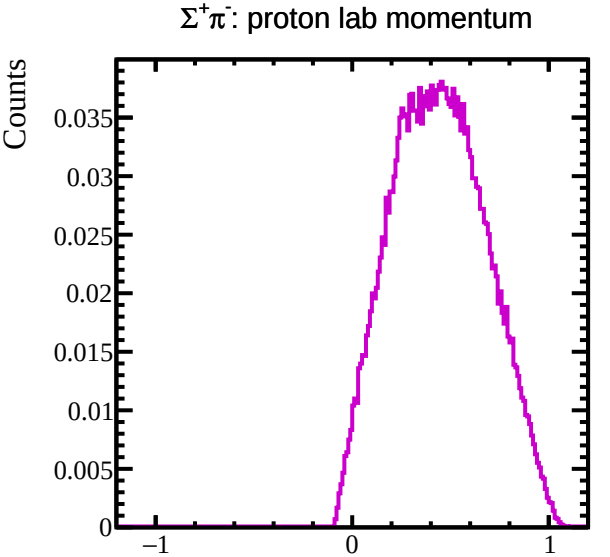

Σ⁰π⁰: proton lab momentum
Counts
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_y,p [GeV/c]

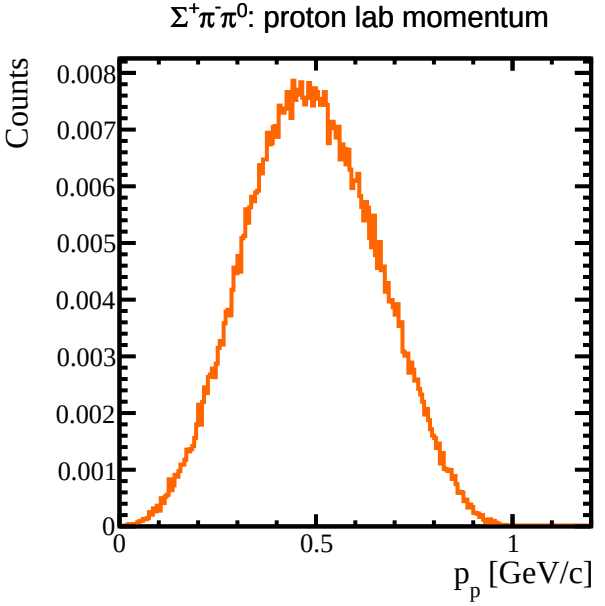

Σ⁰π⁰: proton lab momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_z,p [GeV/c]

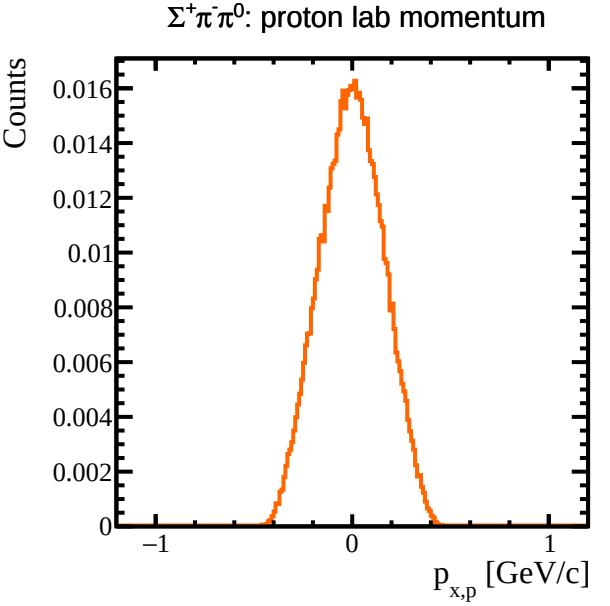

Σ+π-: proton lab momentum
Counts
0.024
0.022
0.02
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
0
0.5
1
p_p [GeV/c]

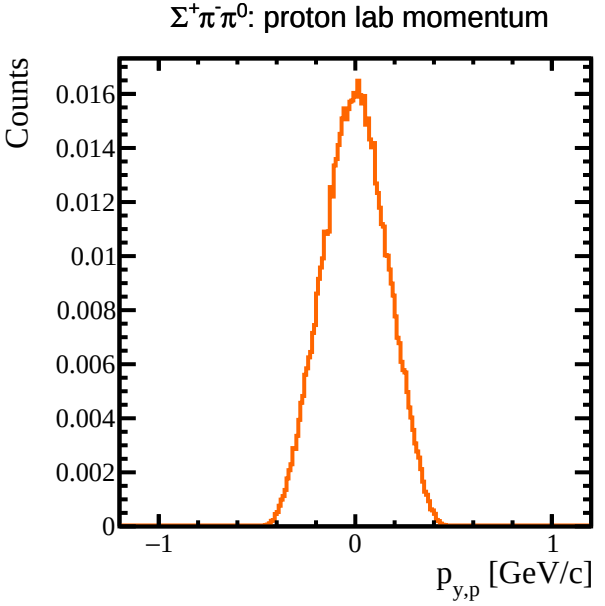

Σ+π-: proton lab momentum
Counts
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_x,p [GeV/c]

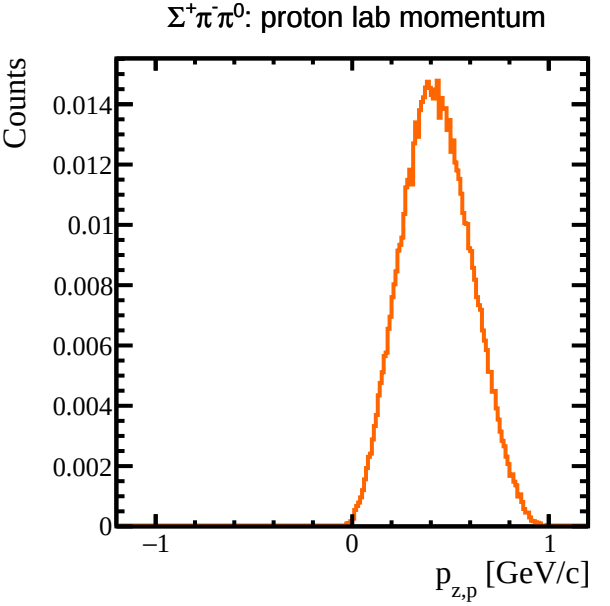

Σ+π-: proton lab momentum
Counts
0.045
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_y,p [GeV/c]

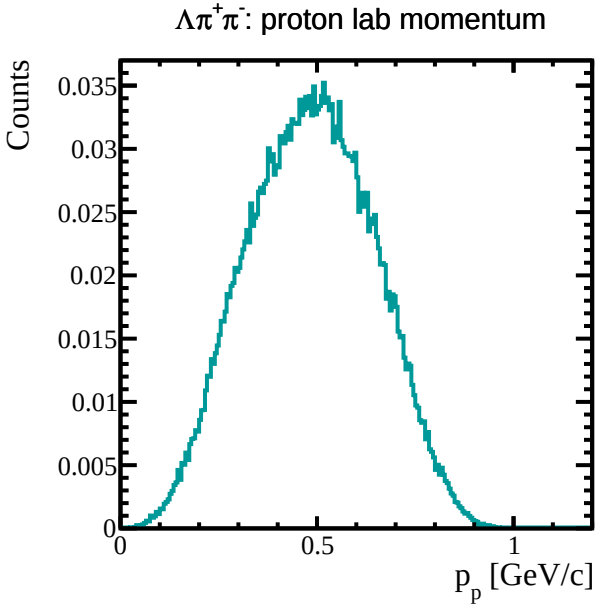

Σ+π-: proton lab momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_z,p [GeV/c]

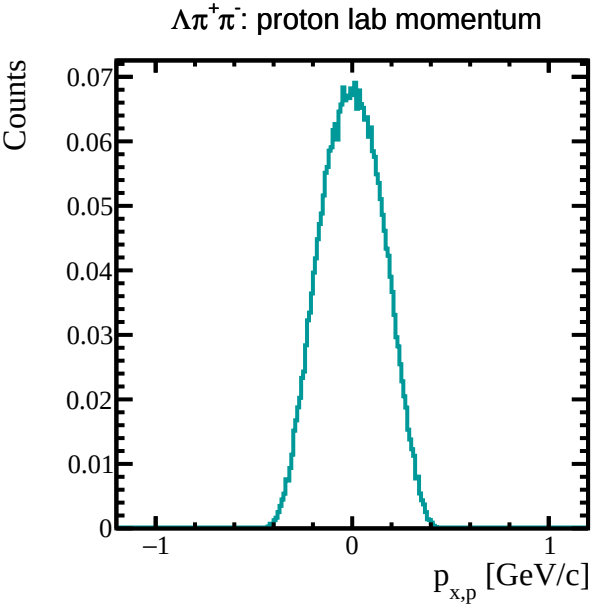

Σ+π-π0: proton lab momentum
Counts
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0
0
0.5
1
p_p [GeV/c]

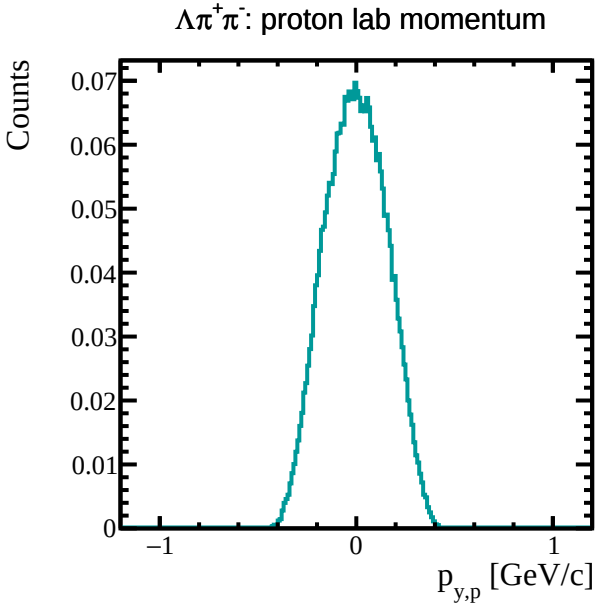

Σ+π-π0: proton lab momentum
Counts
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
p_x,p [GeV/c]

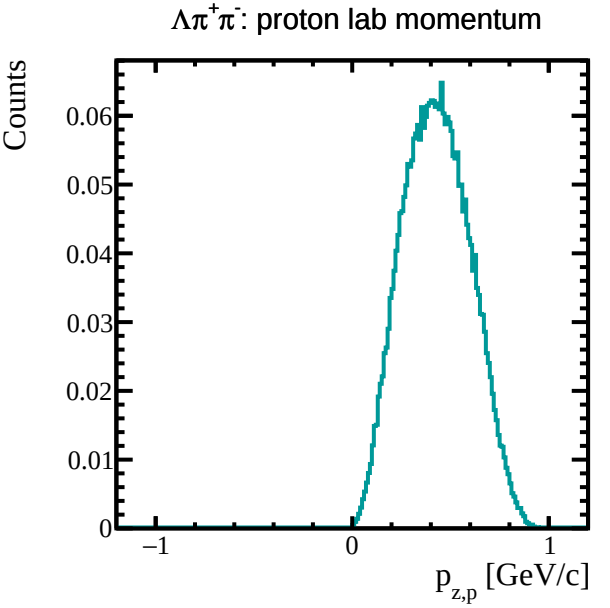

Σ+π-π0: proton lab momentum
Counts
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
p_y,p [GeV/c]

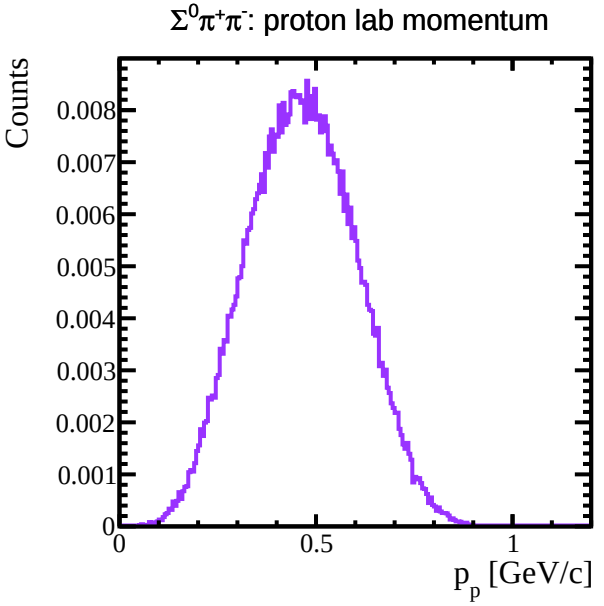

Σ+π-π0: proton lab momentum
Counts
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
p_z,p [GeV/c]

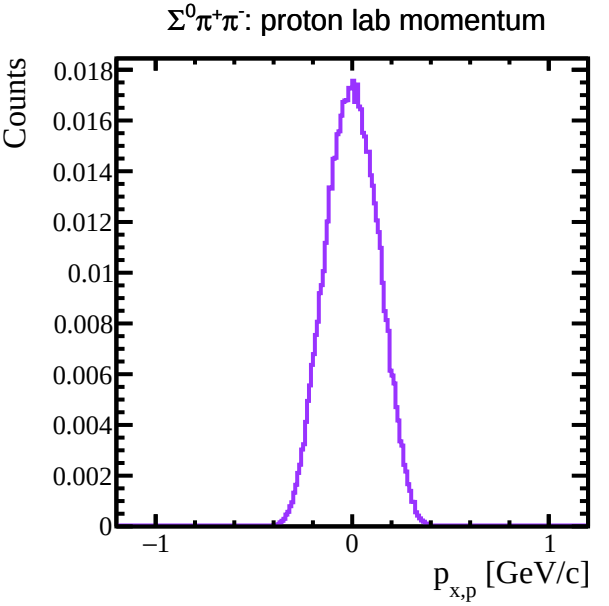

Λπ+π-: proton lab momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
0
0.5
1
p_p [GeV/c]

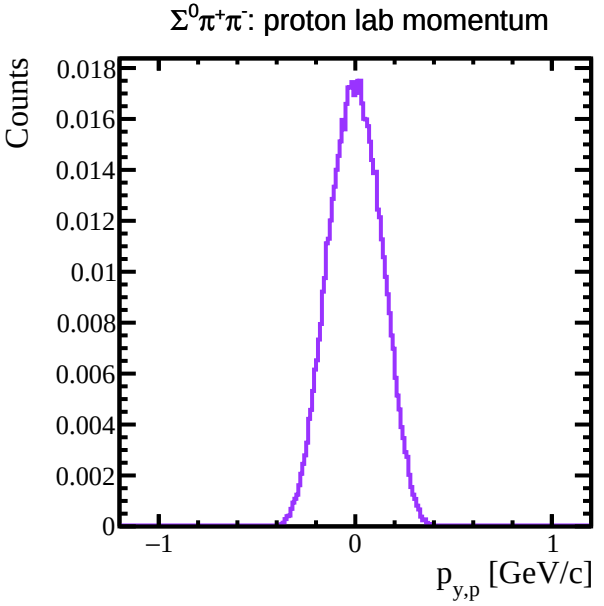

Λπ+π-: proton lab momentum
Counts
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_x,p [GeV/c]

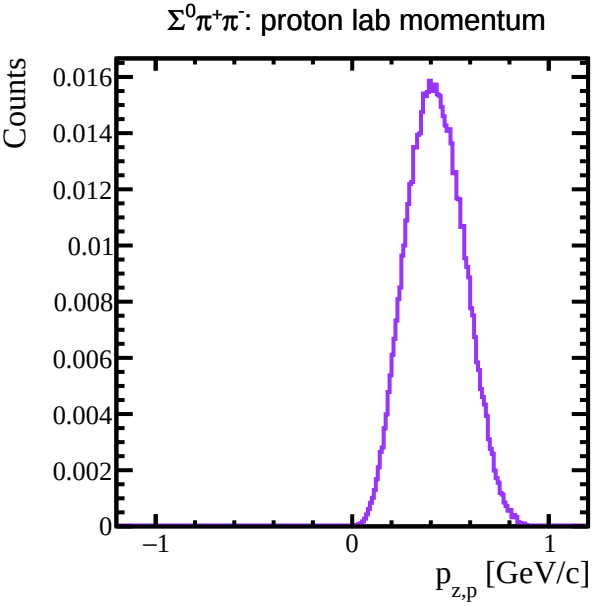

Λπ+π-: proton lab momentum
Counts
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_y,p [GeV/c]

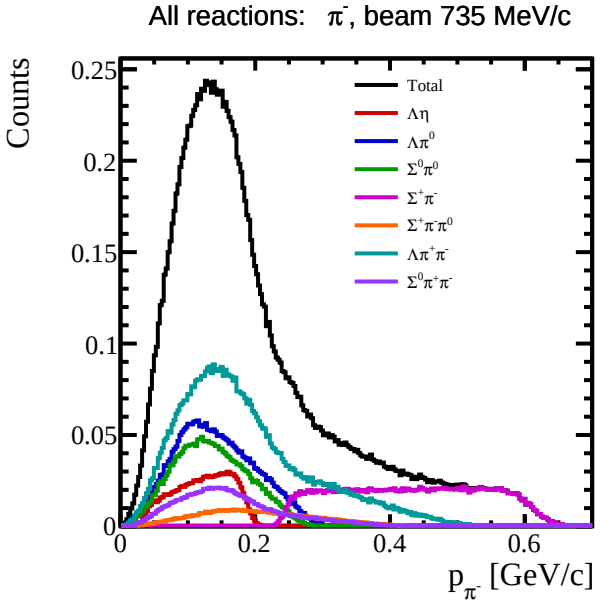

Λπ+π-: proton lab momentum
Counts
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_z,p [GeV/c]

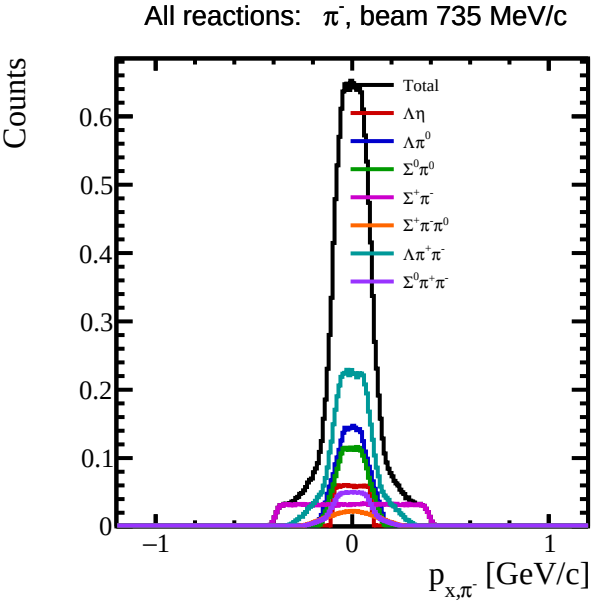

$\Sigma^0\pi^+\pi^-$: proton lab momentum
Counts
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0
0
0.5
1
p_p [GeV/c]

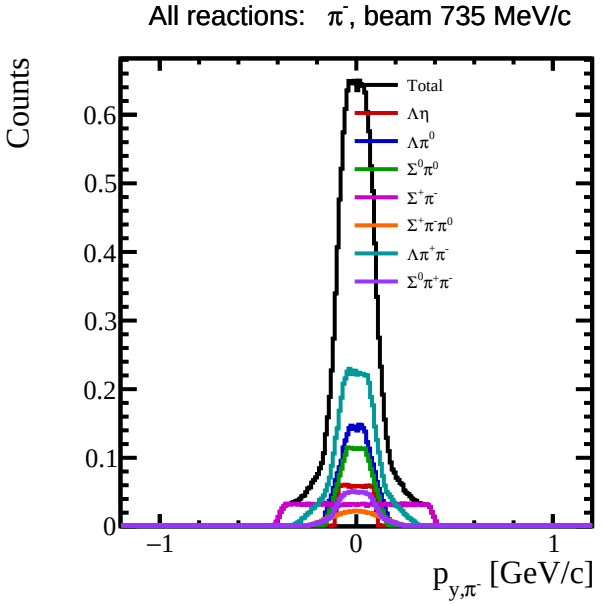

$\Sigma^0\pi^+\pi^-$: proton lab momentum
Counts
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
$p_{x,p}$ [GeV/c]

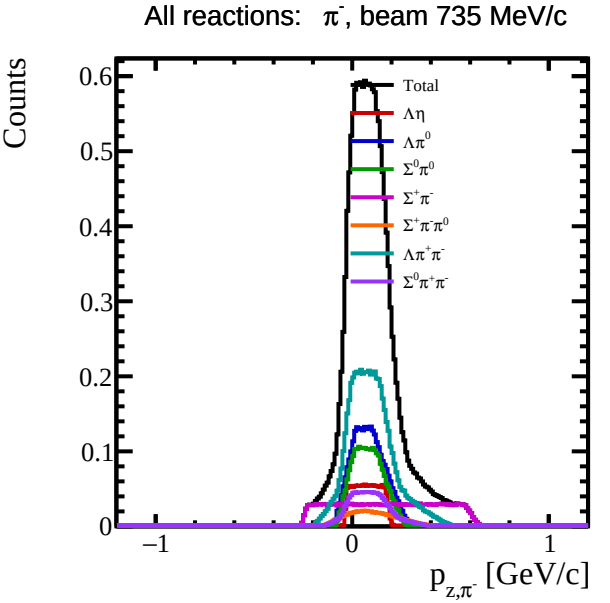

$\Sigma^0\pi^+\pi^-$: proton lab momentum
Counts
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
$p_{y,p}$ [GeV/c]

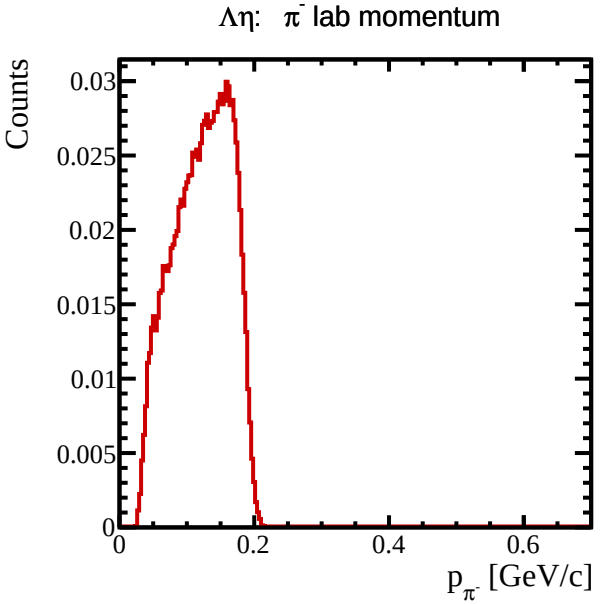

$\Sigma^0\pi^+\pi^-$: proton lab momentum
Counts
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
$p_{z,p}$ [GeV/c]

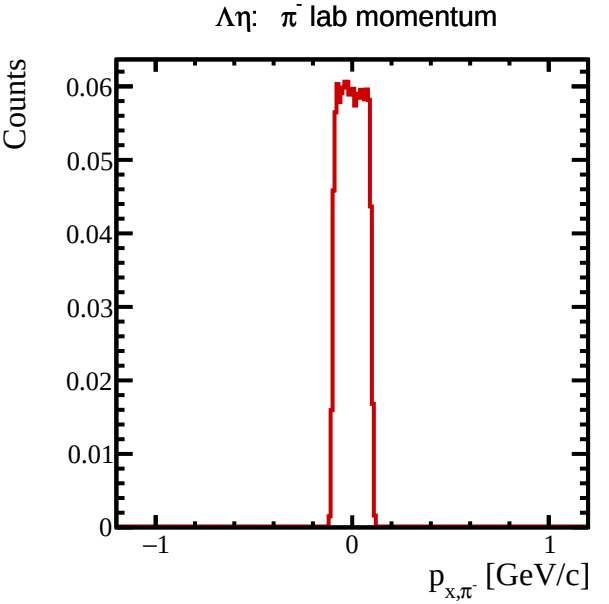

All reactions: π⁻, beam 735 MeV/c
Counts
Total
Λη
Λπ⁰
Σ⁰π⁰
Σ⁺π⁻
Σ⁺π⁻π⁰
Λπ⁺π⁻
Σ⁰π⁺π⁻
p_π⁻ [GeV/c]

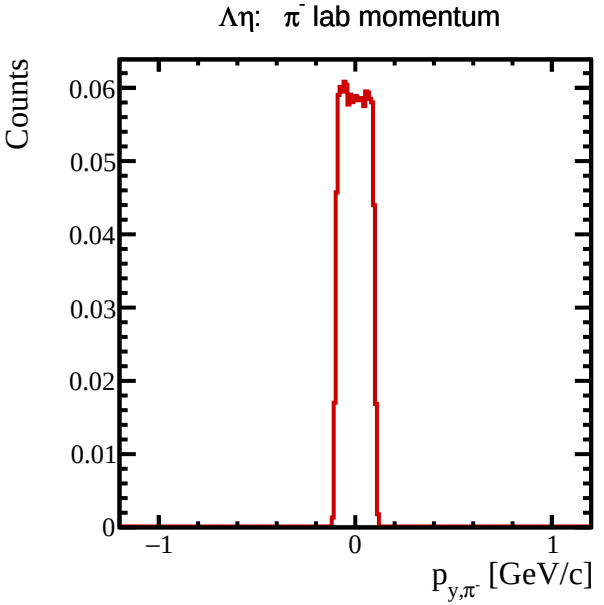

All reactions: π⁻, beam 735 MeV/c
Counts
Total
Λη
Λπ⁰
Σ⁰π⁰
Σ⁺π⁻
Σ⁺π⁻π⁰
Λπ⁺π⁻
Σ⁰π⁺π⁻
p_x,π⁻ [GeV/c]

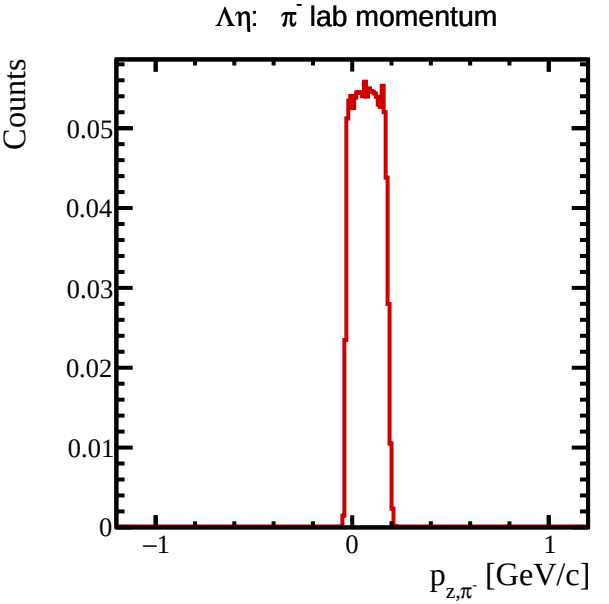

All reactions: π⁻, beam 735 MeV/c
Counts
Total
Λη
Λπ⁰
Σ⁰π⁰
Σ⁺π⁻
Σ⁺π⁻π⁰
Λπ⁺π⁻
Σ⁰π⁺π⁻
p_y,π⁻ [GeV/c]

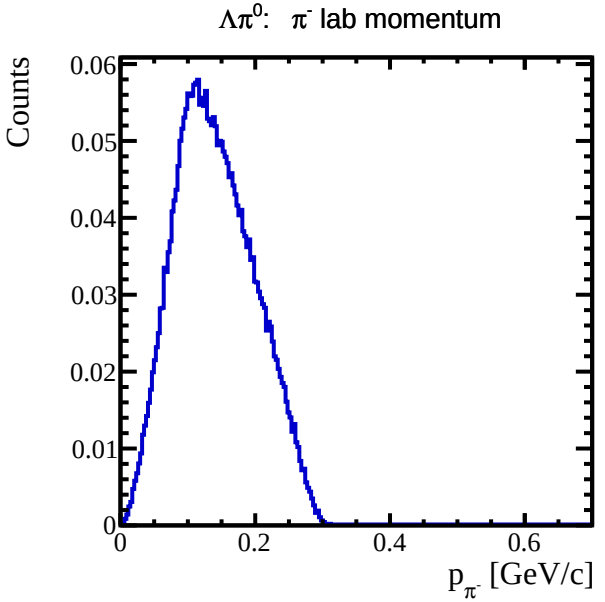

All reactions: π⁻, beam 735 MeV/c
Counts
Total
Λη
Λπ⁰
Σ⁰π⁰
Σ⁺π⁻
Σ⁺π⁻π⁰
Λπ⁺π⁻
Σ⁰π⁺π⁻
p_z,π⁻ [GeV/c]

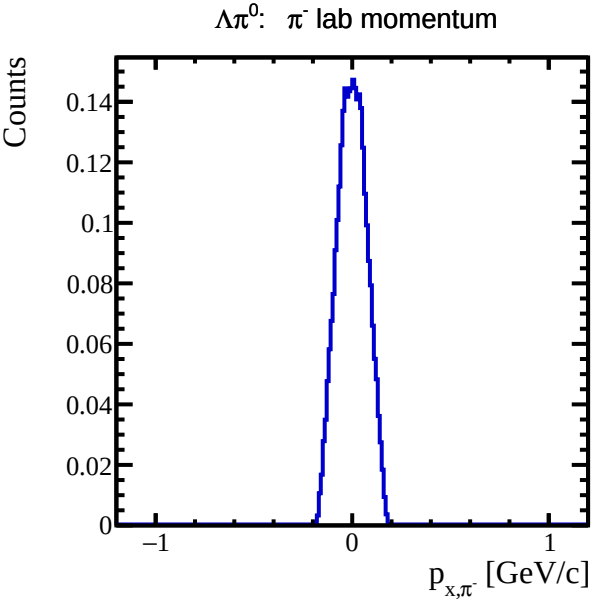

Λη: π⁻ lab momentum
Counts
0.03
0.025
0.02
0.015
0.01
0.005
0
0
0.2
0.4
0.6
p_π⁻ [GeV/c]

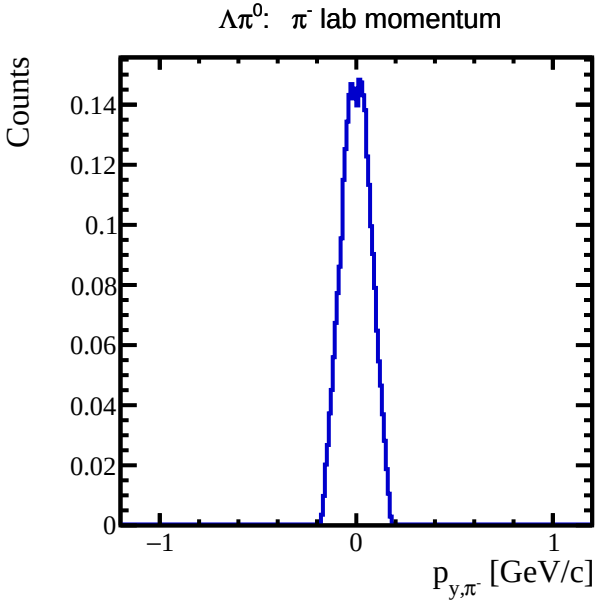

Λη: π⁻ lab momentum
Counts
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_x,π⁻ [GeV/c]

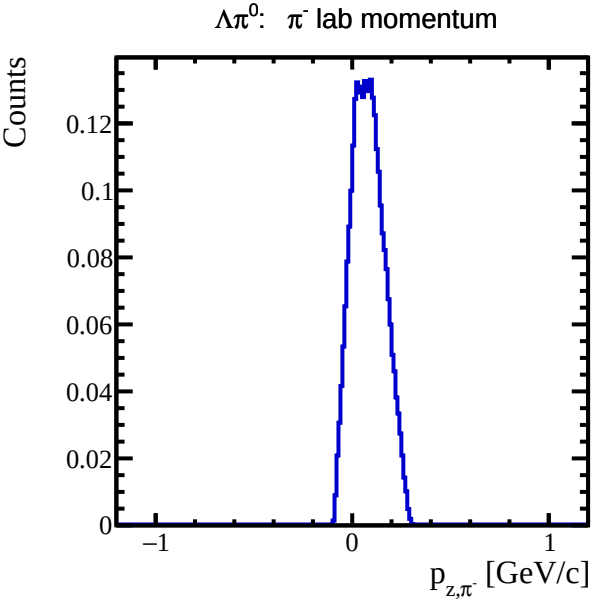

Λη: π⁻ lab momentum
Counts
0.06
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_y,π⁻ [GeV/c]

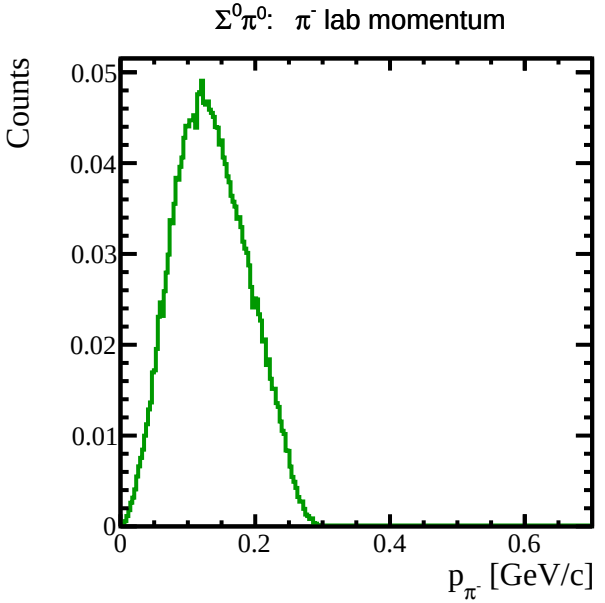

Λη: π⁻ lab momentum
Counts
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_z,π⁻ [GeV/c]

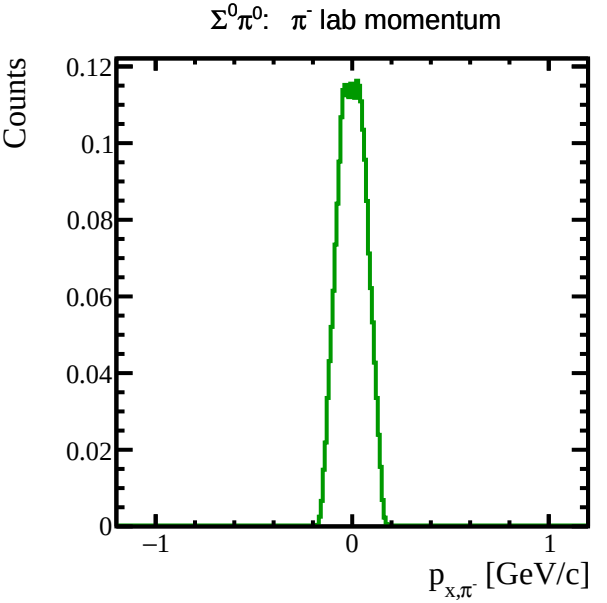

$\Lambda\pi^0$: π^- lab momentum
Counts
0.06
0.05
0.04
0.03
0.02
0.01
0
0
0.2
0.4
0.6
p_{π^-} [GeV/c]

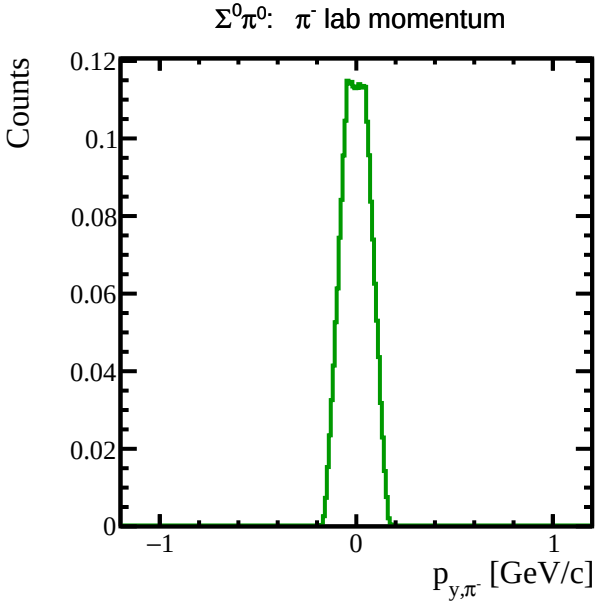

$\Lambda\pi^0$: π^- lab momentum
Counts
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
p_{x,π^-} [GeV/c]

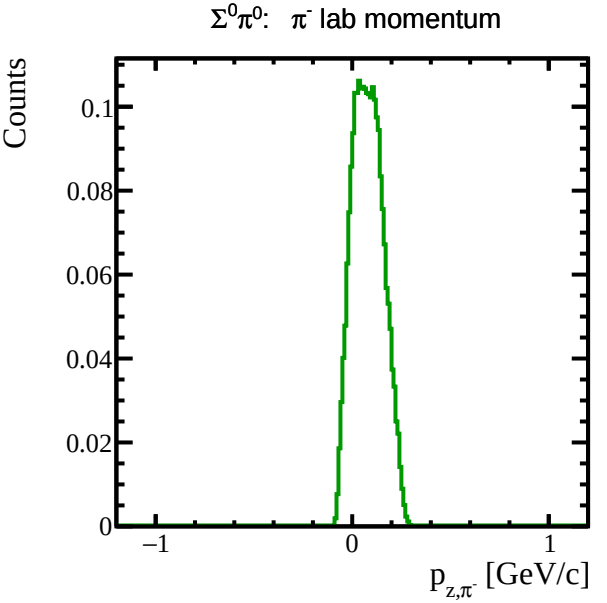

$\Lambda\pi^0$: π^- lab momentum
Counts
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
p_{y,π^-} [GeV/c]

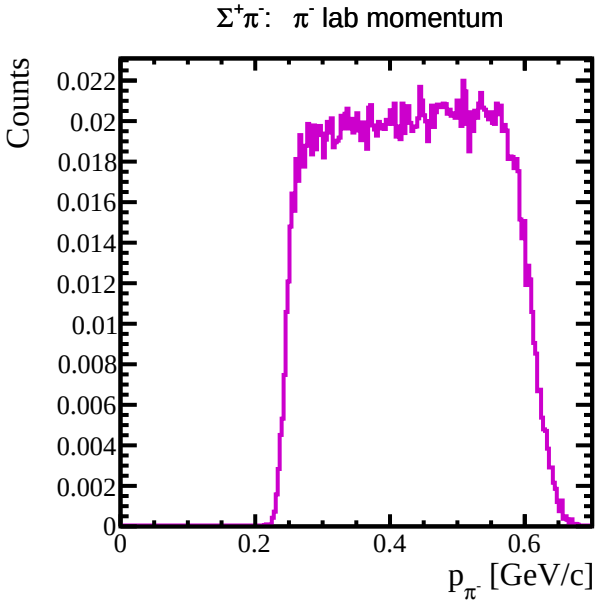

$\Lambda\pi^0$: π^- lab momentum
Counts
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
p_{z,π^-} [GeV/c]

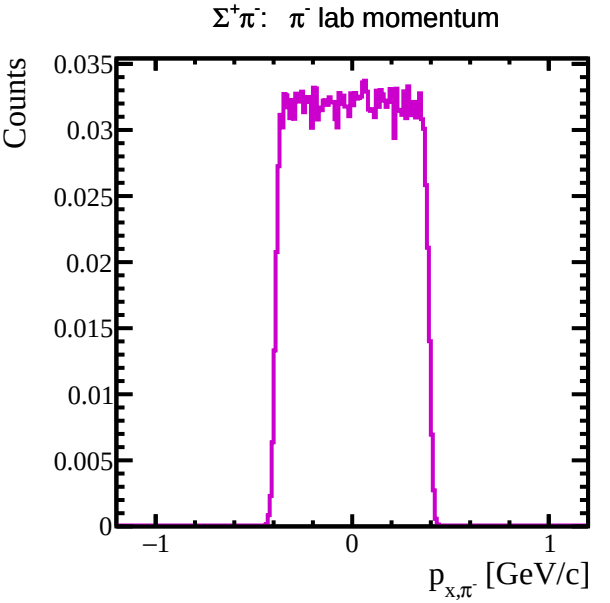

$\Sigma^0\pi^0$: π^- lab momentum
Counts
0.05
0.04
0.03
0.02
0.01
0
0
0.2
0.4
0.6
p_{π^-} [GeV/c]

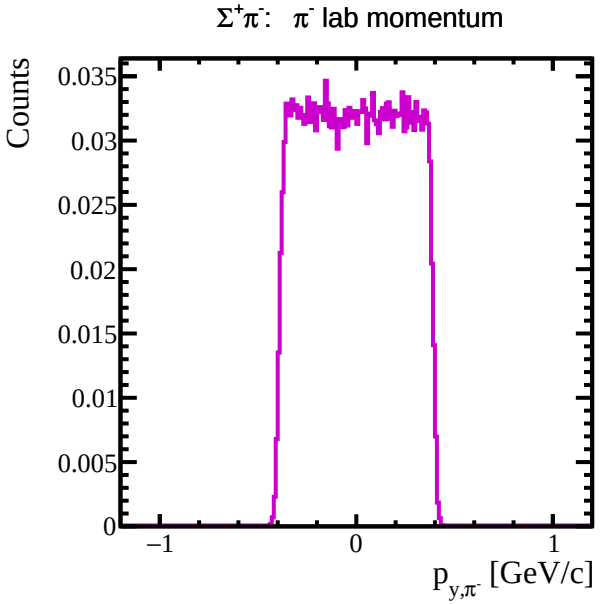

$\Sigma^0\pi^0$: π^- lab momentum
Counts
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
p_{x,π^-} [GeV/c]

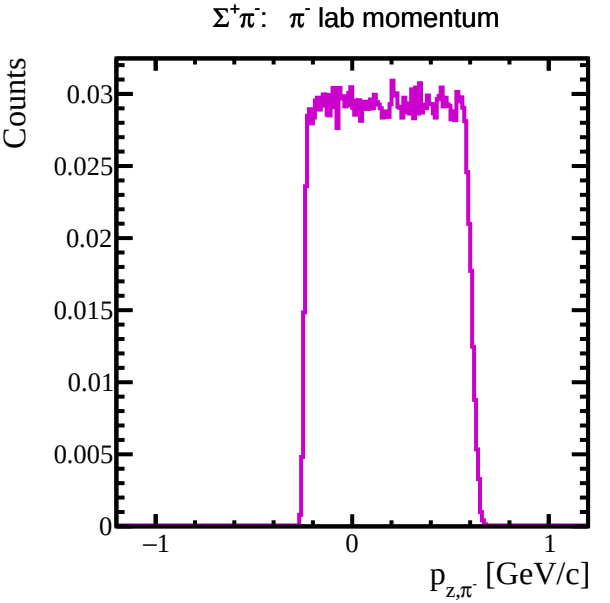

$\Sigma^0\pi^0$: π^- lab momentum
Counts
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
p_{y,π^-} [GeV/c]

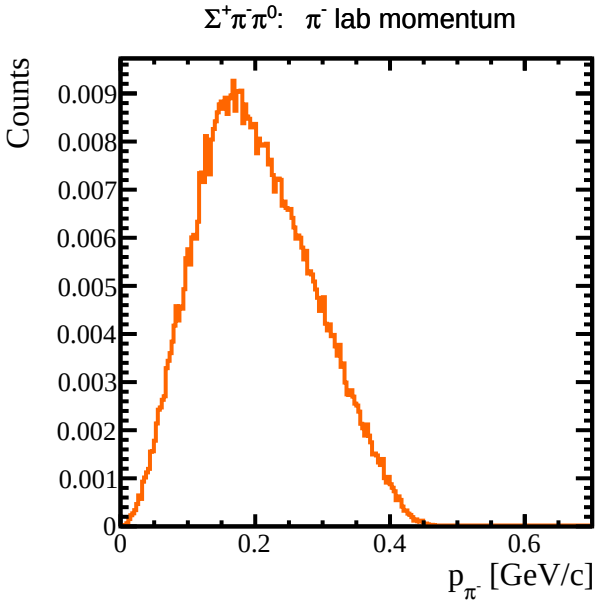

$\Sigma^0\pi^0$: π^- lab momentum
Counts
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
p_{z,π^-} [GeV/c]

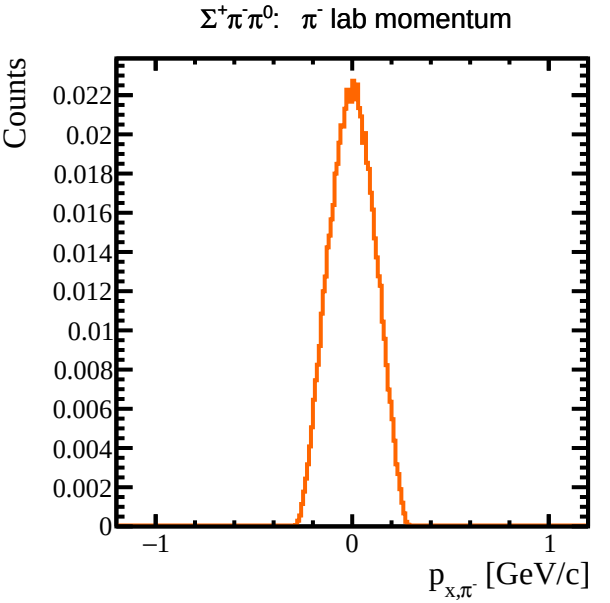

Σ+π-: π- lab momentum
Counts
0.022
0.02
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
0
0.2
0.4
0.6
pπ- [GeV/c]

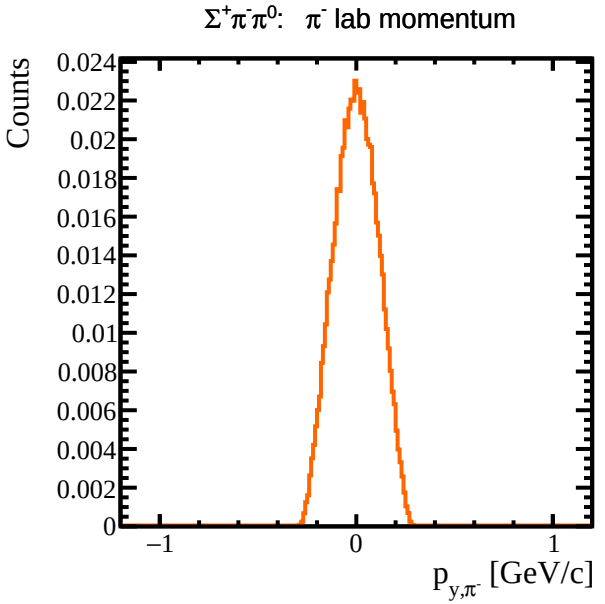

Σ+π-: π- lab momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
px,π- [GeV/c]

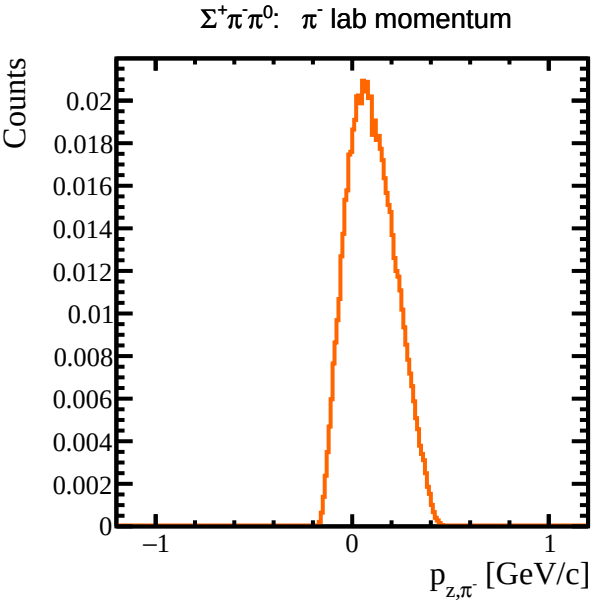

Σ+π-: π- lab momentum
Counts
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
py,π- [GeV/c]

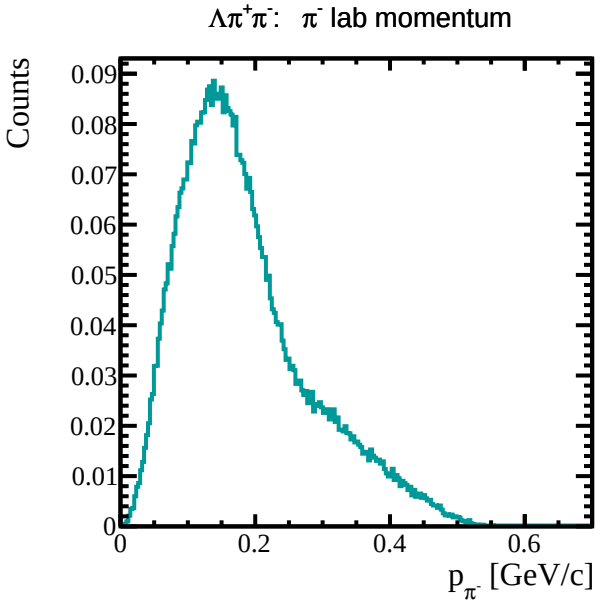

Σ+π-: π- lab momentum
Counts
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
pz,π- [GeV/c]

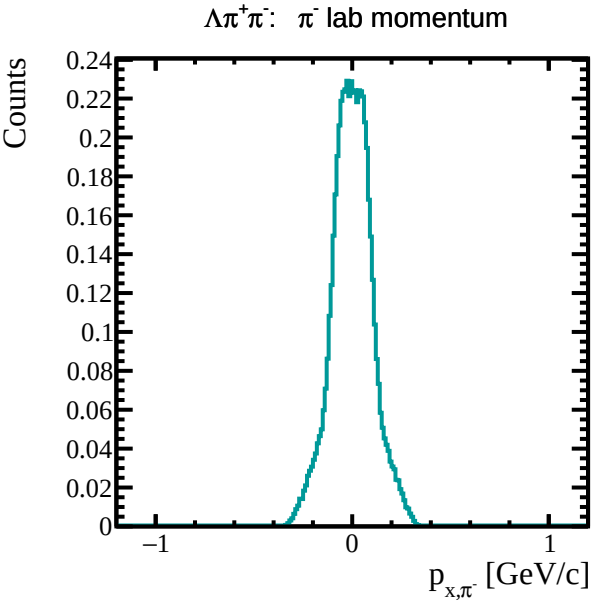

$\Sigma^+\pi^-\pi^0$: π^- lab momentum
Counts
0.009
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0
0
0.2
0.4
0.6
p_{π^-} [GeV/c]

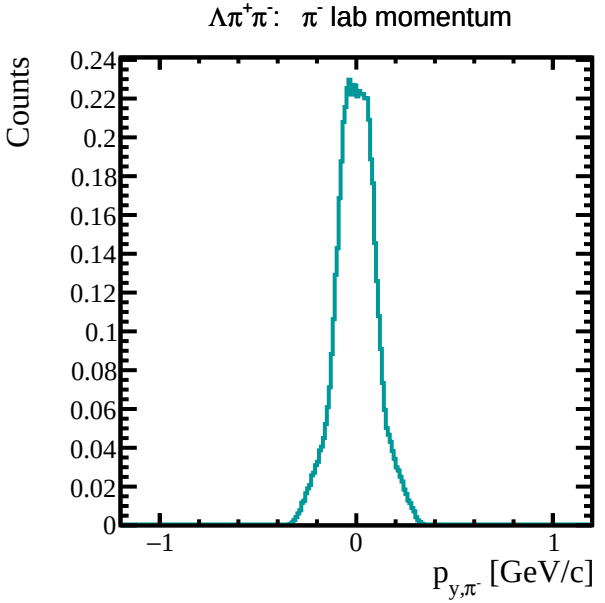

$\Sigma^+\pi^-\pi^0$: π^- lab momentum
Counts
0.022
0.02
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
p_{x,π^-} [GeV/c]

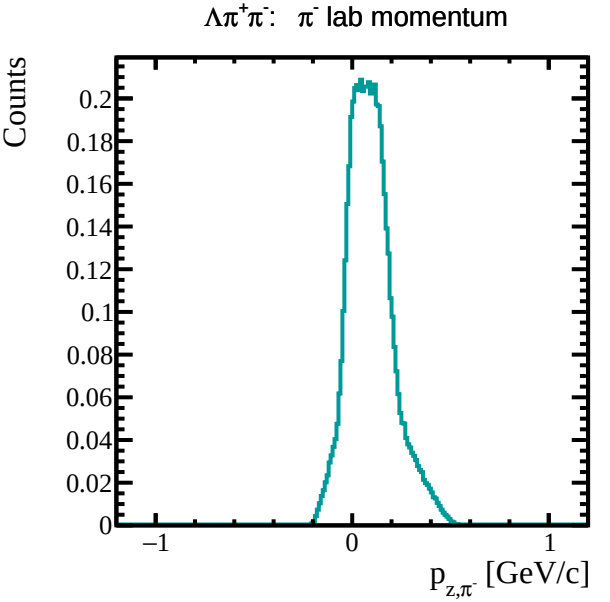

$\Sigma^+\pi^-\pi^0$: π^- lab momentum
Counts
0.024
0.022
0.02
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
p_{y,π^-} [GeV/c]

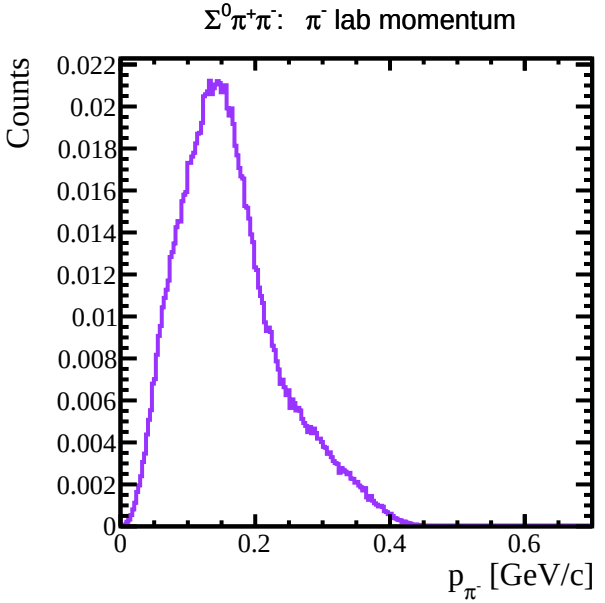

$\Sigma^+\pi^-\pi^0$: π^- lab momentum
Counts
0.02
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
−1
0
1
p_{z,π^-} [GeV/c]

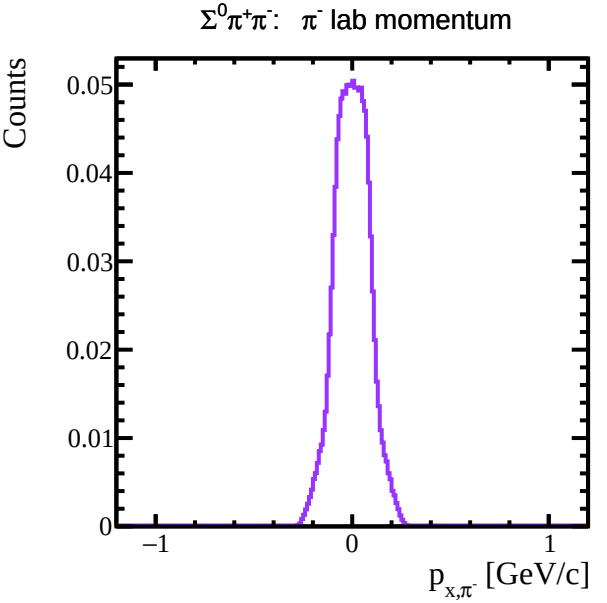

Λπ+π-: π- lab momentum
Counts
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
0
0.2
0.4
0.6
pπ- [GeV/c]

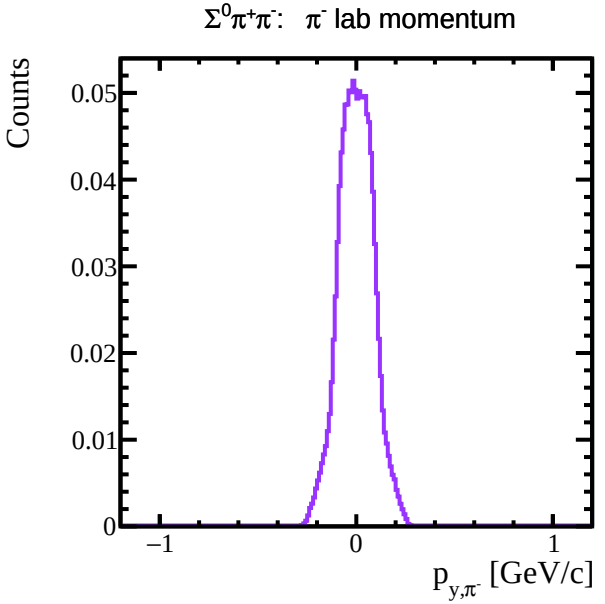

Λπ+π-: π- lab momentum
Counts
0.24
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
px,π- [GeV/c]

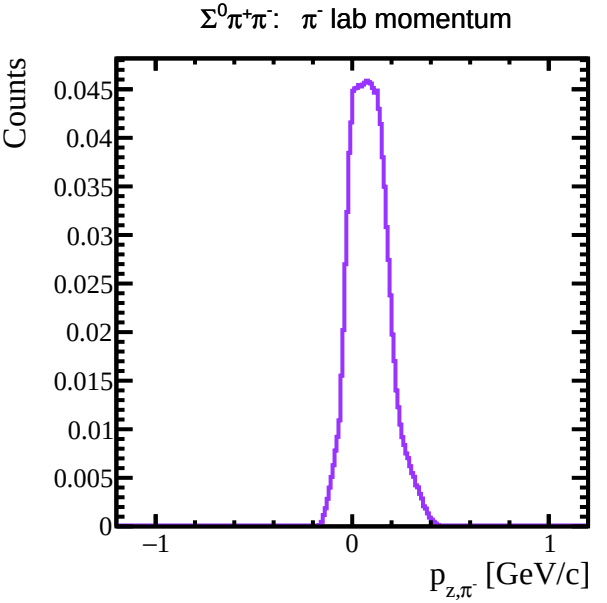

Λπ+π-: π- lab momentum
Counts
0.24
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
py,π- [GeV/c]

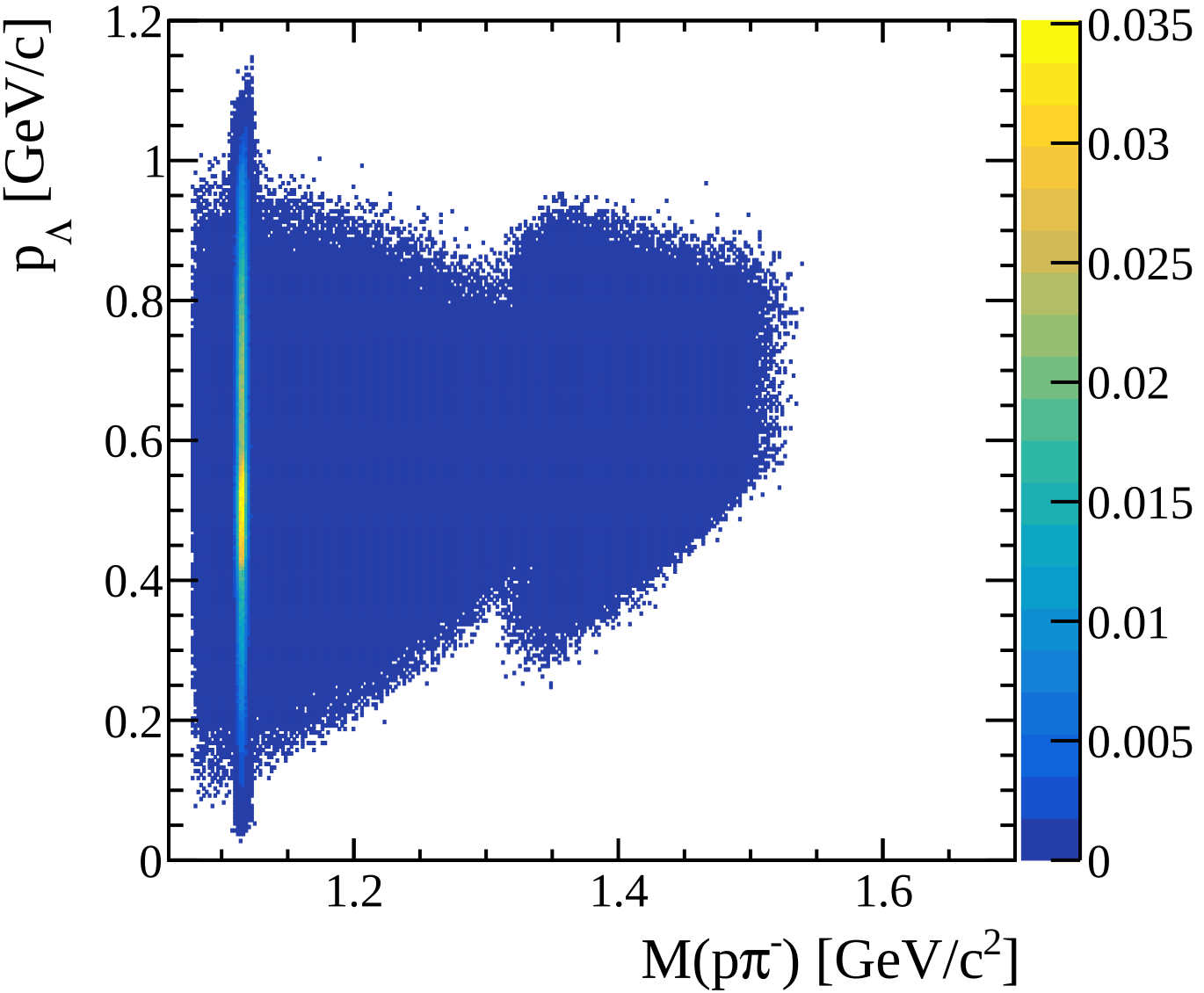

Λπ+π-: π- lab momentum
Counts
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
−1
0
1
pz,π- [GeV/c]

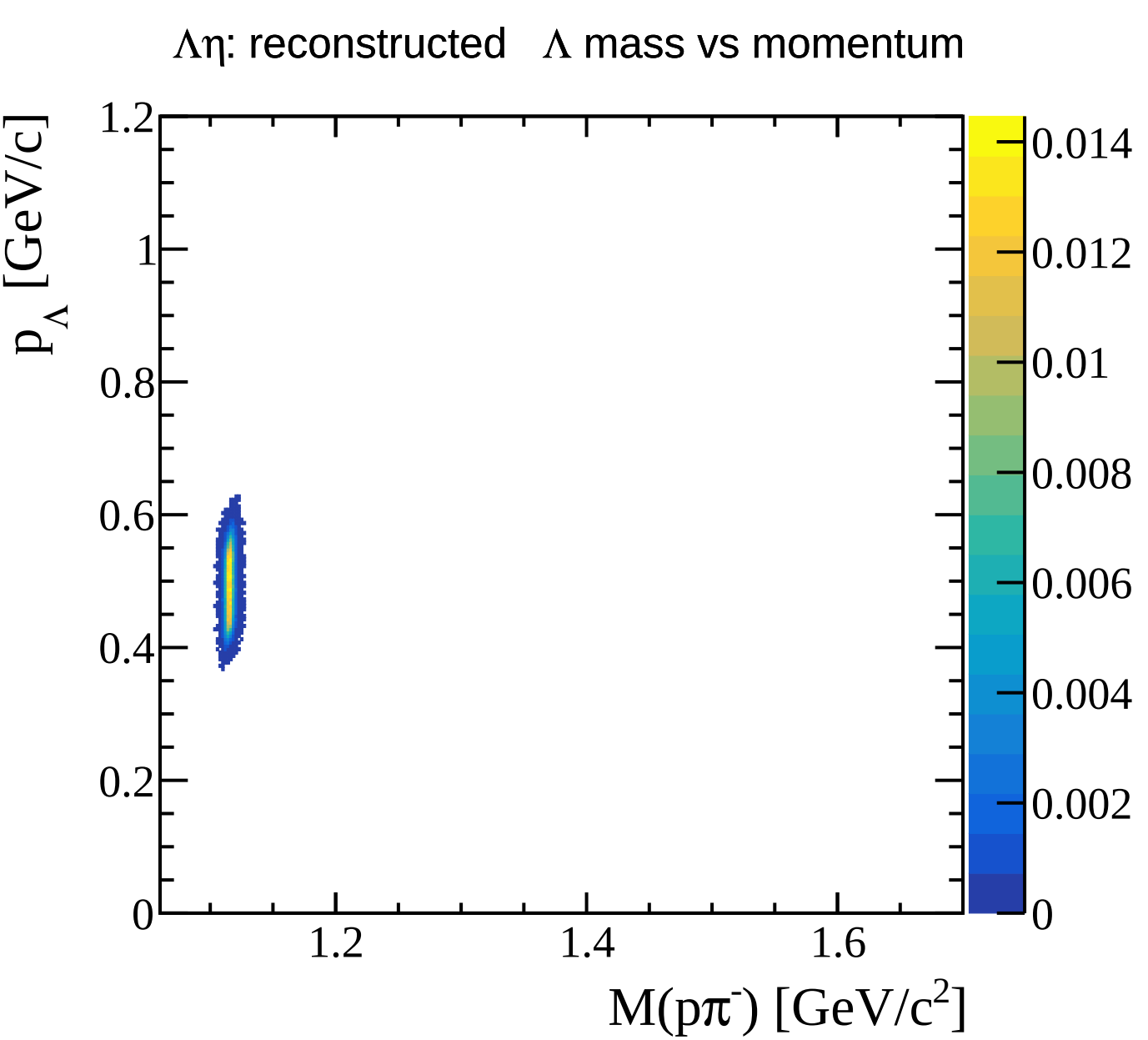

$\Sigma^0\pi^+\pi^-$: π^- lab momentum
Counts
0.022
0.02
0.018
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0
0
0.2
0.4
0.6
p_{π^-} [GeV/c]

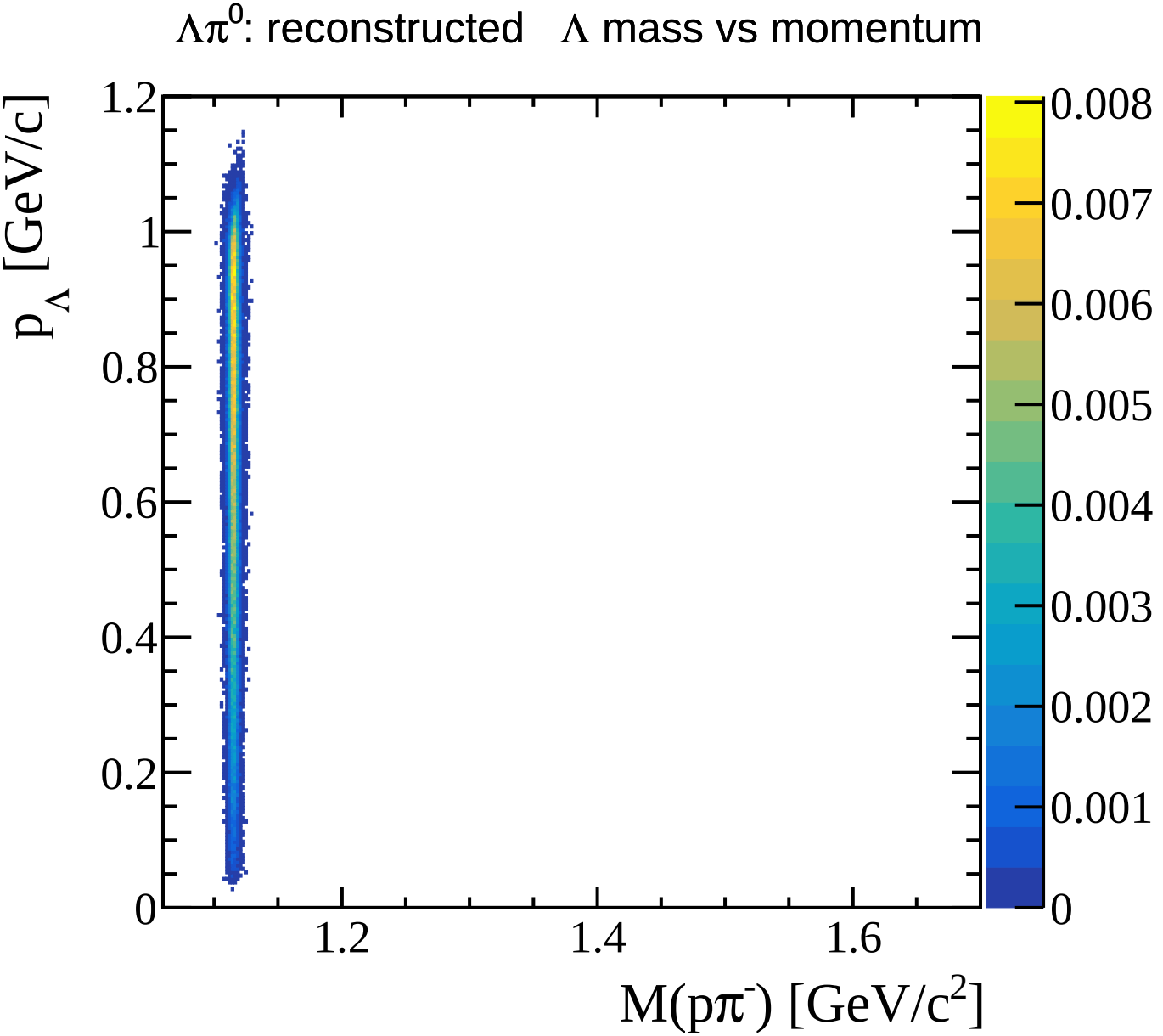

$\Sigma^0\pi^+\pi^-$: π^- lab momentum
Counts
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_{x,π^-} [GeV/c]

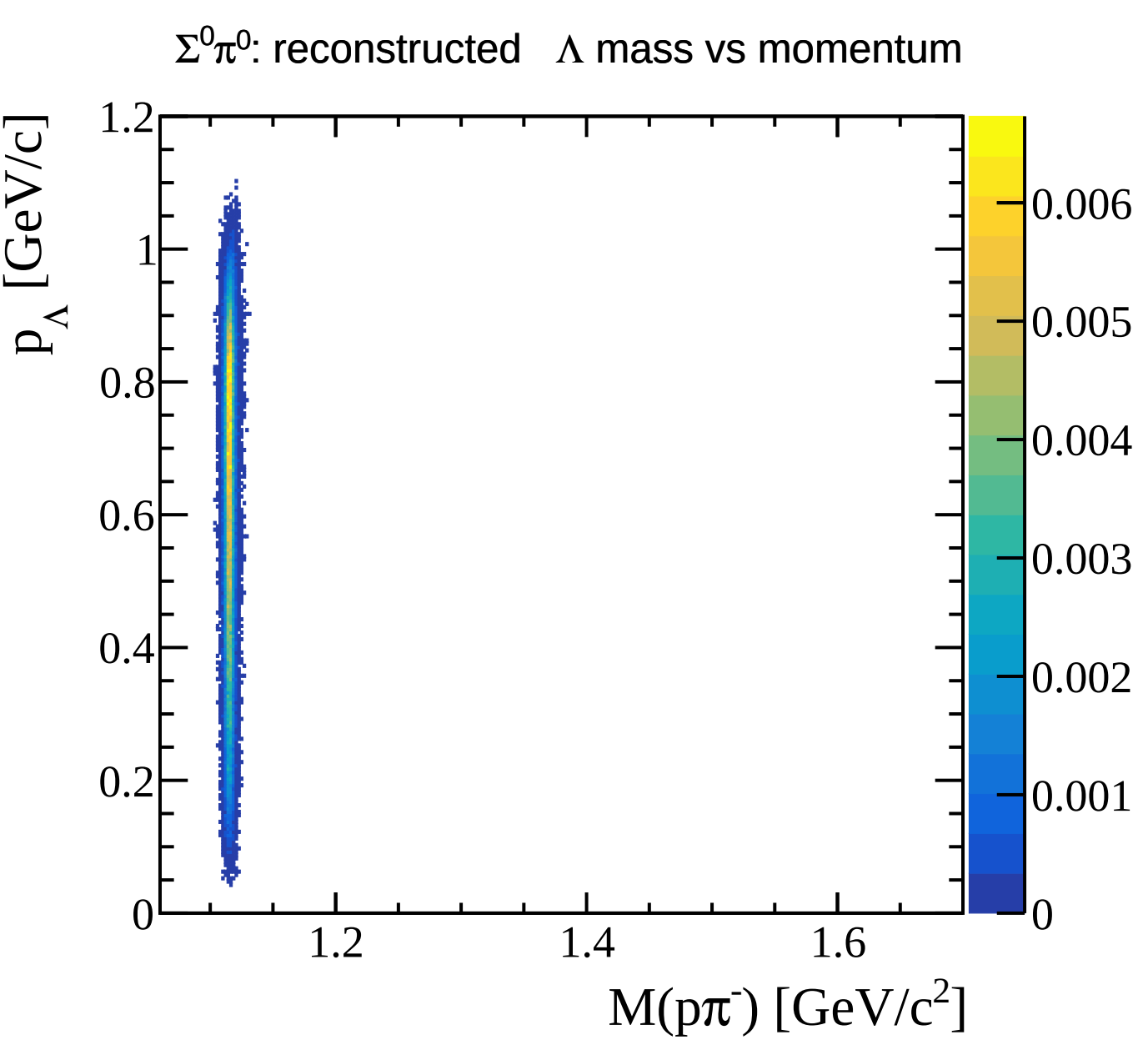

$\Sigma^0\pi^+\pi^-$: π^- lab momentum
Counts
0.05
0.04
0.03
0.02
0.01
0
−1
0
1
p_{y,π^-} [GeV/c]

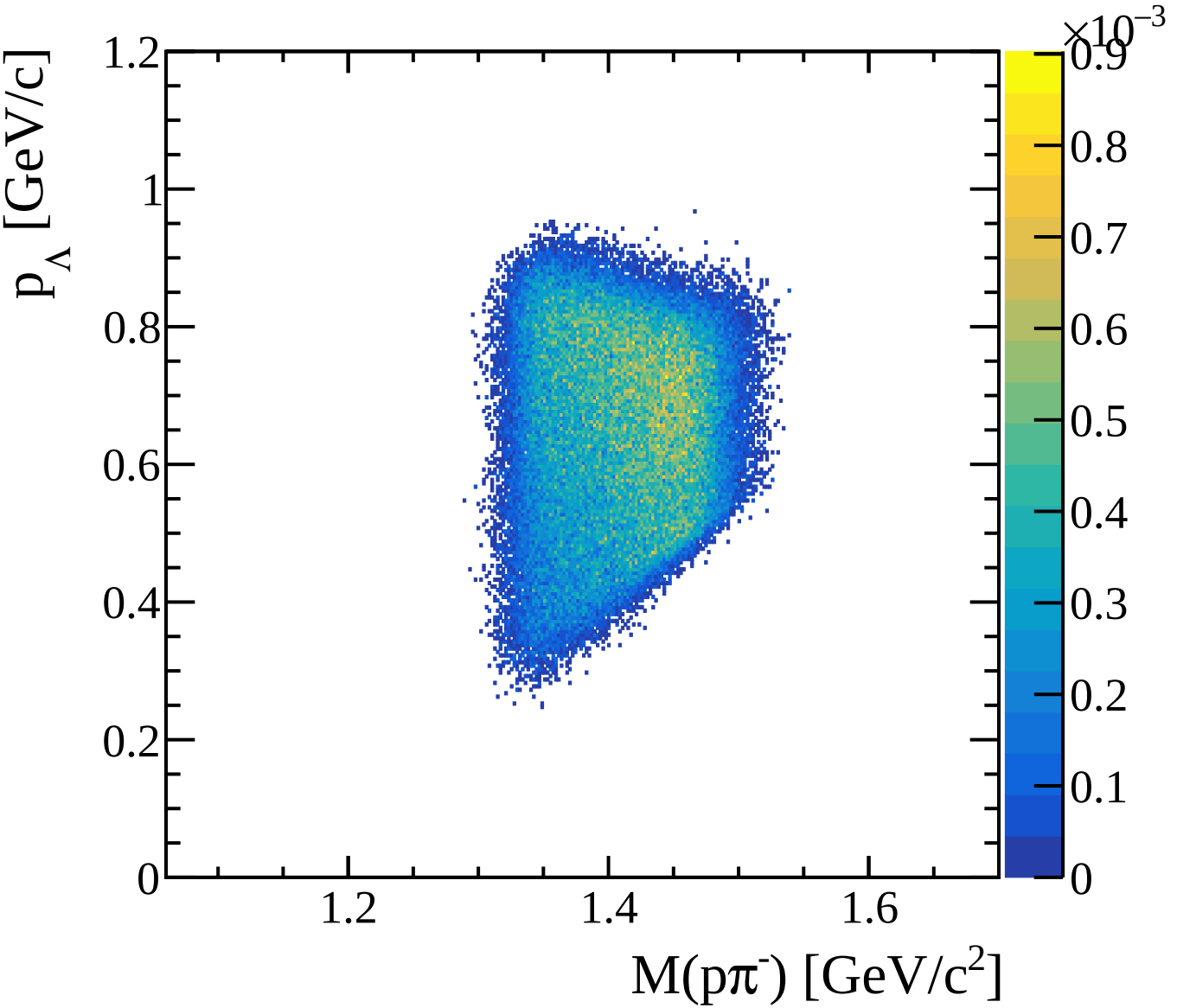

$\Sigma^0\pi^+\pi^-$: π^- lab momentum
Counts
0.045
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−1
0
1
p_{z,π^-} [GeV/c]

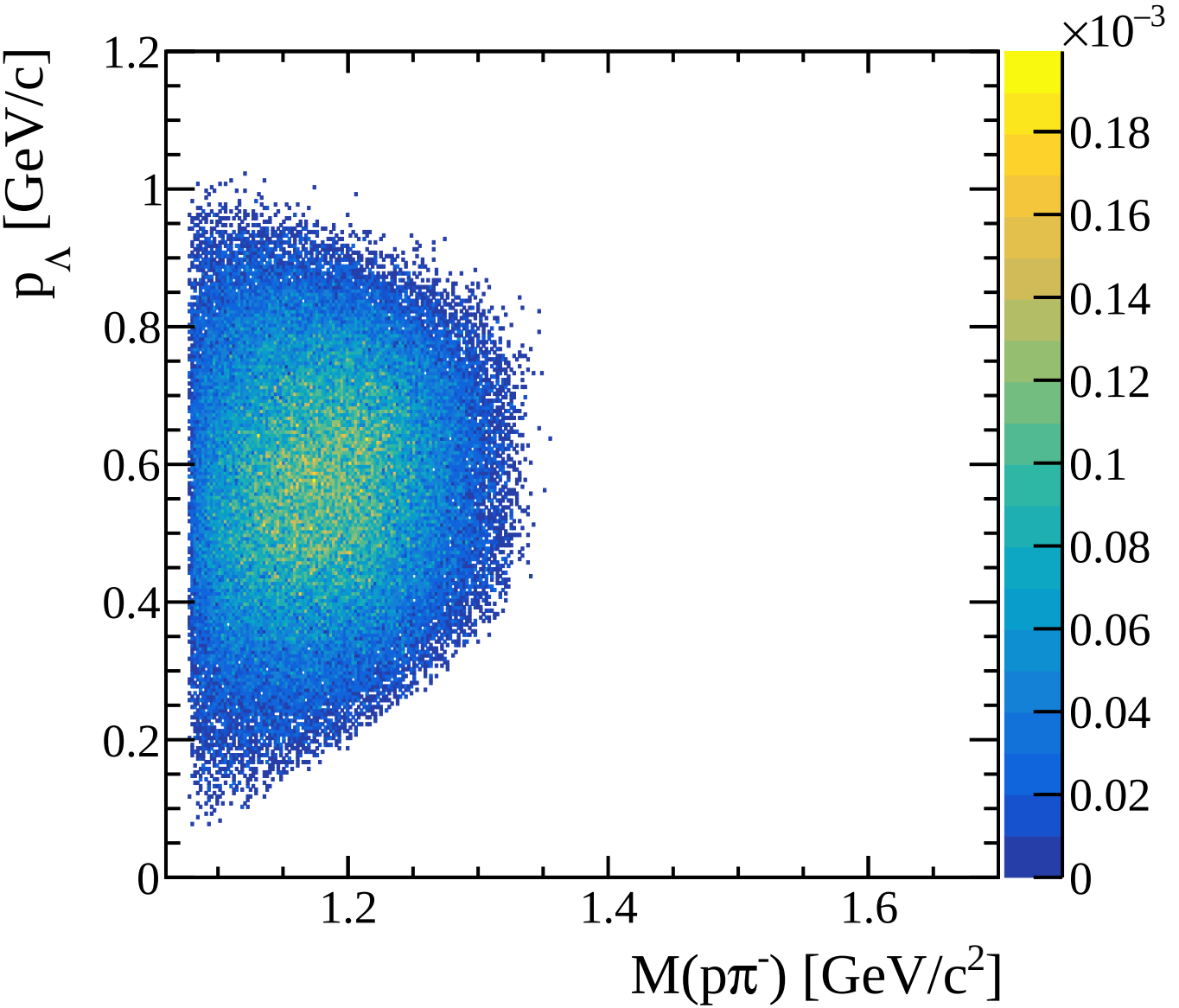

All reactions: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ⁻) [GeV/c²]
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0

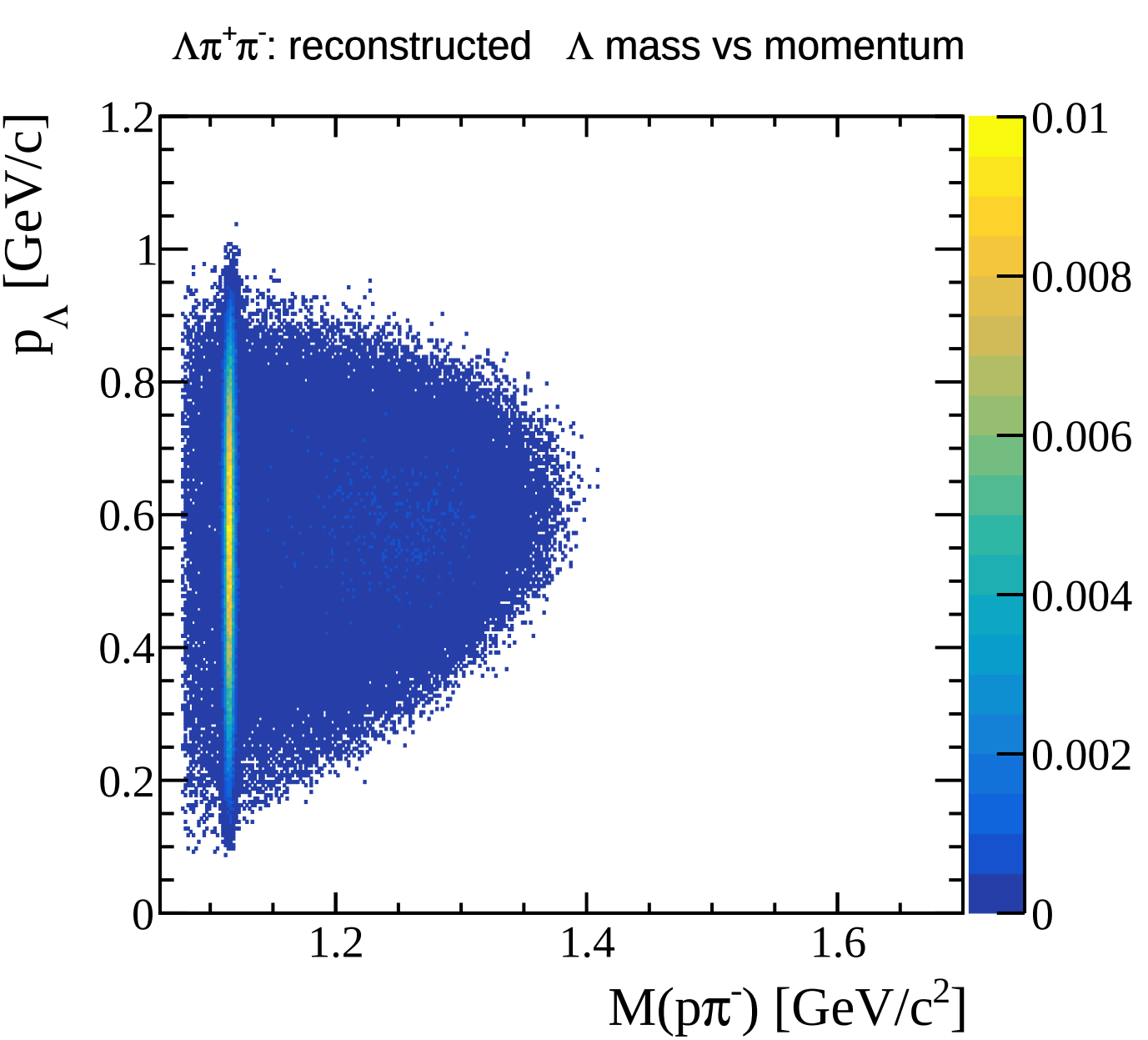

Λη: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ⁻) [GeV/c²]
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0

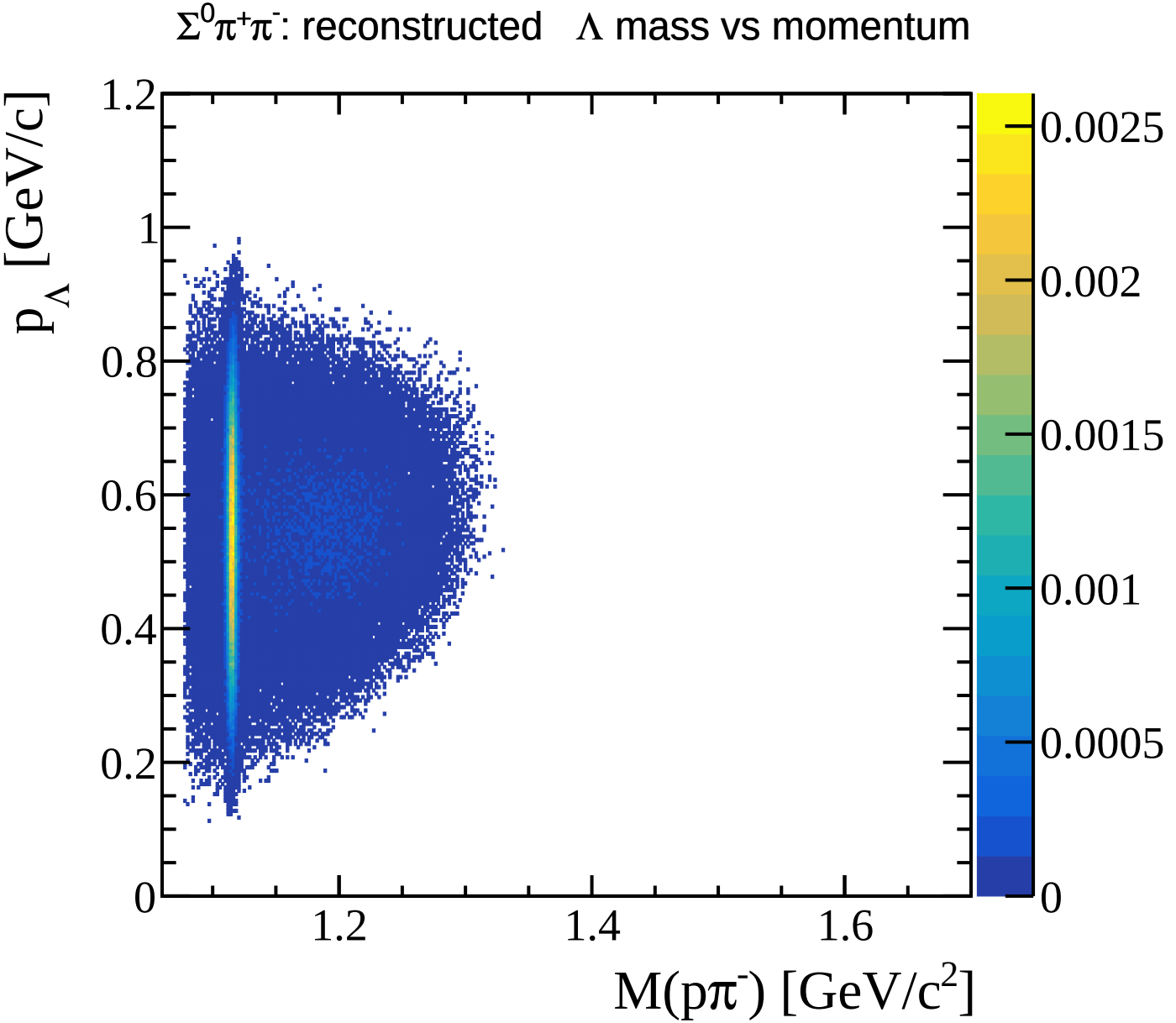

Λπ0: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ-) [GeV/c2]
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0

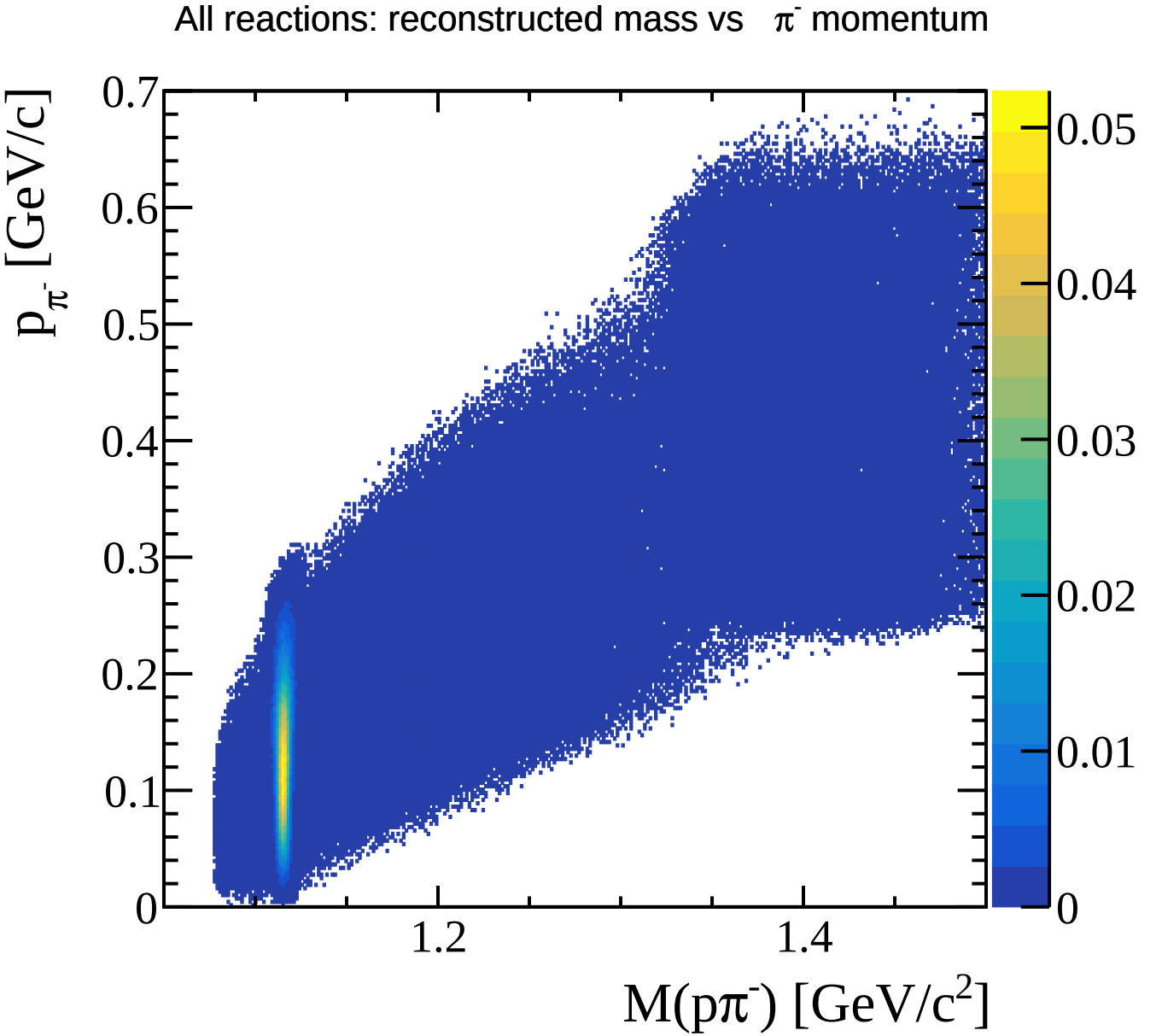

$\Sigma^0\pi^0$: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ^-) [GeV/c^2]
0.006
0.005
0.004
0.003
0.002
0.001
0

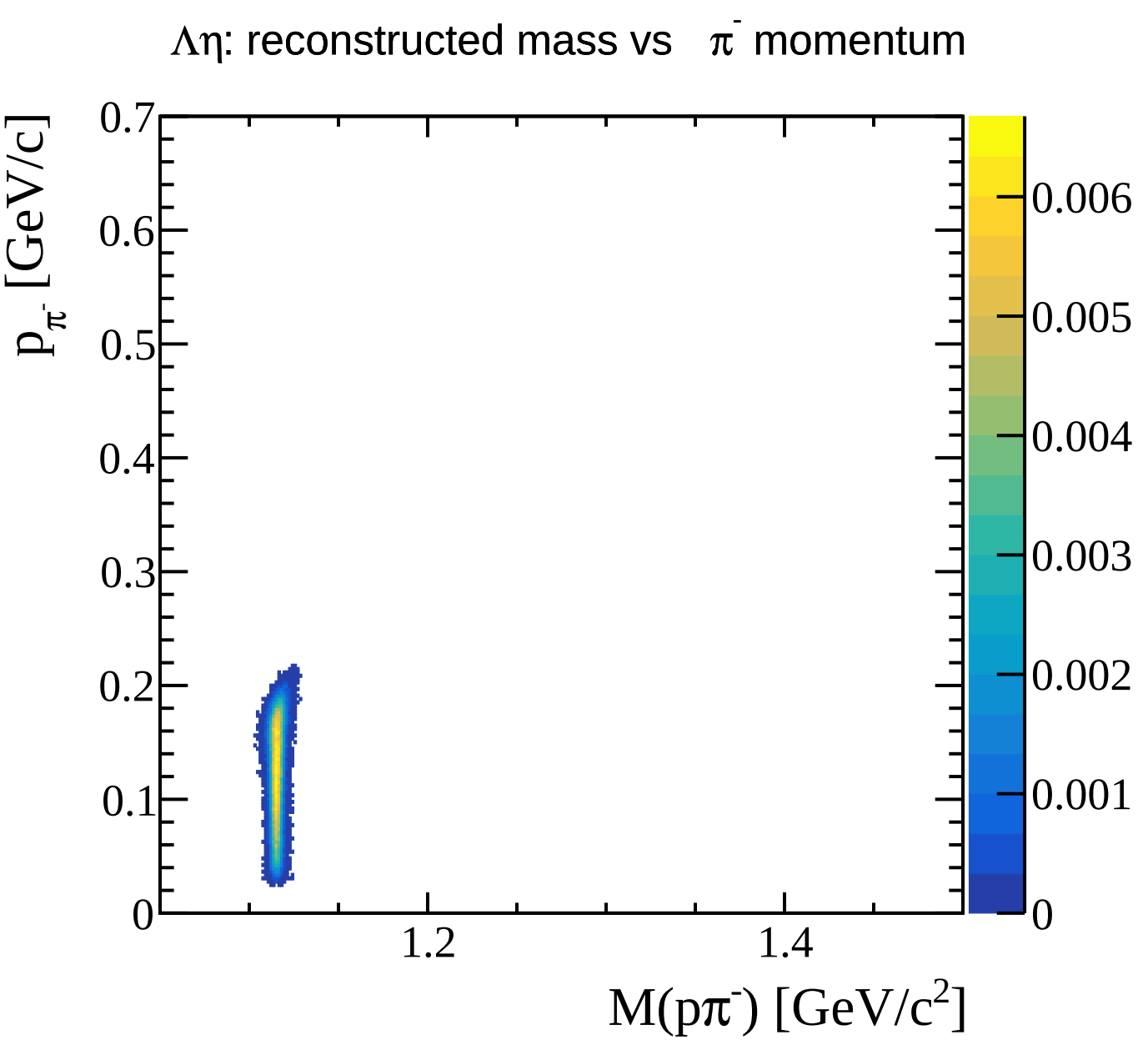

$\Sigma^+\pi^-$: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ^-) [GeV/c^2]
$\times 10^{-3}$
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0

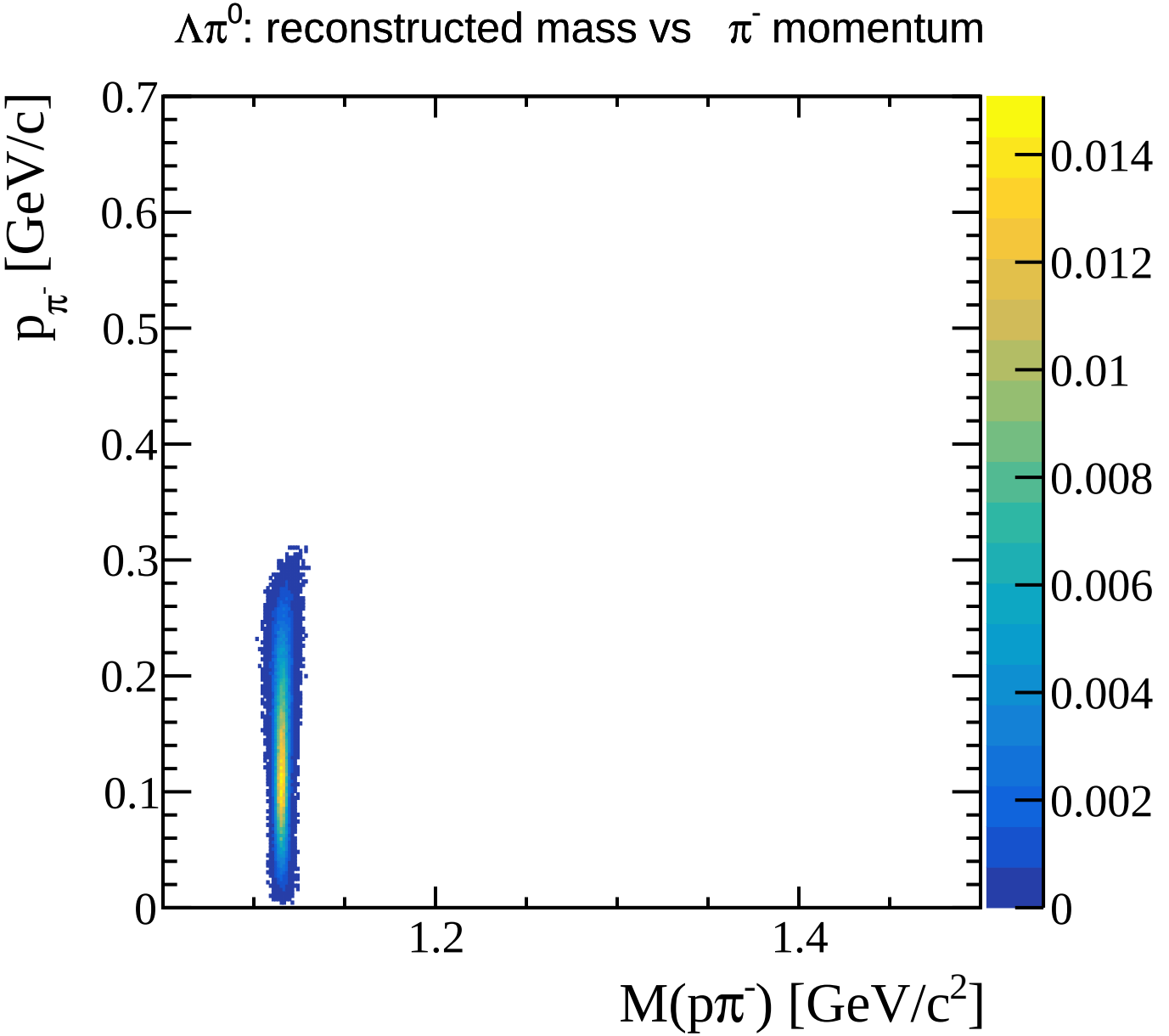

$\Sigma^+\pi^-\pi^0$: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ^-) [GeV/c^2]
$\times 10^{-3}$
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0

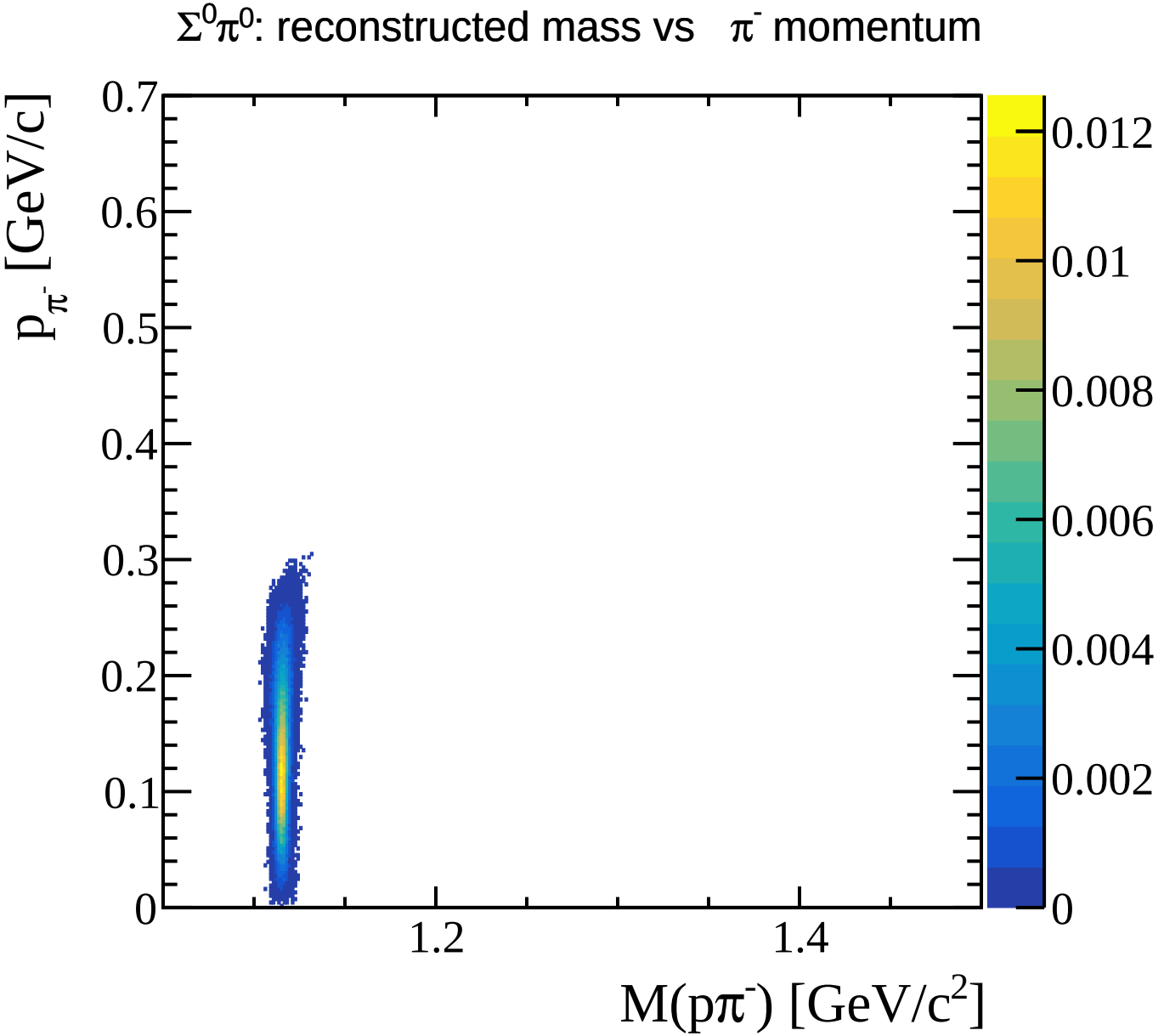

Λπ+π-: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
M(pπ-) [GeV/c2]
0.01
0.008
0.006
0.004
0.002
0

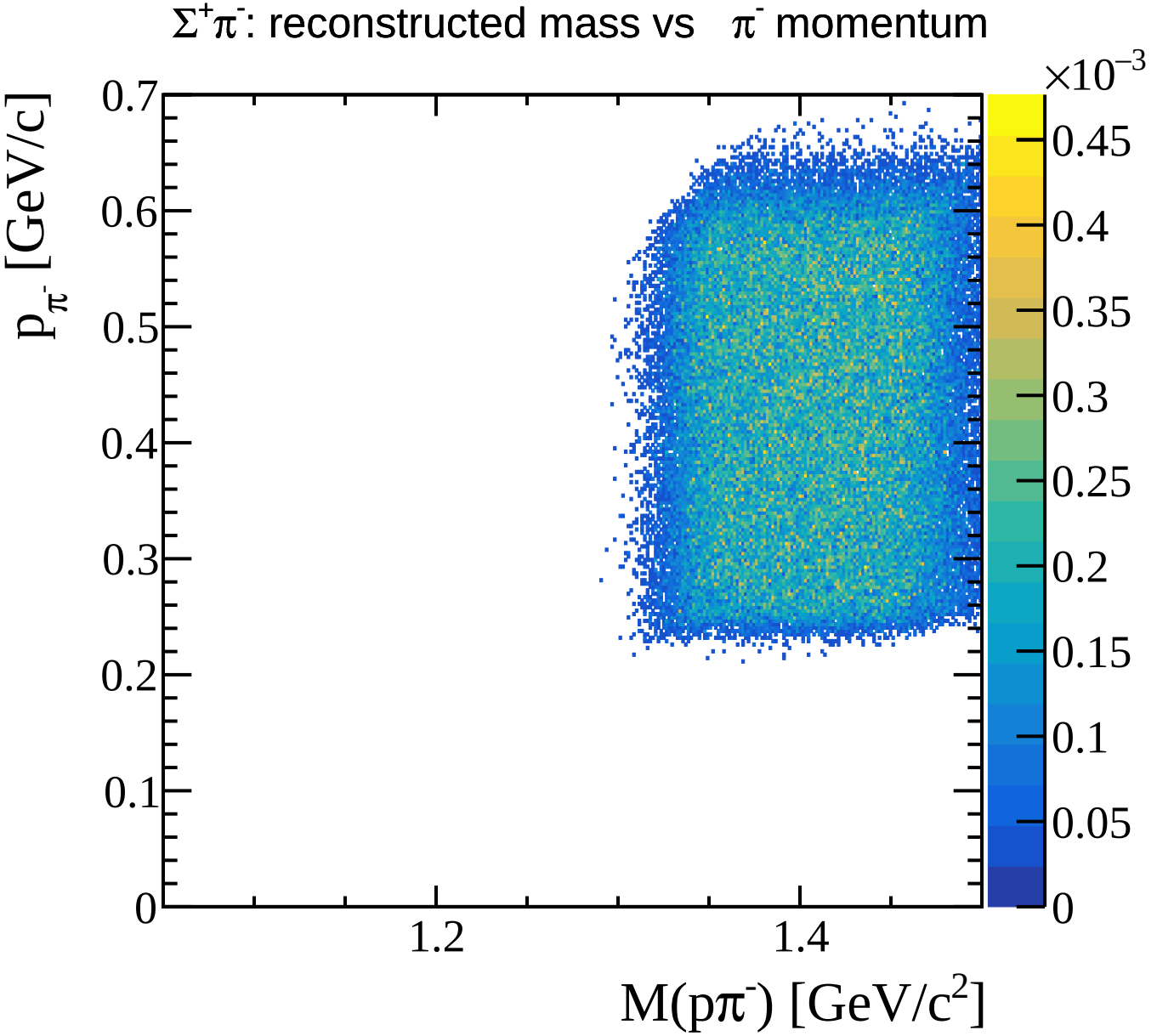

$\Sigma^0\pi^+\pi^-$: reconstructed Λ mass vs momentum
p_Λ [GeV/c]
1.2
1
0.8
0.6
0.4
0.2
0
1.2
1.4
1.6
$M(p\pi^-)$ [GeV/c^2]
0.0025
0.002
0.0015
0.001
0.0005
0

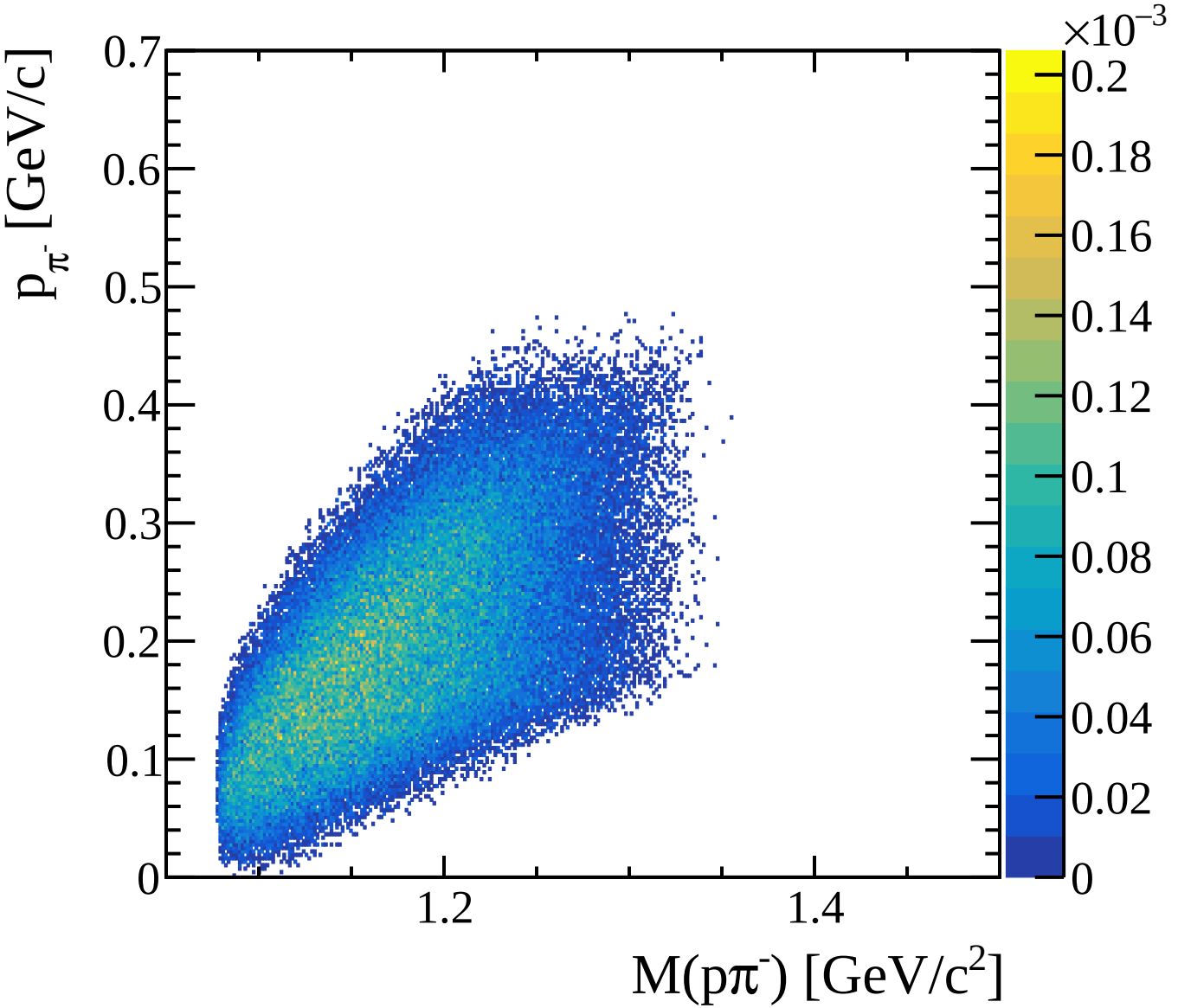

All reactions: reconstructed mass vs π- momentum
p_π- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c²]
0.05
0.04
0.03
0.02
0.01
0

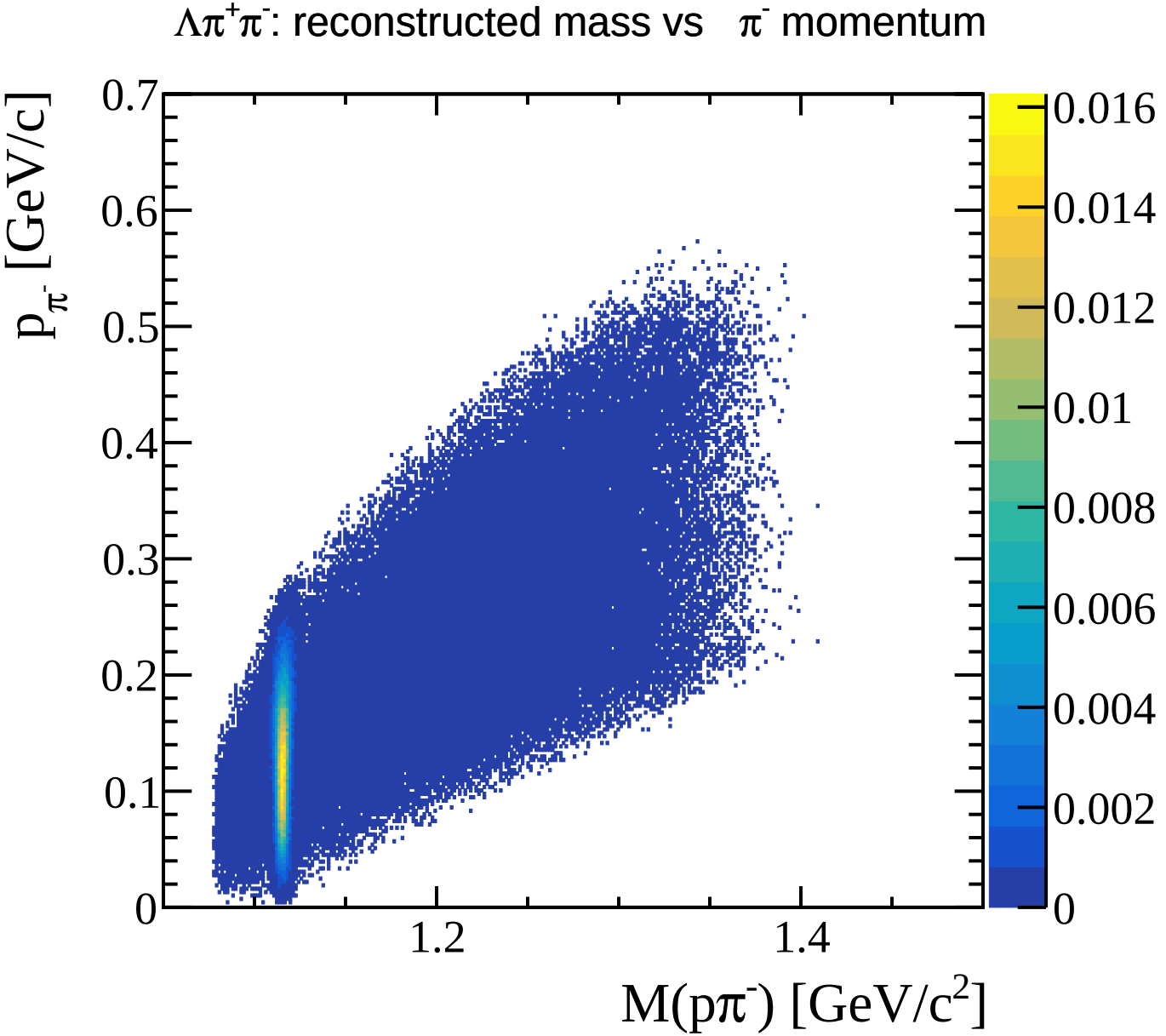

Λη: reconstructed mass vs π- momentum
p_π- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c2]
0.006
0.005
0.004
0.003
0.002
0.001
0

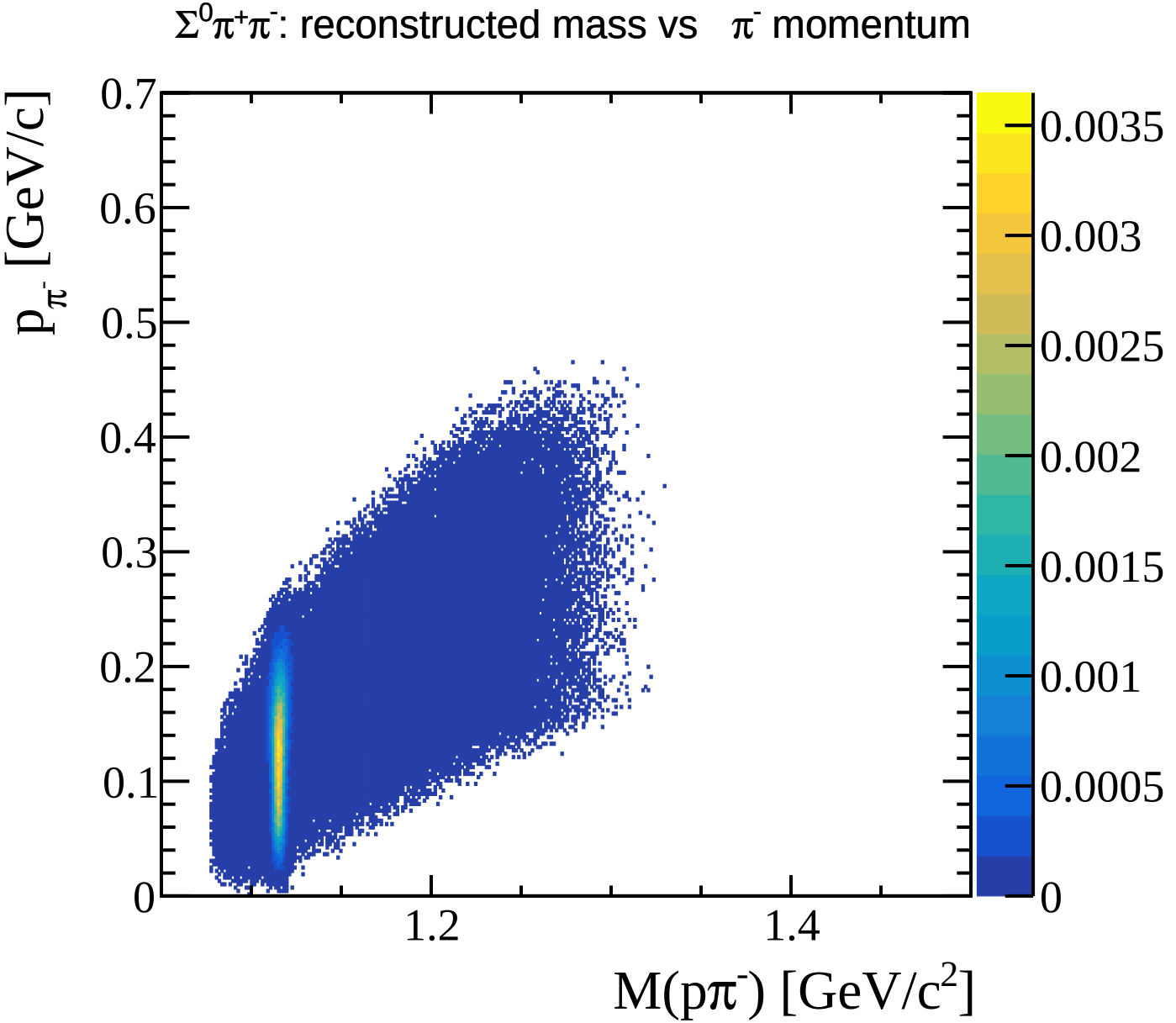

Λπ0: reconstructed mass vs π- momentum
p π- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c2]
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0

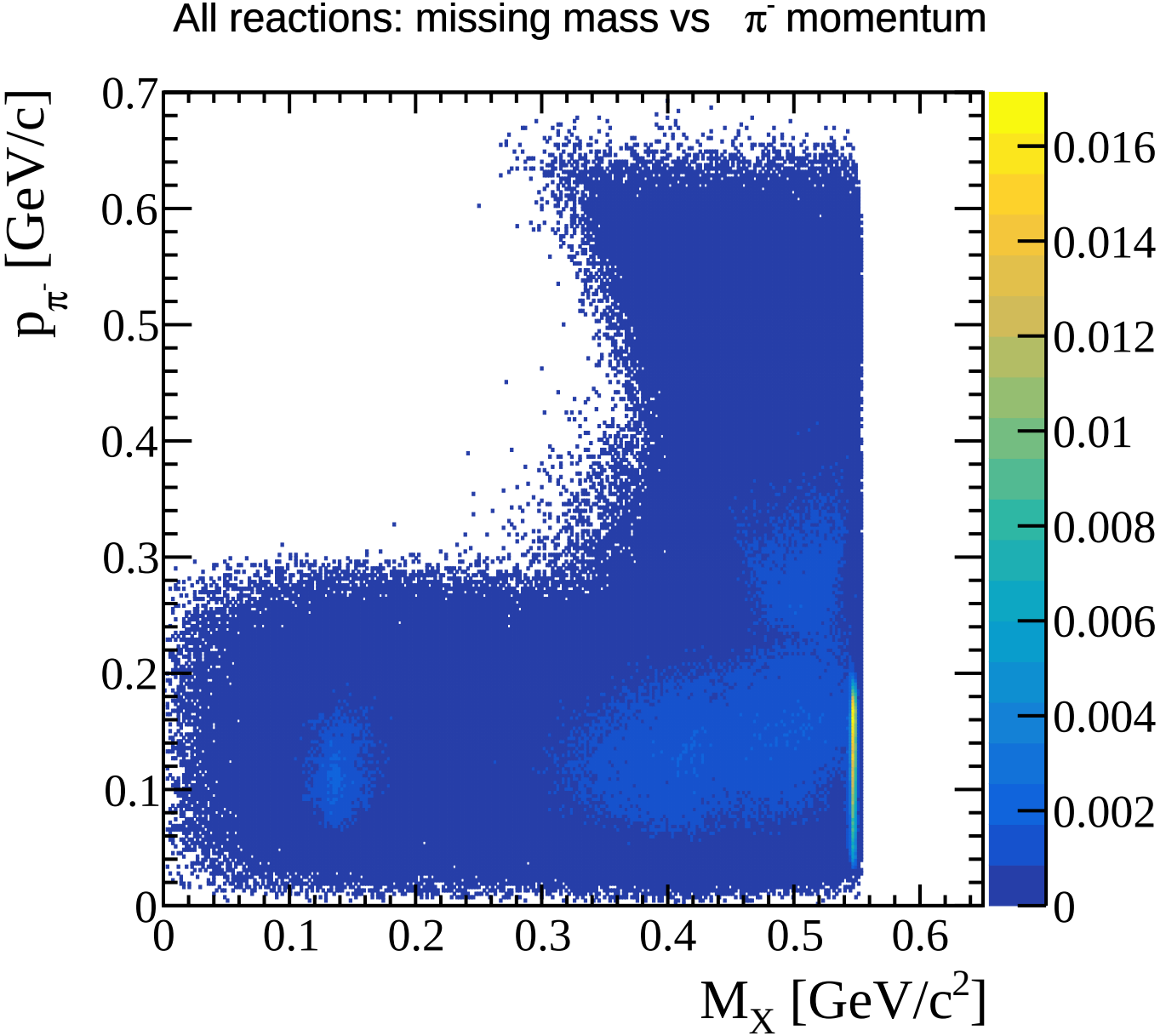

$\Sigma^0\pi^0$: reconstructed mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ^-) [GeV/c^2]
0.012
0.01
0.008
0.006
0.004
0.002
0

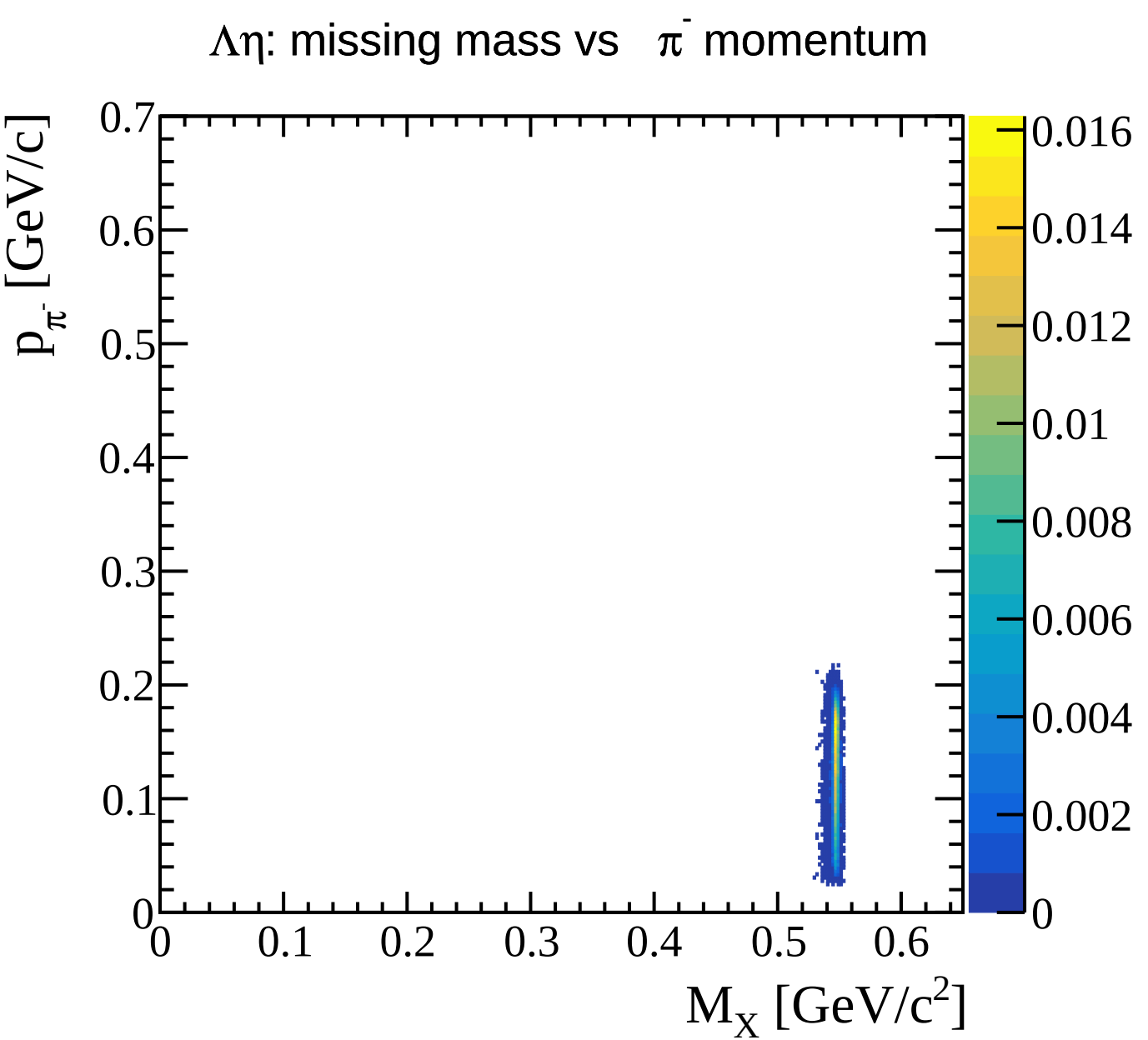

Σ+π-: reconstructed mass vs π- momentum
p_π- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c2]
×10−3
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0

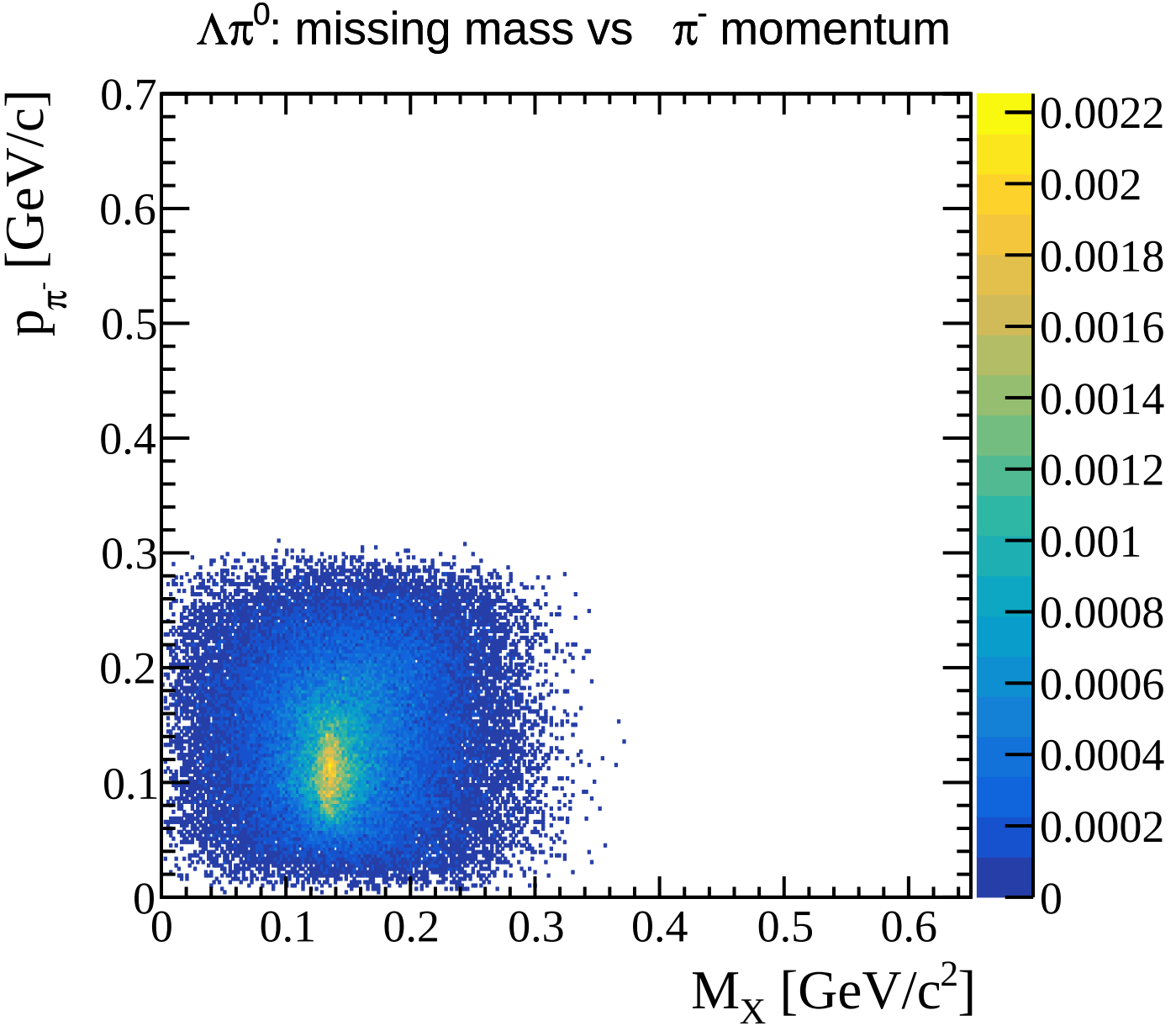

Σ+π-π0: reconstructed mass vs π- momentum
pπ- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c2]
×10−3
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0

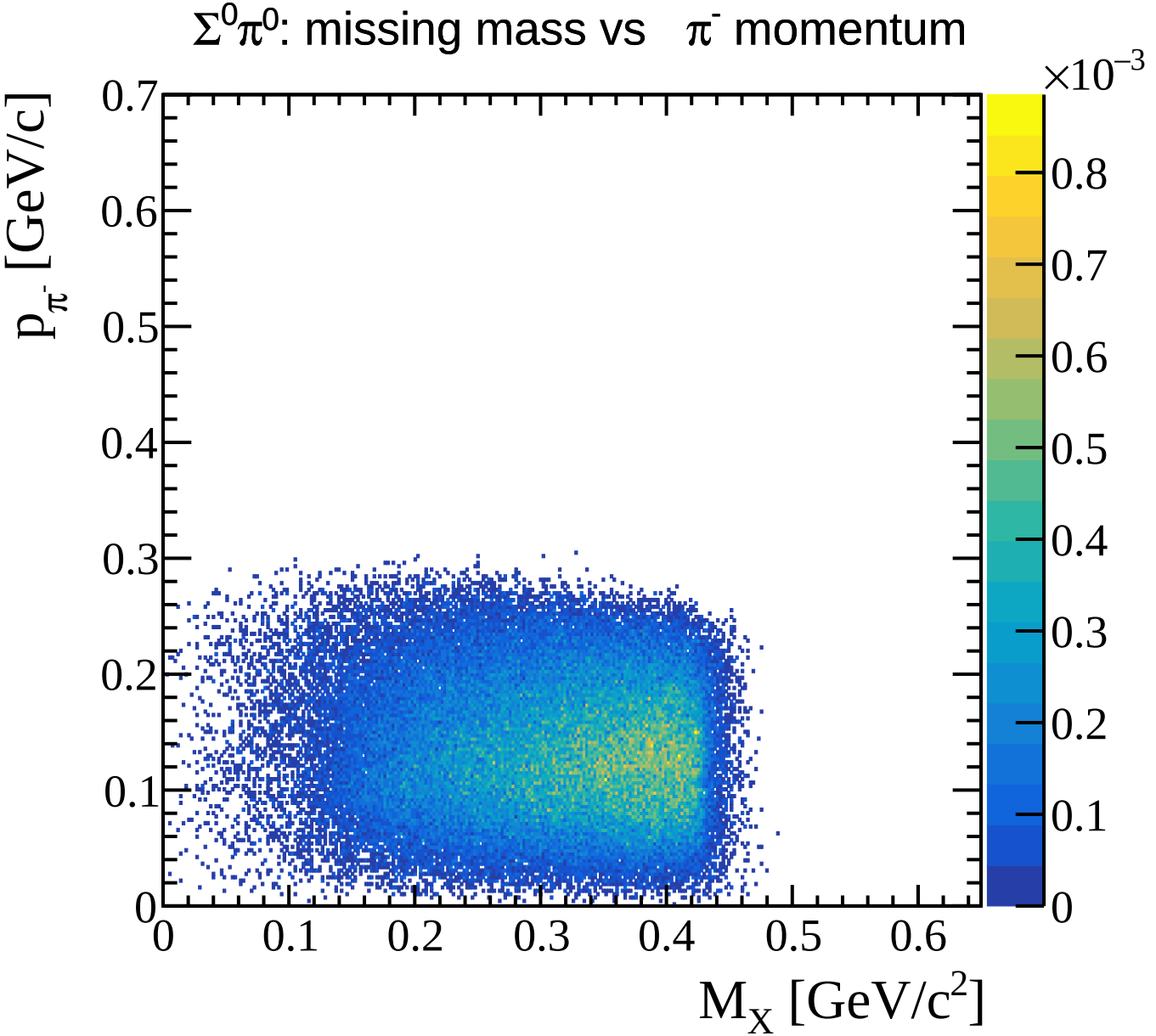

Λπ+π-: reconstructed mass vs π- momentum
p_π- [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
M(pπ-) [GeV/c2]
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0

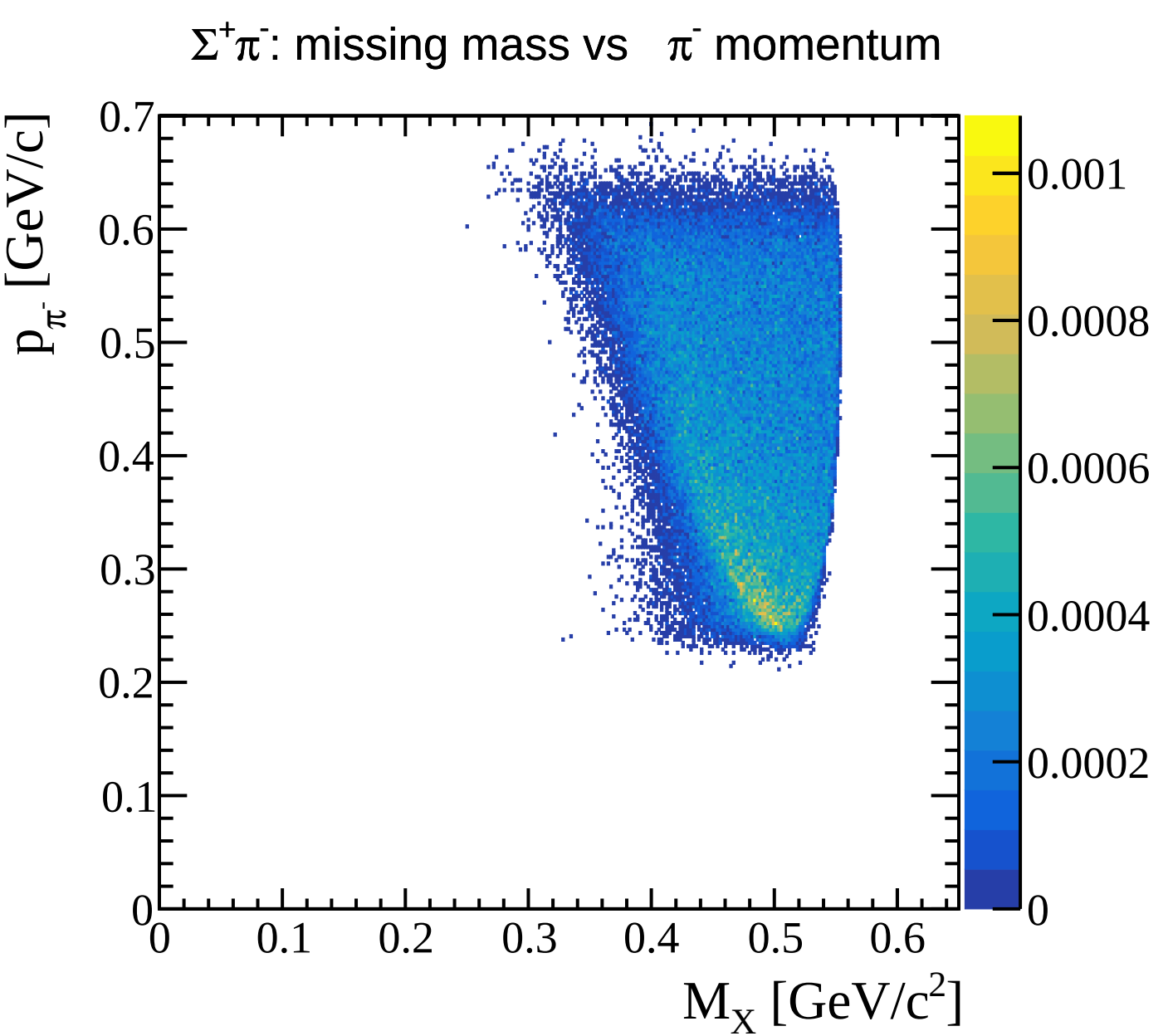

$\Sigma^0\pi^+\pi^-$: reconstructed mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1.2
1.4
$M(p\pi^-)$ [GeV/c^2]
0.0035
0.003
0.0025
0.002
0.0015
0.001
0.0005
0

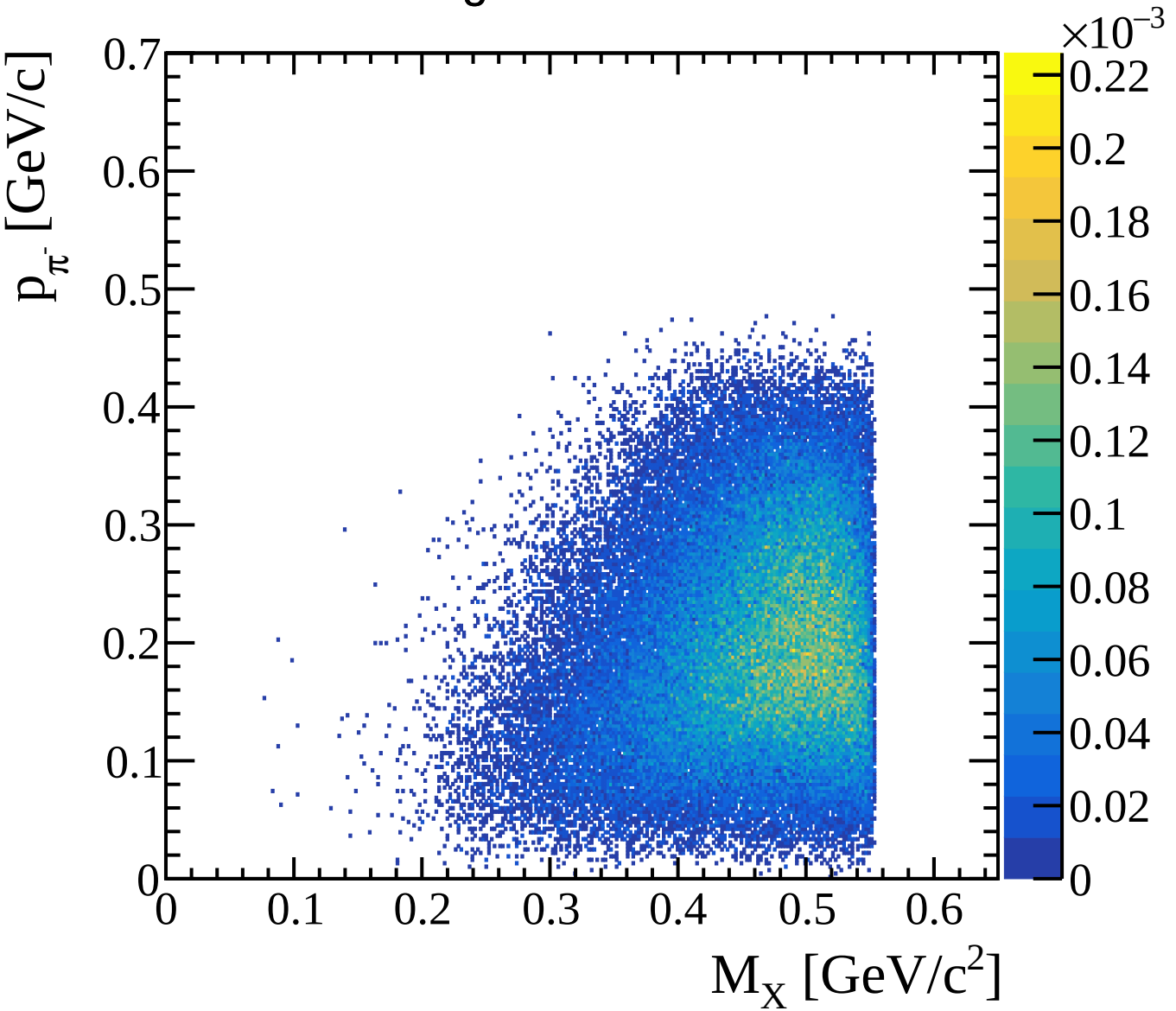

All reactions: missing mass vs π⁻ momentum
p_π⁻ [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c²]
0.016
0.014
0.012
0.01
0.008
0.006
0.004
0.002
0

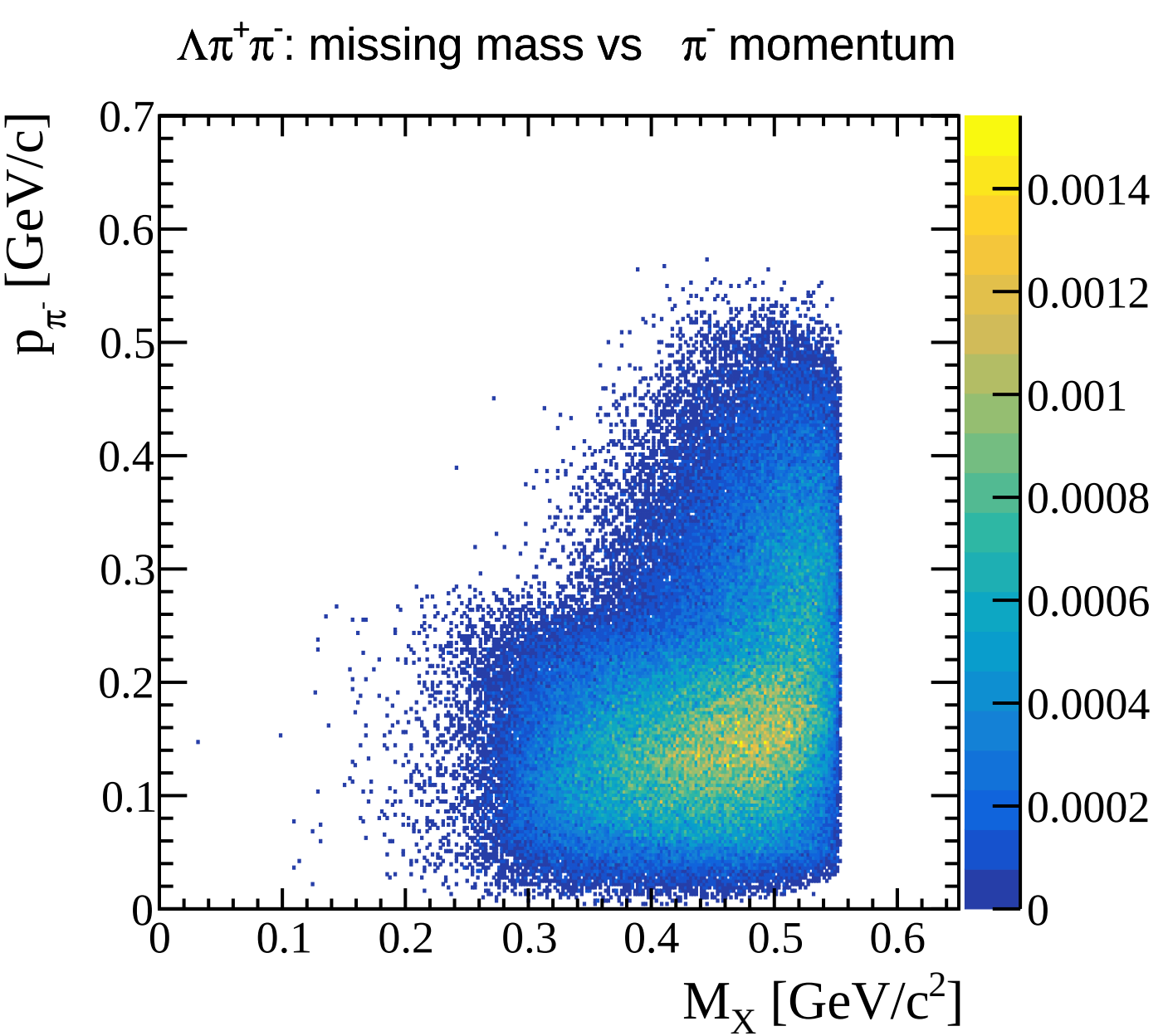

Λη: missing mass vs π⁻ momentum
p_π⁻ [GeV/c]
M_X [GeV/c²]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.002
0.004
0.006
0.008
0.01
0.012
0.014
0.016

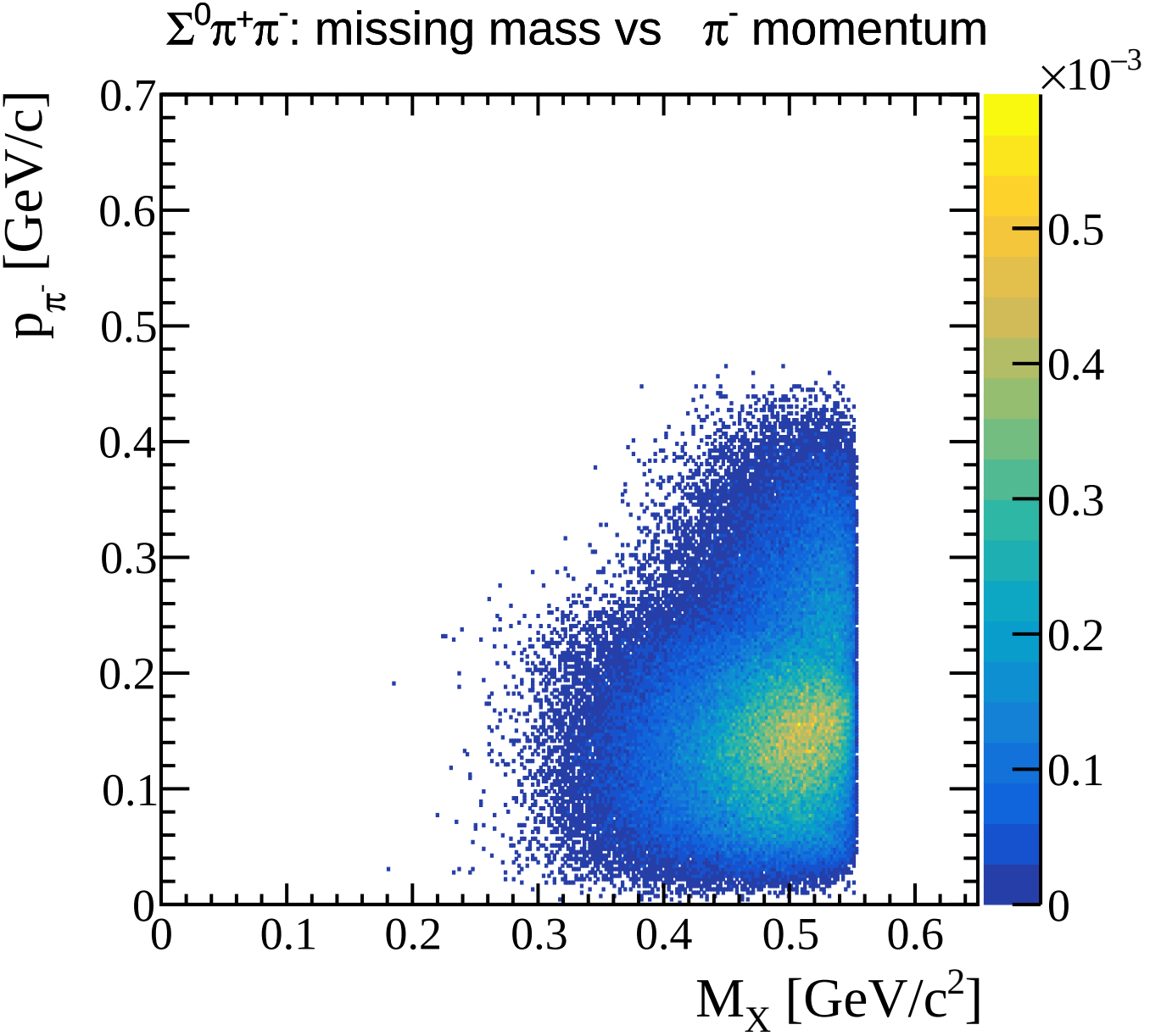

$\Lambda\pi^0$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
0.0022
0.002
0.0018
0.0016
0.0014
0.0012
0.001
0.0008
0.0006
0.0004
0.0002
0

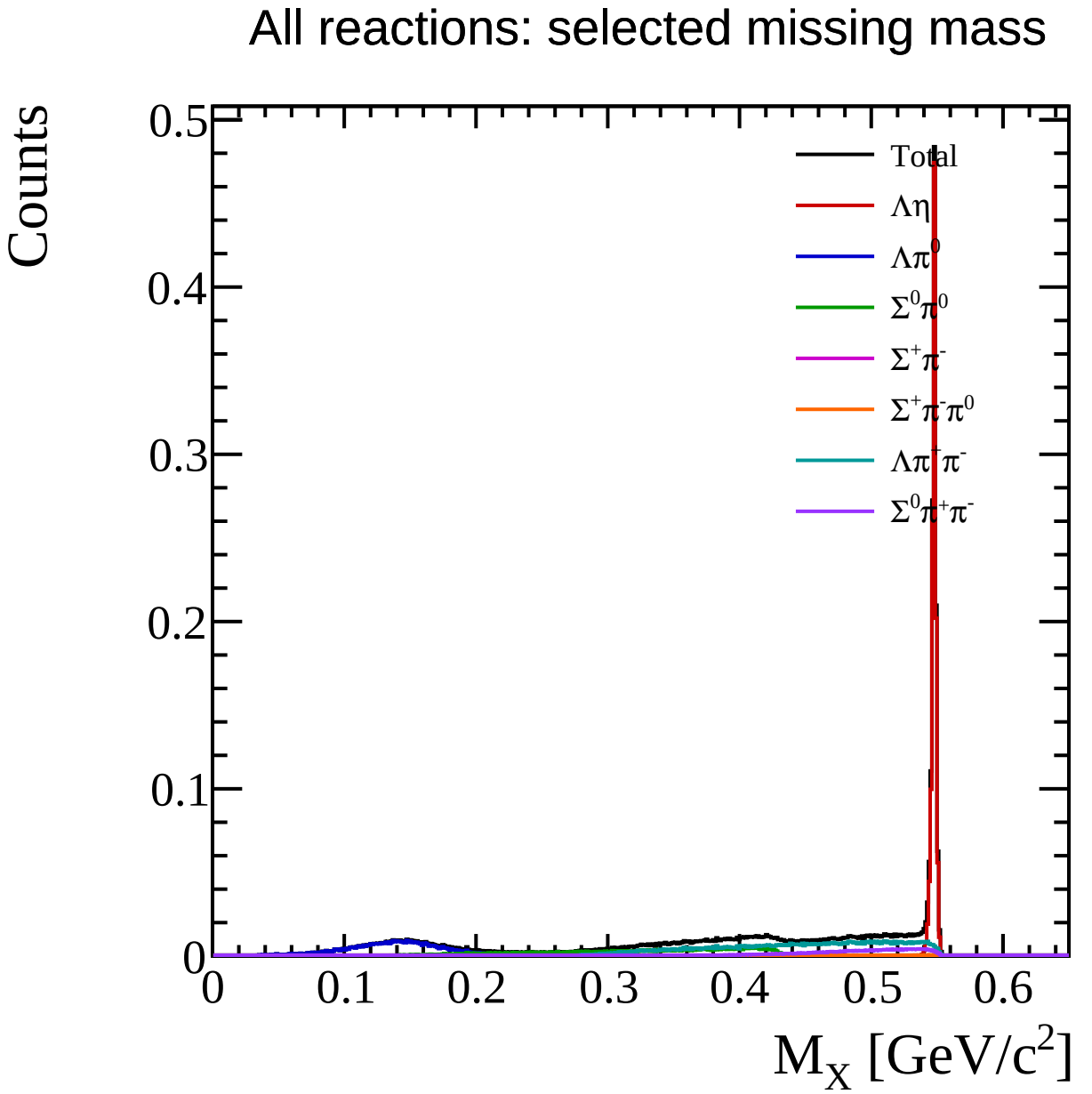

$\Sigma^0\pi^0$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
M_X [GeV/c^2]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
$\times 10^{-3}$
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8

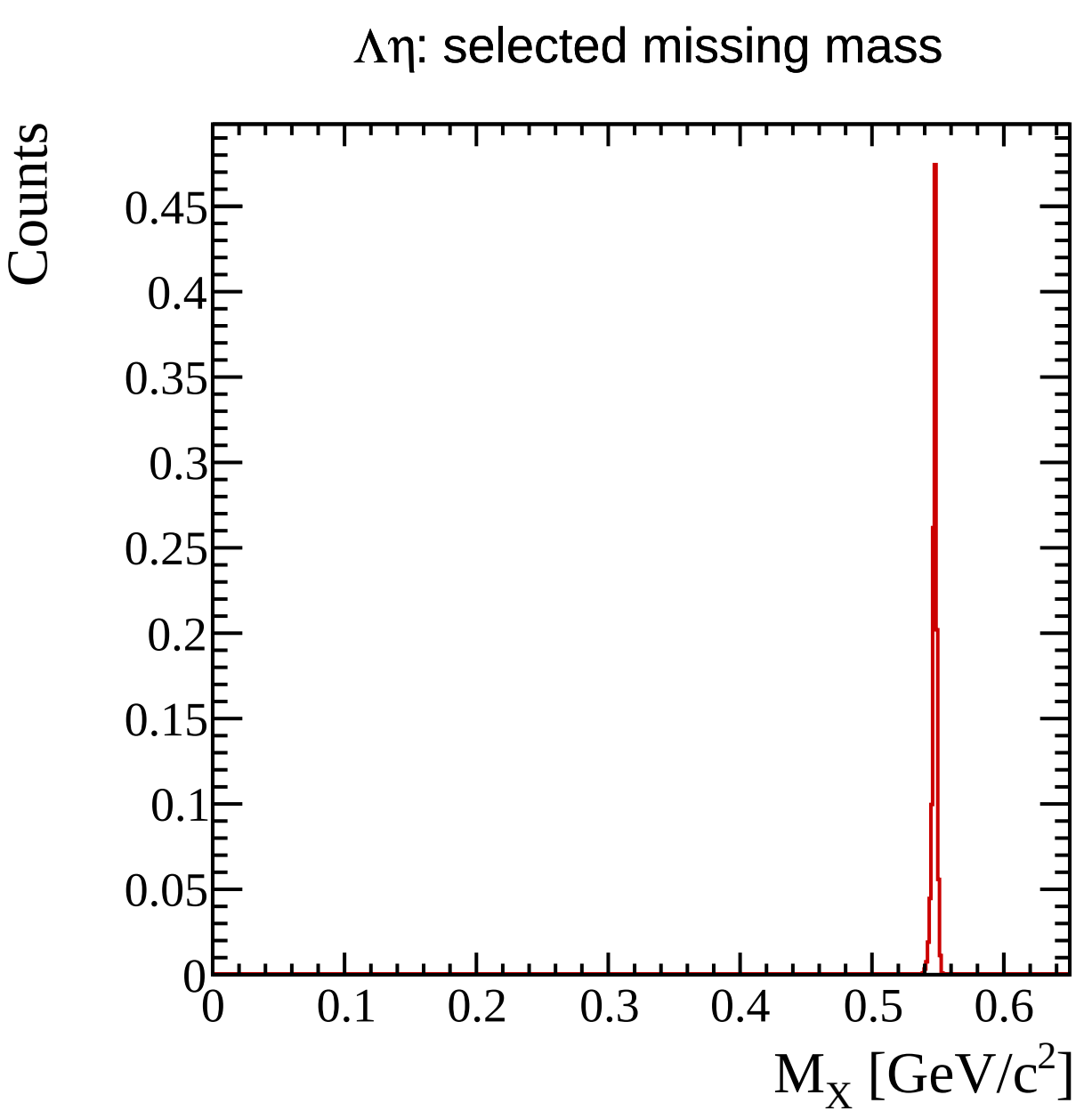

$\Sigma^+\pi^-$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
0.001
0.0008
0.0006
0.0004
0.0002
0

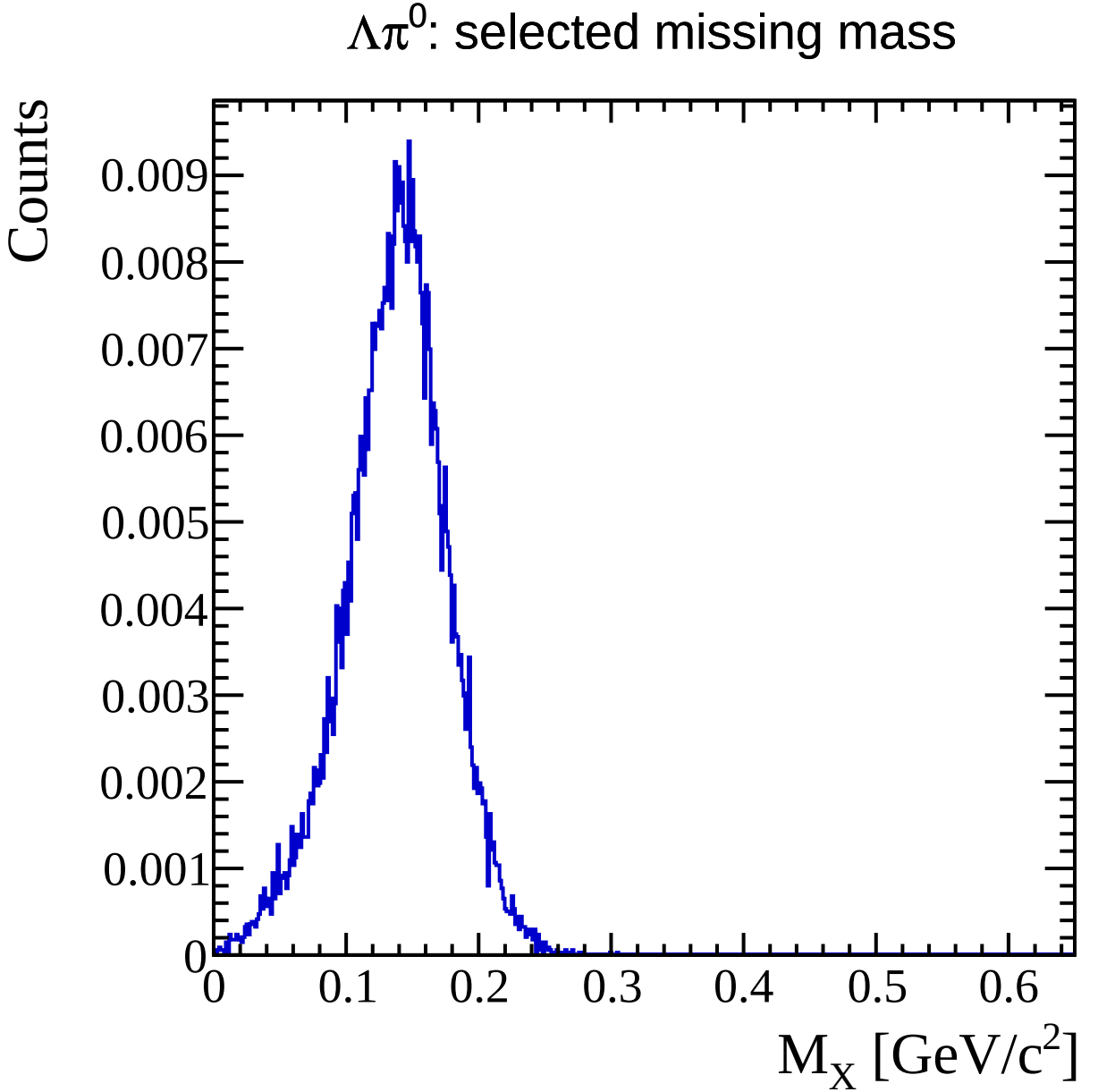

$\Sigma^+\pi^-\pi^0$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
M_X [GeV/c^2]
$\times 10^{-3}$
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.22
0.2
0.18
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0

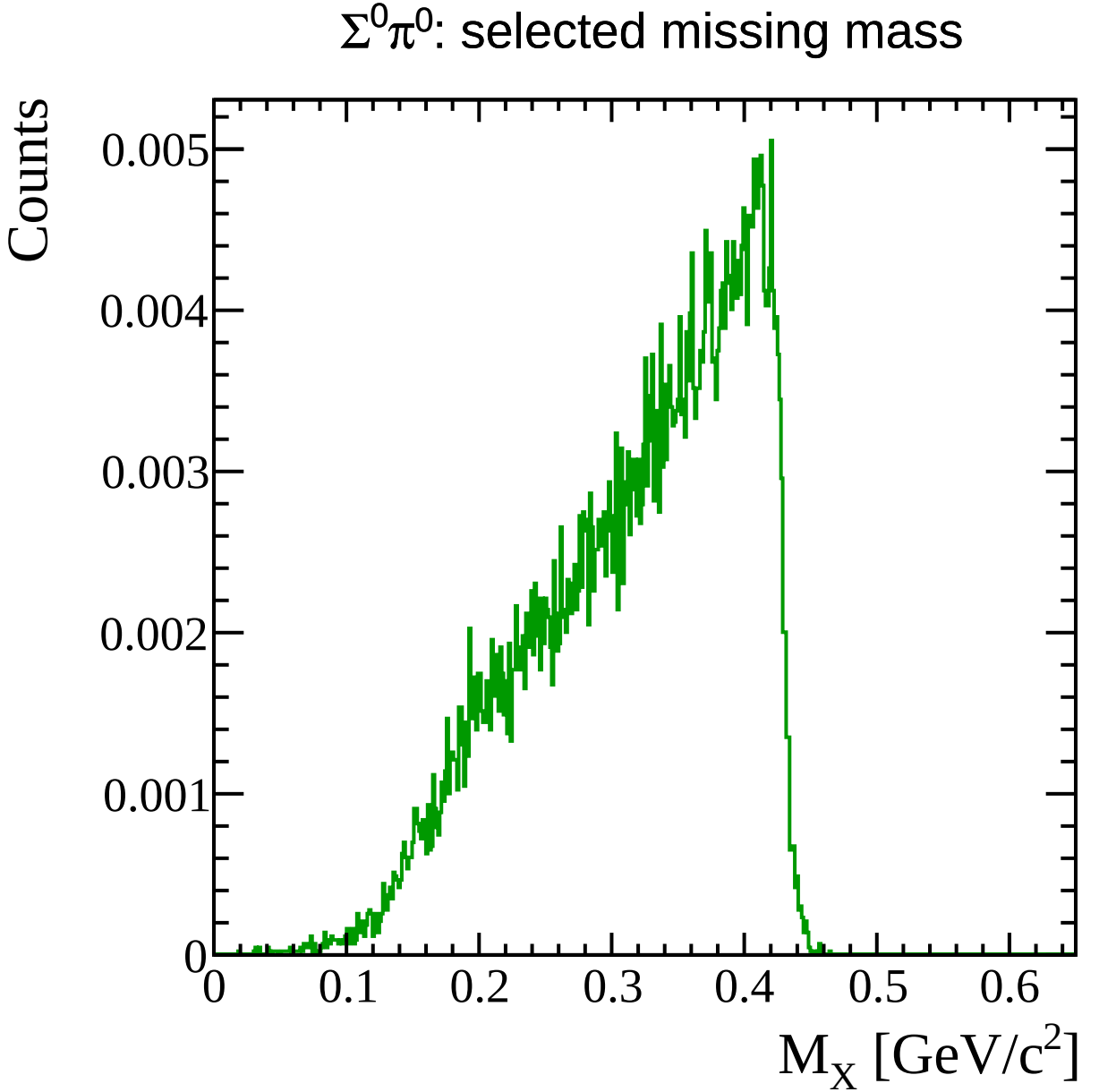

$\Lambda\pi^+\pi^-$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
0.0014
0.0012
0.001
0.0008
0.0006
0.0004
0.0002
0

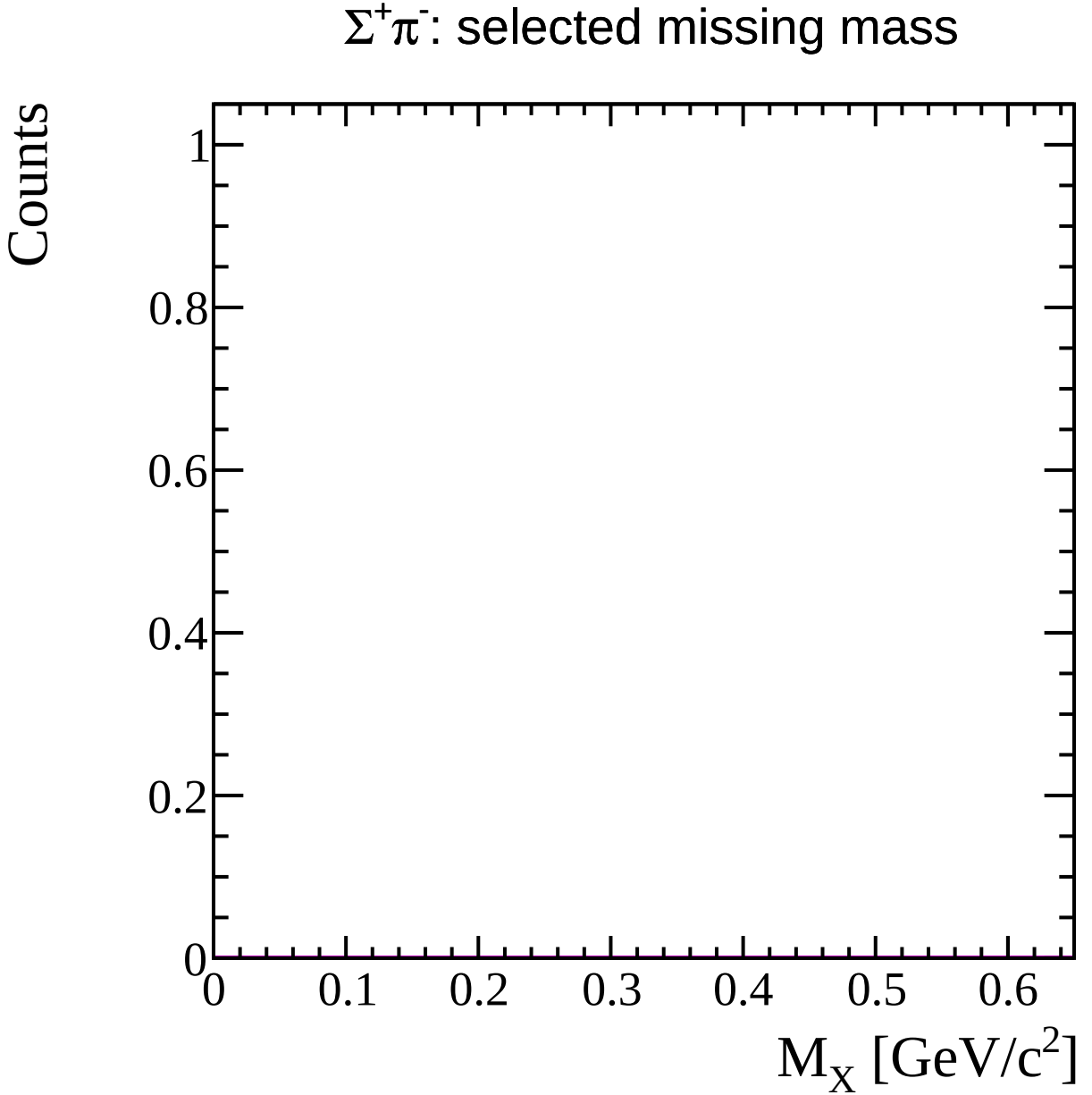

$\Sigma^0\pi^+\pi^-$: missing mass vs π^- momentum
p_{π^-} [GeV/c]
M_X [GeV/c^2]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
$\times 10^{-3}$

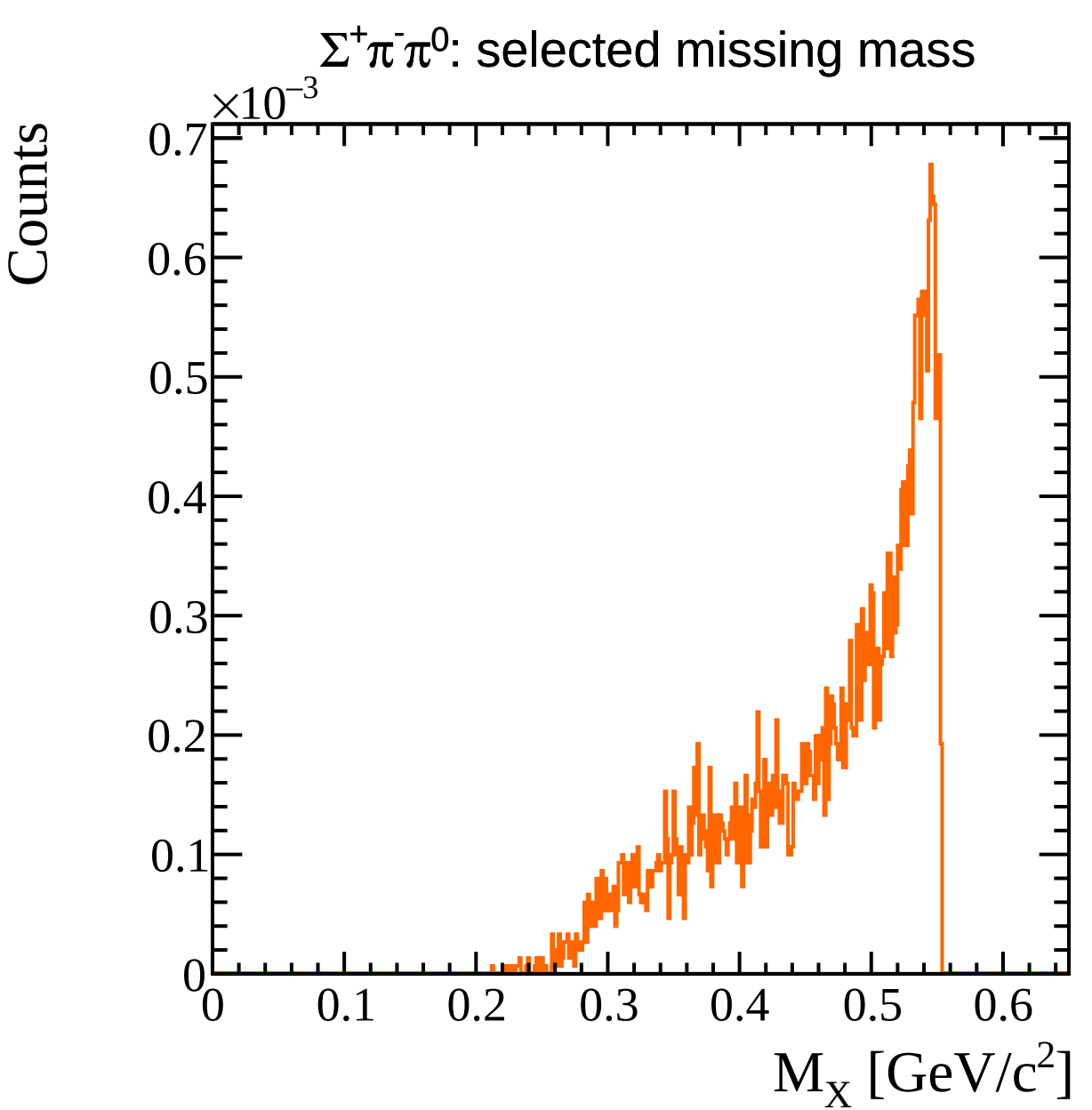

All reactions: selected missing mass
Counts
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]
Total
Λη
$\Lambda\pi^0$
$\Sigma^0\pi^0$
$\Sigma^+\pi^-$
$\Sigma^+\pi^-\pi^0$
$\Lambda\pi^+\pi^-$
$\Sigma^0\pi^+\pi^-$

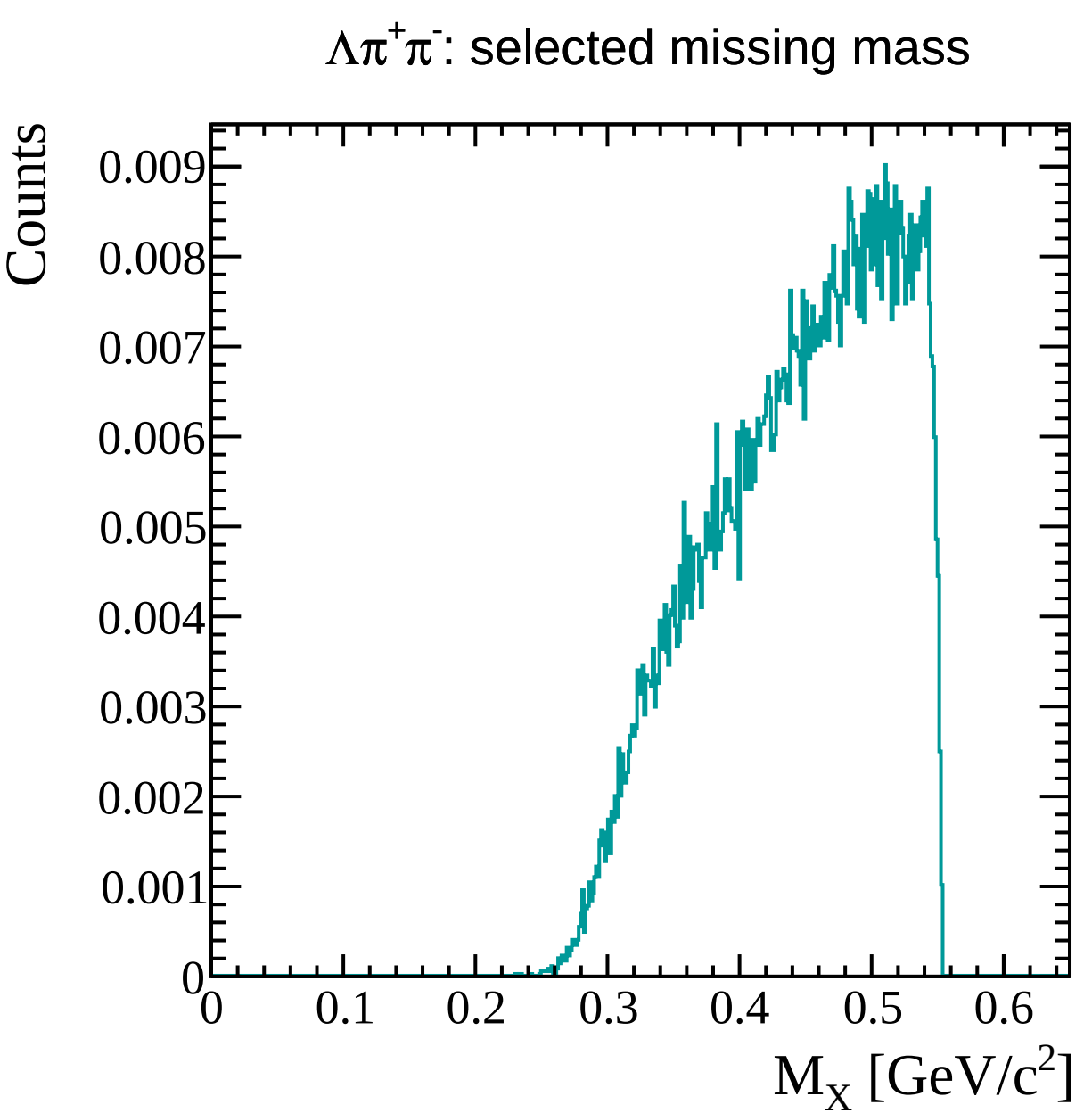

Λη: selected missing mass
Counts
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]

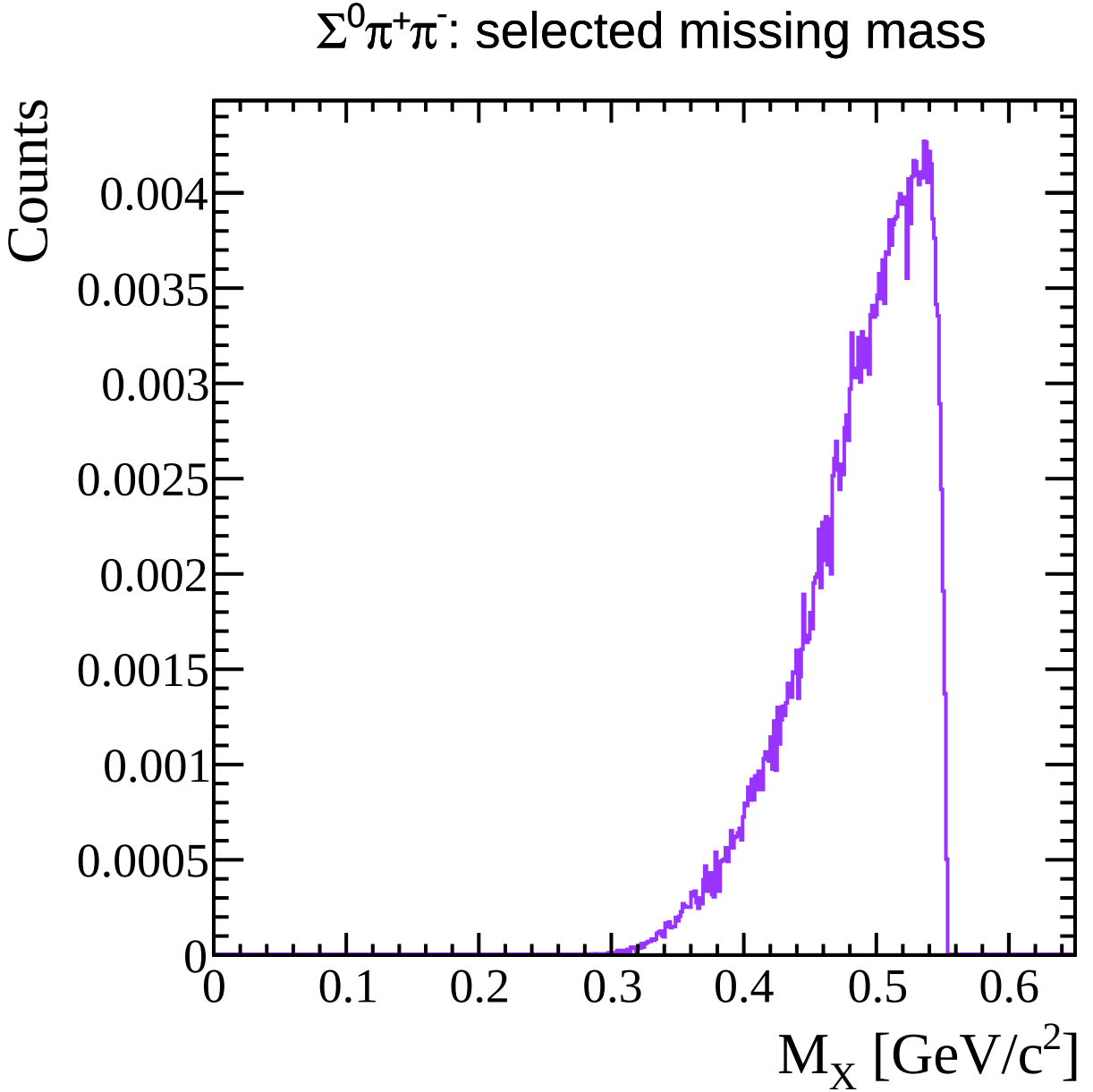

$\Lambda\pi^0$: selected missing mass
Counts
0.009
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]

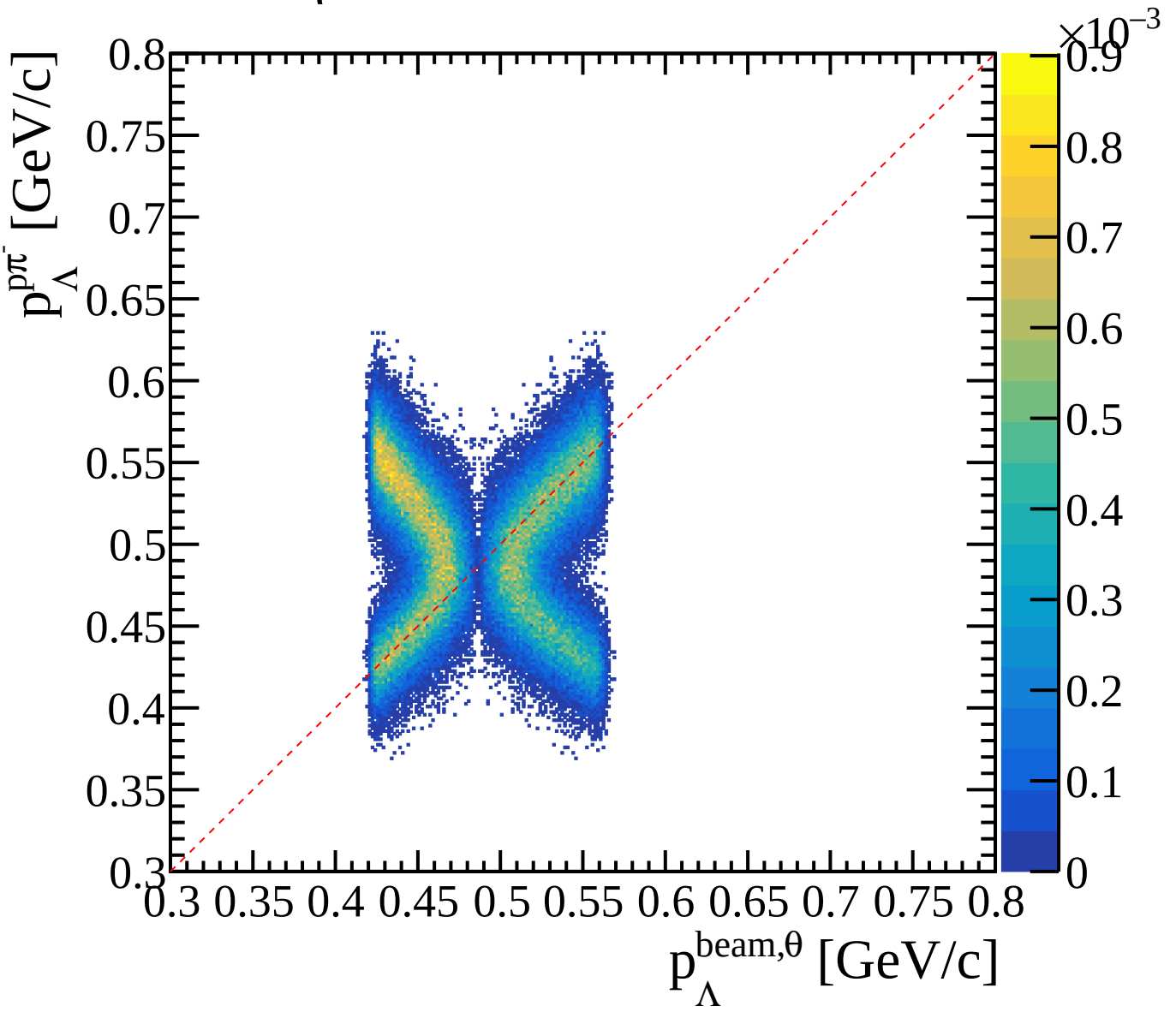

$\Sigma^0\pi^0$: selected missing mass
Counts
0.005
0.004
0.003
0.002
0.001
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]


$\Sigma^+\pi^-$: selected missing mass
Counts
1
0.8
0.6
0.4
0.2
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]


$\Sigma^+\pi^-\pi^0$: selected missing mass
Counts
$\times 10^{-3}$
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]


Λπ+π-: selected missing mass
Counts
0.009
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0
0
0.1
0.2
0.3
0.4
0.5
0.6
MX [GeV/c2]


$\Sigma^0\pi^+\pi^-$: selected missing mass
Counts
0.004
0.0035
0.003
0.0025
0.002
0.0015
0.001
0.0005
0
0
0.1
0.2
0.3
0.4
0.5
0.6
M_X [GeV/c^2]


Λη: both momentum solutions
$p_\Lambda^{p\pi^-}$ [GeV/c]
$p_\Lambda^{beam,\theta}$ [GeV/c]
$\times 10^{-3}$
0.8
0.75
0.7
0.65
0.6
0.55
0.5
0.45
0.4
0.35
0.3
0.3 0.35 0.4 0.45 0.5 0.55 0.6 0.65 0.7 0.75 0.8
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0